

SCOTT YUAN RESEARCH

ALPHABET INC.

GOOG / GOOGL

FULL EQUITY RESEARCH REPORT

Search resilience, Cloud acceleration, AI monetization and the economics of the infrastructure build-out

Independent Research | Institutional Black & White Edition

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Independent research on technology, AI infrastructure, capital markets and valuation.

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Detailed supporting tables, formulas, assumptions and sensitivities are retained within each section.

Measurement convention: percentages (%) denote absolute levels, revenue mix and growth rates; bps denote changes or differences in margins, rates and yields. 100 bps = 1.0 percentage point.

Data-availability convention: N/A = not available or not applicable; N/M = not meaningful; No published consensus indicates that a reliable market estimate was not available for the specific line item.

Alphabet Q2 2026 Earnings Scorecard

1. Headline Financial Results

Metric	Q2 Actual	Consensus	Beat / Miss	YoY Change	QoQ Change	Assessment
Total Revenue	\$119.80B	\$117.02B	+\$2.78B / +2.4%	+24.2%	+9.0%	Meaningful beat
Revenue ex-TAC	\$103.62B	\$101.07B	+\$2.55B / +2.5%	+26.8%	+9.5%	Meaningful beat
Operating Income	\$40.77B	\$40.55B	+\$0.22B / +0.5%	+30.4%	+2.7%	Modest beat
Operating Margin	34.0%	Approximately 34.7%	Approximately -70 bps	+160 bps	-210 bps	Strong revenue, but mixed margin performance
Net Income	\$112.19B	Not meaningful	N/M	+297.9%	+79.3%	Materially inflated by investment gains
GAAP EPS	\$9.11	Approximately \$2.89	Large apparent beat	+294%	+78%	Not directly comparable
Adjusted EPS	Approximately \$2.85	\$2.89	-\$0.04 / -1.4%	+27.8%	+3.3%	Core earnings slightly below expectations

The reported operating margin consensus is estimated by dividing expected operating income by expected revenue.

Adjusted EPS excludes major equity investment gains and other non-operating items. Alphabet does not provide a fully standardized non-GAAP EPS measure in its Form 10-Q.

The most important distinction is that Alphabet's reported GAAP EPS of \$9.11 should not be interpreted as recurring operating earnings. Other income and expense reached approximately \$97.98 billion, including roughly \$99.03 billion of net equity investment gains, largely associated with valuation increases in private-company investments.

2. Revenue Segment Scorecard

Business Segment	Q2 Actual	Consensus	Beat / Miss	YoY Growth	QoQ Growth	Assessment
Google Services	\$94.54B	\$94.32B	+\$0.22B / +0.2%	+14.5%	+5.5%	Broadly in line

Business Segment	Q2 Actual	Consensus	Beat / Miss	YoY Growth	QoQ Growth	Assessment
Google Advertising	\$81.63B	\$81.12B	+\$0.51B / +0.6%	+14.4%	+5.7%	Modest beat
Google Search & Other	\$63.27B	\$63.28B	-\$0.01B	+16.8%	+4.8%	Essentially in line
YouTube Advertising	\$11.06B	\$10.81B	+\$0.25B / +2.3%	+12.9%	+11.9%	Positive beat
Google Network	\$7.30B	No published consensus	N/A	-0.7%	+4.8%	Remains structurally weak
Subscriptions, Platforms and Devices	\$12.91B	No published consensus	N/A	+15.2%	+4.3%	Solid growth
Google Cloud	\$24.77B	\$22.46B	+\$2.31B / +10.3%	+81.8%	+23.7%	Largest upside surprise
Other Bets	\$0.38B	No published consensus	N/A	+2.4%	-7.1%	Still immaterial to consolidated revenue

Approximately 83% of Alphabet's total revenue beat was attributable to Google Cloud:

Google Cloud share of total revenue beat = \$2.31B / \$2.78B = 83.1%

The revenue composition can therefore be summarized as follows:

- Google Search was broadly in line with expectations.
- YouTube advertising modestly exceeded expectations.
- Google Cloud materially outperformed expectations.

The appropriate operating interpretation is:

Alphabet's advertising businesses remained resilient, while Google Cloud delivered a significant upward inflection.

3. Segment Profitability Scorecard

Metric	Q2 Actual	Consensus	YoY Change	QoQ Change	Assessment
Google Services Operating Income	\$39.54B	No published consensus	+19.6%	-2.6%	Strong YoY growth, but lower sequentially

Metric	Q2 Actual	Consensus	YoY Change	QoQ Change	Assessment
Google Services Operating Margin	41.8%	No published consensus	+177 bps	-345 bps	Pressured by legal costs and investment spending
Google Cloud Operating Income	\$8.81B	Approximately \$6.92B	+211.9%	+33.6%	Significant outperformance
Google Cloud Operating Margin	35.6%	30.8%	+1,484 bps	+264 bps	Material margin beat
Other Bets Operating Loss	-\$1.80B	No published consensus	Loss widened by \$0.55B	Loss narrowed by \$0.30B	Remains in a high-investment phase
Alphabet-Level Activities Loss	-\$5.79B	No published consensus	Loss widened by \$2.42B	Loss widened by \$0.40B	Shared AI and corporate costs increased

The Google Cloud operating income consensus is estimated using expected Cloud revenue of \$22.46 billion multiplied by the approximately 30.8% expected operating margin.

Google Cloud generated approximately \$8.81 billion of operating income and achieved a 35.6% operating margin, substantially above market expectations. The combination of a large revenue beat and strong margin expansion makes Google Cloud the highest-quality component of the quarter.

However, Alphabet's consolidated operating margin was only 34.0%, down approximately 210 bps from 36.1% in Q1. The positive Cloud operating leverage was partially offset by:

- Higher shared AI research and development expenses;
- Increased depreciation and technical infrastructure costs;
- Approximately \$2.1 billion of PriceRunner-related legal expenses;
- Higher employee compensation and promotional spending.

4. Cash Flow and Capital Expenditure Scorecard

Metric	Q2 Actual	Consensus	Beat / Miss	YoY Change	QoQ Change	Assessment
Operating Cash Flow	\$39.07B	No published consensus	N/A	+40.8%	-14.7%	Core cash generation remained strong

Metric	Q2 Actual	Consensus	Beat / Miss	YoY Change	QoQ Change	Assessment
Capital Expenditures	\$44.92B	\$44.15B	\$0.77B above expectations	+100.1%	+25.9%	Extremely high investment intensity
Free Cash Flow	-\$5.86B	Approximately +\$0.09B	Approximately \$5.95B below expectations	Declined by \$11.16B	Declined by \$15.97B	Largest financial negative
Free Cash Flow Margin	-4.9%	Approximately 0%	Approximately -490 bps	-1,039 bps	-1,409 bps	Significant deterioration in cash conversion
FY2026 CapEx Guidance	\$195B-\$205B	Previous guidance of \$180B-\$190B	Midpoint raised by \$15B	N/A	N/A	Material upward revision

Alphabet generated approximately \$39.1 billion of operating cash flow during the quarter. However, capital expenditures reached approximately \$44.9 billion, resulting in negative free cash flow of approximately \$5.9 billion.

Management also raised its full-year 2026 capital expenditure guidance from \$180-\$190 billion to \$195-\$205 billion.

Higher-than-expected capital expenditure should not be interpreted as a conventional positive earnings beat. While elevated infrastructure spending may benefit Alphabet's suppliers and increase future computing capacity, it has an immediate negative effect on Alphabet's free cash flow:

Higher CapEx -> Lower FCF -> Greater near-term valuation pressure

Overall Earnings Scorecard

Dimension	Rating	Rationale
Total Revenue	Strong	Revenue exceeded expectations by approximately 2.4%
Google Search	Neutral to Positive	Strong growth, but broadly in line with expectations
YouTube	Positive	Revenue exceeded expectations and accelerated sequentially
Google Cloud Revenue	Very Strong	Approximately 10% above consensus
Google Cloud Margin	Very Strong	35.6%, approximately 480 bps above expectations

Dimension	Rating	Rationale
Consolidated Operating Income	Neutral to Positive	Modest dollar beat, but weaker-than-expected margin
Core EPS	Slightly Weak	Adjusted earnings were modestly below expectations
Earnings Quality	Weak	GAAP earnings were dominated by unrealized investment gains
Free Cash Flow	Very Weak	Free cash flow declined to approximately negative \$5.9 billion
Capital Expenditure Outlook	Negative / High Risk	Full-year guidance was raised materially

Bottom-Line Assessment

Alphabet delivered a quarter characterized by exceptionally strong Google Cloud performance, resilient advertising revenue, modest underlying earnings performance, and substantial free cash flow pressure.

The quarter can be summarized as:

Quarter summary: Google Cloud outperformance + Stable Search - CapEx and FCF deterioration

This should not be described as a uniformly strong earnings report. Revenue growth and Google Cloud profitability were clearly positive, but adjusted earnings were less impressive and free cash flow deteriorated materially.

The central investment question following the quarter is:

Can the growth in Google Cloud and Search operating profits continue to outpace the growth in capital expenditures, depreciation, infrastructure operating expenses, and financing costs?

2. Revenue and Segment Analysis

2.1 Revenue Overview

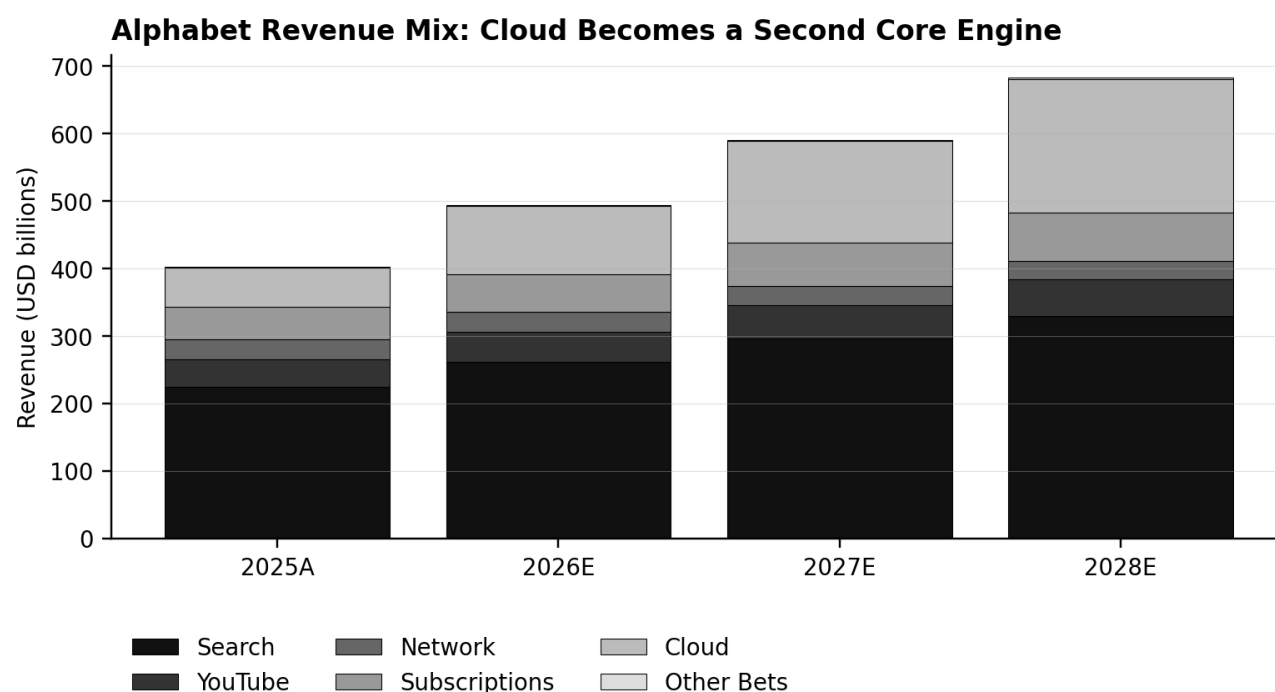


Figure 1. Alphabet revenue mix: Google Cloud becomes a second core growth engine. Source: company filings and analyst estimates.

Alphabet generated \$119.8 billion of revenue in Q2 2026, representing growth of 24.2% year over year and 9.0% sequentially. Revenue increased approximately 23% on a constant-currency basis, indicating that foreign exchange contributed approximately 100 bps to reported growth.

Revenue Category	Q2 2025	Q1 2026	Q2 2026	YoY Growth	QoQ Growth	Q2 2026 Revenue Mix
Google Search & Other	\$54.19B	\$60.40B	\$63.27B	+16.8%	+4.8%	52.8%
YouTube Advertising	\$9.80B	\$9.88B	\$11.06B	+12.9%	+11.9%	9.2%
Google Network	\$7.35B	\$6.97B	\$7.30B	-0.7%	+4.8%	6.1%
Subscriptions, Platforms and Devices	\$11.20B	\$12.38B	\$12.91B	+15.2%	+4.3%	10.8%
Google Services	\$82.54B	\$89.64B	\$94.54B	+14.5%	+5.5%	78.9%
Google Cloud	\$13.62B	\$20.03B	\$24.77B	+81.8%	+23.7%	20.7%
Other Bets	\$0.37B	\$0.41B	\$0.38B	+2.4%	-7.1%	0.3%
Hedging Gains	-\$0.11B	-\$0.18B	\$0.11B	N/M	N/M	0.1%
Total Alphabet Revenue	\$96.43B	\$109.90B	\$119.80B	+24.2%	+9.0%	100.0%

Q1 2026 figures are calculated by subtracting the reported Q2 figures from Alphabet's first-half 2026 results. Revenue mix and growth rates are derived from Alphabet's reported figures.

The quarter reflected a meaningful shift in Alphabet's revenue mix toward Google Cloud. Cloud represented approximately **20.7% of consolidated revenue**, up from approximately **14.1% in Q2 2025**. Google Services remained the largest business, but its share of Alphabet revenue declined from approximately 85.6% to 78.9% as Cloud expanded substantially faster.

Contribution to Year-over-Year Revenue Growth

Alphabet added approximately **\$23.37 billion** of revenue compared with Q2 2025.

Revenue Category	YoY Revenue Increase	Contribution to Total Revenue Growth
Google Search & Other	+\$9.08B	38.9%
YouTube Advertising	+\$1.26B	5.4%
Google Network	-\$0.05B	-0.2%
Subscriptions, Platforms and Devices	+\$1.71B	7.3%
Google Cloud	+\$11.14B	47.7%
Other Bets	+\$0.01B	Approximately 0%

Google Cloud and Search together generated approximately **86.6% of Alphabet's total year-over-year revenue increase**. This represents a more balanced growth structure than in prior periods, when Alphabet depended more heavily on advertising.

The principal revenue conclusion is:

Alphabet's growth profile is transitioning from an advertising-dominated model toward a two-engine model supported by Search monetization and rapidly expanding enterprise AI and Cloud demand.

2.2 Google Search & Other

Google Search & Other revenue increased **16.8% year over year** to **\$63.27 billion**, accounting for approximately 52.8% of Alphabet's consolidated revenue and 67% of Google Services revenue.

The increase was driven by:

- Higher search-query volume;
- Increased mobile usage and user adoption;
- Growth in advertiser spending;
- Improvements in advertising formats and delivery;
- Strength in the retail and financial-services verticals.

Search Monetization Bridge

Alphabet reported the following year-over-year changes in Search monetization metrics:

Search Monetization Metric	Q2 2026 YoY Change
Paid Clicks	+13%

Search Monetization Metric	Q2 2026 YoY Change
Cost per Click	+3%
Search & Other Revenue	+16.8%

The combination of higher paid-click volume and higher cost per click explains most of the reported Search revenue growth.

Conceptually:

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Search revenue growth ≈ Paid-click growth + Cost-per-click growth + Mix effects

The 13% increase in paid clicks indicates that growth was principally volume-driven rather than dependent entirely on advertiser pricing. The additional 3% increase in cost per click suggests healthy advertiser demand and continued improvements in ad relevance and conversion performance.

Impact of AI on Search

Management stated that the integration of AI Overviews and AI Mode continued to increase overall Search usage and generate incremental queries. AI Mode surpassed one billion monthly active users, while engineering and hardware optimization reduced the cost of generating AI Mode responses during the quarter.

The advertising implications are important. Management indicated that monetization on queries displaying AI Overviews remained encouraging, including as AI Overviews expanded into more commercially oriented searches. Users of AI-powered Search products are also asking longer and more specific questions, potentially creating additional opportunities to deliver highly relevant advertisements.

Google is applying Gemini across three primary components of its advertising infrastructure:

1. **Ad quality and query understanding:** Gemini improves Google’s ability to interpret complex searches and identify relevant commercial intent.
2. **Advertiser tools:** AI Max and Performance Max automate campaign targeting, creative generation and bidding.
3. **New AI advertising formats:** Google is experimenting with contextual advertisements, sponsored links and direct offers within AI Mode.

Management stated that advertisers using AI Max or Performance Max achieved, on average, approximately 15% more conversions or value at a similar return on advertising spending. AI Max had been adopted by approximately 500,000 advertisers.

Search Revenue Quality

The quality of Search revenue growth was strong for three reasons:

- Growth was supported by both higher engagement and higher pricing;
- Paid clicks increased substantially, suggesting continued user activity;
- AI functionality appears to be expanding query volume rather than immediately cannibalizing traditional Search usage.

However, the company does not disclose separate revenue, query volume, cost per query or advertising yield for AI Overviews and AI Mode. It therefore remains impossible to determine precisely whether AI-generated searches have the same unit economics as traditional searches.

A more complete assessment requires monitoring:

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AI Search contribution spread = Revenue per search - AI inference cost per search

Even when AI increases total queries, shareholder value will depend on whether incremental advertising revenue exceeds the additional computing and infrastructure cost required to serve those queries.

Search Outlook

The Search outlook remains positive, but growth comparisons are expected to become more difficult. Management stated that Alphabet will begin lapping the acceleration in Search growth that began in Q3 2025. The company also expects a slight foreign-exchange headwind in Q3 2026, compared with an approximately 100 bps currency benefit in Q2, primarily affecting Search and YouTube advertising.

Search Assessment

Factor	Assessment
Query and click growth	Strong
Advertising pricing	Positive
AI-related engagement	Positive
AI monetization evidence	Encouraging but incomplete
Competitive position	Resilient
Near-term comparison base	More difficult
Revenue sustainability	High, subject to AI unit economics

Search remains Alphabet's primary cash-generating franchise. Q2 results provide evidence that generative AI is currently expanding Search activity rather than causing material revenue disruption, although the long-term economics of AI-powered queries remain insufficiently disclosed.

2.3 YouTube Advertising

YouTube advertising revenue increased **12.9% year over year** and **11.9% sequentially** to **\$11.06 billion**.

Growth was driven by both:

- Direct Response advertising;
- Brand advertising;
- Continued strength in connected television and living-room viewing;
- Advertising activity related to the 2026 FIFA World Cup.

The World Cup provided a meaningful engagement and advertising catalyst during the quarter. Management reported that more than 1.7 billion unique viewers watched World Cup-related videos on YouTube, including more than 550 million viewers watching on television screens.

Because the World Cup contribution is event-driven, some of Q2's acceleration should be treated as non-recurring. However, the broader structural YouTube growth drivers extend beyond the tournament.

Primary YouTube Advertising Drivers

Connected Television and the Living Room

YouTube continues to expand as a television platform rather than solely a mobile-video platform. This gives the company access to traditional television brand-advertising budgets while also allowing Google to apply digital targeting and performance measurement.

Management identified connected television as an ongoing growth opportunity across both Brand and Direct Response advertising. New commerce features, including the ability to complete purchases through Google Pay on connected televisions, are intended to convert television viewing into a measurable performance-advertising channel.

Direct Response and Demand Gen

Direct Response remained the leading contributor to YouTube advertising growth, followed by Brand advertising. Demand Gen campaigns allow advertisers to distribute visual advertising across YouTube, Shorts and other Google properties. Management indicated that small and medium-sized advertisers continued to support Direct Response growth.

Shorts Monetization

Shorts remains a potential source of incremental monetization. The strategic opportunity is to narrow the revenue gap between short-form engagement and traditional long-form YouTube videos through:

- More effective advertising formats;
- Improved targeting;
- Shoppable advertisements;
- Creator partnerships;
- Greater adoption by small and medium-sized businesses.

Management did not disclose standalone Shorts revenue or revenue per viewing hour, limiting the ability to measure the pace of monetization improvement.

Creator Ecosystem

YouTube is using creator relationships to differentiate itself from conventional television and competing social-video platforms. Creator-led programming, sponsorships, affiliate commerce and YouTube Shopping could allow the platform to capture both advertising and transaction-related revenue.

YouTube Revenue Quality

The quarter demonstrated a broad combination of brand, performance and connected-TV demand. However, several factors should be distinguished:

Revenue Driver	Nature of Driver
World Cup advertising	Primarily event-driven
Connected-TV growth	Structural
Direct Response and Demand Gen	Structural
Shorts monetization	Structural but still developing
Creator sponsorships	Structural
Commerce and shopping formats	Early-stage opportunity

The principal risk is that increasing engagement does not automatically translate into proportional revenue growth. Monetization depends on the viewing surface, video format, geographic mix and advertiser demand. Shorts, international users and ad-supported television content may have different revenue yields.

YouTube Outlook

Future acceleration could be supported by:

- Continued connected-TV adoption;
- Improved Shorts monetization;
- Expansion of shoppable advertising;
- Stronger Direct Response products;
- Creator-led premium content;
- Growth among small and medium-sized advertisers.

Management also stated that YouTube subscription revenue was growing faster than YouTube advertising revenue, creating a more diversified monetization structure across the platform.

YouTube Advertising Assessment

Factor	Assessment
Advertising growth	Strong
Direct Response demand	Strong
Brand advertising	Positive
Connected-TV opportunity	Significant
Shorts monetization	Improving but undisclosed
Q2 event benefit	Material
Long-term revenue sustainability	Positive

YouTube is increasingly becoming a full-funnel advertising platform spanning brand awareness, performance advertising, connected television, short-form video and commerce. Q2 growth was partly enhanced by the World Cup, but the broader monetization opportunity appears structural.

2.4 Google Network

Google Network revenue declined **0.7% year over year** to **\$7.30 billion**, although it increased approximately 4.8% sequentially.

The year-over-year decline was primarily attributable to lower AdSense revenue, partially offset by growth in AdMob.

Alphabet reported:

Google Network Metric	Q2 2026 YoY Change
Ad Impressions	-12%
Cost per Impression	+13%
Network Revenue	-0.7%

The 13% increase in cost per impression almost fully offset the 12% decline in advertising impressions. This indicates that advertiser pricing remained healthy, but the underlying volume of Network advertising inventory continued to contract.

The decline in impressions could reflect:

- Product and policy changes;
- Reduced AdSense inventory;
- Shifts in web traffic;
- Changes in property and geographic mix;
- Ongoing pressure on the open-web advertising ecosystem.

Network Revenue Quality

Google Network is the weakest component of Alphabet's advertising portfolio. Unlike Search and YouTube, it does not benefit as directly from Google's owned-and-operated user experiences.

The business remains more exposed to:

- Third-party publisher traffic;
- Privacy and regulatory changes;
- Open-web traffic trends;
- Publisher economics;
- Competition from alternative monetization platforms.

The shift away from Google Network toward Search also helped reduce Alphabet's overall traffic acquisition cost rate from 20.6% to 19.8%, because Network revenue generally carries a higher TAC burden.

Google Network Assessment

Factor	Assessment
Revenue growth	Weak
Impression volume	Negative
Advertising pricing	Strong
Revenue visibility	Lower than Search
Margin contribution	Relatively weaker
Strategic importance	Declining

Network revenue is becoming a smaller part of Alphabet's consolidated growth profile. Although better pricing is offsetting lower impressions, the business appears structurally mature and should not be treated as a major future growth engine.

2.5 Subscriptions, Platforms and Devices

Subscriptions, Platforms and Devices revenue increased **15.2% year over year** to **\$12.91 billion**.

Alphabet attributed the increase primarily to paid-subscription growth across:

- YouTube Music and Premium;
- Other YouTube subscription products;
- Google One;
- AI-related Google One plans.

The reporting category also includes:

- YouTube TV;
- NFL Sunday Ticket;
- Google Play sales and in-app purchase fees;
- Pixel devices;
- Other consumer products and services.

Because Alphabet does not disclose revenue for each component, the precise contributions from YouTube subscriptions, Google One, Google Play and hardware cannot be independently determined.

Subscription Growth and Revenue Quality

Subscriptions generally provide higher revenue visibility than advertising because revenue is recognized over the subscription period and is less directly dependent on quarterly advertiser budgets.

The category is strategically important because it enables Alphabet to monetize its installed user base through recurring consumer payments.

The strongest disclosed drivers were:

1. **YouTube Music and Premium:** Management stated that subscription growth was especially strong in these services.
2. **Google One:** Demand for AI plans contributed to growth, suggesting that Gemini access is beginning to generate direct consumer revenue.
3. **YouTube’s broader subscription ecosystem:** YouTube subscriptions grew faster than YouTube advertising during the quarter.

Limitations of the Reporting Category

Subscriptions, Platforms and Devices combines businesses with materially different economics:

Component	Typical Revenue Characteristics
Consumer subscriptions	Recurring and relatively predictable
Google Play	Transaction- and usage-driven
Pixel hardware	Seasonal and lower-margin
NFL Sunday Ticket	Content-intensive
Google One AI plans	Recurring, but potentially compute-intensive

The category’s reported 15% growth does not necessarily imply equivalent profit growth. AI subscriptions may generate attractive recurring revenue, but their contribution margin depends on inference costs and customer usage. Hardware and sports-content products may also carry materially lower margins than digital subscriptions.

Strategic Interpretation

This segment provides Alphabet with an important second layer of consumer monetization beyond advertising:

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Large consumer user base -> Premium features -> Recurring subscription revenue

The emerging investment question is whether Gemini and Google One can become a meaningful paid-consumer-AI franchise rather than merely supporting engagement in Google’s advertising businesses.

Subscriptions, Platforms and Devices Assessment

Factor	Assessment
Revenue growth	Strong
Recurring revenue contribution	Increasing
YouTube subscription momentum	Strong
Google One AI monetization	Encouraging
Disclosure quality	Limited
Margin transparency	Low
Long-term strategic value	High

The category is developing into a meaningful recurring-revenue business. However, Alphabet's aggregated disclosure makes it difficult to separate high-quality subscription growth from hardware, content and platform revenue.

2.6 Google Cloud

Google Cloud was the strongest revenue segment in Q2 2026. Revenue increased **81.8% year over year** and **23.7% sequentially** to **\$24.77 billion**.

Cloud contributed approximately:

- 20.7% of Alphabet's consolidated revenue;
- 47.7% of Alphabet's year-over-year revenue increase;
- Approximately 83% of the total revenue beat relative to consensus.

The reported growth was driven primarily by Google Cloud Platform, which grew faster than the overall Cloud segment. Management identified core GCP, AI infrastructure and enterprise AI solutions as the principal drivers.

Google Cloud Revenue Components

Google Cloud revenue consists of:

- Google Cloud Platform**
 - Compute and storage;
 - AI infrastructure;
 - Vertex AI;
 - Gemini Enterprise;
 - Data and analytics;
 - Cybersecurity;
 - Other infrastructure and platform services.
- Google Workspace**
 - Gmail;
 - Docs;
 - Drive;
 - Meet;
 - Gemini-integrated productivity tools.
- TPU System Sales**
 - Hardware;
 - Software;

- Installation;
- Support;
- Extended warranty services.

4. Other Enterprise Services

Core GCP Growth

Alphabet stated that GCP grew faster than Google Cloud overall and that revenue growth accelerated meaningfully even when TPU system sales were excluded. This is important because it indicates that the 82% growth rate was not solely the result of recognizing hardware revenue.

The core drivers appear to include:

- Higher consumption of AI infrastructure;
- Increased demand for enterprise AI models;
- Greater adoption of Gemini Enterprise;
- Growth in data analytics;
- Cybersecurity demand;
- Increased usage among existing customers;
- Faster acquisition of new customers.

Management reported that customer-acquisition velocity more than doubled year over year. Existing customers were exceeding their contractual commitments by more than 50%, while transactions through Google Cloud Marketplace increased more than sevenfold year over year.

These indicators suggest that Cloud growth reflects both customer expansion and new-logo acquisition.

Enterprise AI Adoption

Management reported that nearly 90% of the Fortune 100 were using Gemini Enterprise. In addition:

- Approximately 500 Cloud customers had each processed more than one trillion tokens during the preceding year;
- More than 2,000 enterprises had consumed more than 100 billion tokens;
- Google's APIs were processing approximately 22 billion tokens per minute, compared with 16 billion in the prior quarter.

These metrics indicate substantial growth in usage, but Alphabet does not disclose:

- Revenue per token;
- Average pricing per token;
- Gross margin per token;
- The percentage of token usage that is paid;
- The share of consumption supported by discounts or contractual commitments.

Token growth should therefore be treated as a leading usage indicator rather than a direct proxy for revenue or profitability.

TPU System Sales

Alphabet began recognizing revenue from sales of complete TPU systems installed in customer data centers during Q2 2026. The majority of revenue under existing agreements is expected to be recognized in 2027.

This creates both an opportunity and an analytical complication.

Opportunity

TPU system sales expand Google Cloud's addressable market by allowing customers with on-premises or specialized high-scale workloads to purchase Google-designed infrastructure.

Analytical complication

Hardware-system sales may have different:

- Gross margins;
- Revenue-recognition patterns;
- Working-capital requirements;
- Customer-concentration risks;
- Capital requirements;

compared with recurring cloud-consumption revenue.

Future analysis should therefore distinguish:

Cloud revenue mix: Core cloud services revenue vs. TPU product revenue

Alphabet currently does not disclose the size or margin of TPU system sales separately.

Cloud Backlog

Google Cloud backlog reached **\$514 billion**, increasing by more than \$50 billion sequentially. Management expects slightly more than half of the backlog to be recognized as revenue during the next 24 months. The majority of the backlog relates to standard GCP contracts, although TPU system-sales agreements are also included.

Backlog provides significant forward revenue visibility, but it should not be interpreted as guaranteed near-term revenue because:

- Contract periods may extend over multiple years;
- Customer consumption may differ from contractual commitments;
- Some contracts may contain termination or modification provisions;
- TPU revenue recognition depends on system delivery and control transfer;
- Supply constraints may affect the timing of revenue conversion.

A useful conversion metric for future quarters will be:

$$\text{Backlog conversion rate} = \text{Cloud revenue} / \text{Beginning cloud backlog}$$

Supply Constraints

Management stated that Cloud remained supply-constrained, with customer demand exceeding available internal capacity. Alphabet plans to use additional third-party capacity in Q3 as a temporary bridge while internal infrastructure is expanded.

Supply constraints create two competing implications:

Positive implication

Demand is strong enough to exceed currently available capacity, supporting revenue visibility and pricing power.

Negative implication

Alphabet may be unable to fully monetize demand immediately, while third-party capacity could produce lower margins than internally owned infrastructure.

Cloud Revenue Sustainability

The Cloud growth rate is unlikely to remain at 82% indefinitely, particularly as revenue becomes larger. Nevertheless, the combination of:

- Rapid customer acquisition;
- Greater usage among existing customers;
- Enterprise AI adoption;
- A \$514 billion backlog;
- TPU system contracts;
- Continued supply constraints;

supports a strong near- to medium-term revenue outlook.

The major analytical question is no longer whether Google Cloud can grow. It is:

How much of the current growth represents high-margin recurring consumption, and how much reflects hardware sales, acquired revenue or capacity-intensive infrastructure services?

Google Cloud Assessment

Factor	Assessment
Revenue growth	Exceptional
Sequential acceleration	Exceptional
Core GCP demand	Very strong
AI infrastructure demand	Very strong
Enterprise AI adoption	Strong
Backlog visibility	Very high
Supply availability	Constrained
Revenue mix transparency	Limited
Long-term sustainability	Strong, but growth should normalize
Key risk	Capital intensity and mix shift toward hardware

Google Cloud has become Alphabet's second major growth engine. The quarter demonstrated genuine acceleration in core GCP and enterprise AI demand, but investors should avoid treating all Cloud revenue as economically equivalent. Recurring software and consumption revenue, third-party capacity and TPU hardware sales are likely to have different margin and cash-flow characteristics.

2.7 Other Bets

Other Bets generated **\$382 million** of revenue, increasing 2.4% year over year but declining approximately 7.1% sequentially.

Revenue is generated primarily through autonomous-transportation services and internet services. The segment includes businesses such as Waymo and Wing.

The limited revenue growth should be viewed in the context of continued investment and commercialization rather than mature business performance.

Waymo

Management emphasized continued expansion of Waymo's commercial operations and the introduction of a sixth-generation Waymo Driver vehicle. However, Alphabet does not separately disclose:

- Waymo revenue;
- Paid-trip volume;
- Revenue per trip;
- City-level economics;

- Fleet utilization;
- Contribution margin;
- Capital expenditure.

Without these figures, it is not possible to determine whether Waymo is approaching operating breakeven or whether revenue is scaling proportionately with investment.

Wing and Other Businesses

Wing surpassed one million home deliveries and continued expanding through commercial partnerships. Other Bets also includes early-stage businesses in healthcare, life sciences and other emerging technologies.

GFiber Classification

GFiber remained included in Other Bets through the closing of its pending transaction, although its assets had been classified as held for sale. This could affect future year-over-year comparability after the transaction closes and GFiber revenue is no longer consolidated within Other Bets.

Other Bets Assessment

Factor	Assessment
Current revenue contribution	Immaterial
Revenue growth	Limited
Waymo strategic value	Potentially significant
Disclosure quality	Low
Near-term earnings contribution	Negative
Long-term optionality	High
Valuation approach	Sum-of-the-parts

Other Bets should be treated primarily as long-duration strategic optionality rather than a current earnings driver. Waymo may possess meaningful standalone value, but the absence of detailed operational disclosure makes a precise fundamental valuation difficult.

2.8 Geographic Revenue Mix

Alphabet's revenue mix shifted further toward the United States during Q2 2026.

Geography	Q2 2025 Revenue Mix	Q2 2026 Revenue Mix
United States	48%	51%
EMEA	29%	27%
APAC	17%	16%

Geography	Q2 2025 Revenue Mix	Q2 2026 Revenue Mix
Other Americas	6%	6%

The increased U.S. contribution may reflect several factors, including stronger domestic advertising demand, rapid U.S.-based Cloud and AI infrastructure growth, TPU system sales and differences in foreign-exchange movements. Alphabet does not provide sufficient segment-by-geography detail to isolate these effects precisely.

A higher U.S. revenue mix can provide some benefit through:

- Greater exposure to high-value advertising markets;
- Strong enterprise technology spending;
- Reduced currency translation exposure.

However, it may also increase sensitivity to U.S. enterprise spending and advertising cycles.

2.9 Overall Revenue Assessment

Revenue Strength by Business

Segment	Growth Assessment	Revenue Quality	Forward Visibility
Google Search & Other	Strong	High	High
YouTube Advertising	Strong	High, with event benefit	Moderate to High
Google Network	Weak	Moderate to Low	Low
Subscriptions, Platforms and Devices	Strong	Improving	Moderate
Google Cloud	Exceptional	High but increasingly mixed	Very High
Other Bets	Limited	Low at present	Low

Key Positive Findings

1. **Google Cloud has become a genuine second growth engine.**
Cloud generated nearly half of Alphabet's year-over-year revenue increase and now represents more than one-fifth of consolidated revenue.
2. **Search remains resilient despite generative-AI disruption concerns.**
Paid clicks increased 13%, cost per click increased 3%, and management reported incremental query growth from AI-powered Search experiences.
3. **Revenue diversification is improving.**
Cloud and consumer subscriptions are expanding faster than Alphabet's traditional advertising businesses.
4. **YouTube has multiple monetization channels.**
Brand advertising, Direct Response, connected television, Shorts, subscriptions and commerce provide several independent growth levers.

5. **Cloud backlog provides substantial visibility.**

The \$514 billion backlog supports future growth, even though the timing and margin of revenue recognition require further monitoring.

Key Risks and Limitations

1. **Cloud revenue quality is becoming more complex.**

TPU hardware sales, third-party capacity and traditional recurring Cloud services may have materially different margins.

2. **AI Search unit economics are not disclosed.**

Higher query growth is positive, but profitability depends on advertising yield relative to inference costs.

3. **YouTube benefited from a major sporting event.**

The World Cup contribution may create a more difficult year-over-year comparison.

4. **Network advertising remains structurally weak.**

Higher pricing is offsetting declining impression volume rather than restoring underlying growth.

5. **Alphabet provides limited product-level disclosure.**

The company does not separately disclose Gemini subscriptions, Shorts revenue, TPU revenue, Waymo revenue or core Cloud consumption revenue.

Bottom-Line Revenue Conclusion

Alphabet's Q2 revenue performance was strong and increasingly diversified. Search continued to deliver double-digit growth, while Google Cloud produced exceptional acceleration and became the largest source of incremental revenue. YouTube and consumer subscriptions also performed well, although Network advertising remained weak.

The quarter's revenue profile can be summarized as:

Revenue profile: Resilient Search + Accelerating Cloud + Diversifying subscriptions - Weak Network

The most important change in Alphabet's fundamental profile is the growing importance of Google Cloud:

Alphabet is no longer simply an advertising company funding an emerging Cloud business. Google Cloud is becoming a major contributor to consolidated growth, profitability and forward contracted revenue.

The next stage of analysis should determine whether this revenue acceleration is translating into sustainable consolidated margin expansion. That requires a detailed examination of the **Profitability and Margin Bridge**, including traffic acquisition costs, Cloud operating leverage, AI-related depreciation, technical infrastructure costs and shared AI research expenses.

3. Profitability and Margin Bridge

3.1 Executive Assessment

Alphabet delivered substantial operating leverage in Q2 2026, but the composition of the margin expansion was mixed.

Consolidated operating income increased 30.4% year over year to \$40.77 billion, while operating margin expanded approximately 160 bps to 34.0%. The improvement was supported by:

- Strong revenue growth, particularly in Google Cloud;
- A favorable revenue mix shift toward Search and Cloud;
- Lower traffic acquisition costs as a percentage of revenue;
- Significant operating leverage within Google Cloud;
- Continued scalability in Google Services.

These benefits were partially offset by:

- Higher depreciation associated with technical infrastructure;
- Increased energy, equipment and network-capacity costs;

- Higher compensation related to AI and Cloud talent;
- TPU system inventory and product costs;
- Increased shared AI research and development expenses;
- Legal charges and higher marketing investment.

Alphabet's cost structure is therefore moving in two directions simultaneously:

Revenue scale and business-mix improvement → Higher operating leverage

AI infrastructure, model development and capacity investment → Higher depreciation and operating costs

The key conclusion is:

Alphabet's underlying businesses are currently generating enough incremental profit to absorb substantial AI-related spending. However, the reported Google Cloud margin does not include all of Alphabet's shared AI development costs, and future depreciation and third-party capacity costs are likely to create additional pressure.

3.2 Consolidated Profitability Overview

Metric	Q2 2025	Q2 2026	YoY Change	Margin Impact
Revenue	\$96.43B	\$119.80B	+24.2%	Strong revenue base
Gross Profit	\$57.39B	\$73.85B	+28.7%	Positive
Gross Margin	59.5%	61.6%	+213 bps	Major source of expansion
R&D Expense	\$13.81B	\$18.22B	+31.9%	Negative
R&D as % of Revenue	14.3%	15.2%	-89 bps	AI investment pressure
Sales and Marketing	\$7.10B	\$8.40B	+18.3%	Positive leverage
Sales and Marketing as % of Revenue	7.4%	7.0%	+35 bps	Positive
G&A Expense	\$5.21B	\$6.46B	+24.0%	Approximately neutral
G&A as % of Revenue	5.4%	5.4%	Approximately flat	Neutral
Operating Income	\$31.27B	\$40.77B	+30.4%	Strong
Operating Margin	32.4%	34.0%	+160 bps	Positive
Incremental Operating Margin	N/A	40.6%	Current-period metric	Strong operating leverage

Source figures are derived from Alphabet's Q2 2026 Form 10-Q.

Incremental operating margin is calculated as:

Increase in Operating Income ÷ Increase in Revenue

\$9.50B ÷ \$23.37B = 40.6%

This means that approximately 41 cents of each incremental revenue dollar translated into additional operating income.

That is a strong result given the simultaneous expansion in AI infrastructure, technical talent and model-development spending.

3.3 Consolidated Operating Margin Bridge

Year-over-Year Margin Bridge

Margin Driver	Approximate YoY Impact
Q2 2025 Operating Margin	32.4%
Gross-margin expansion	+213 bps
Higher R&D intensity	-89 bps
Sales and marketing leverage	+35 bps
G&A impact	Approximately +1 bp
Q2 2026 Operating Margin	34.0%

The principal source of consolidated margin expansion was the improvement in gross margin. Alphabet's total cost of revenue increased 18%, substantially slower than the 24% increase in revenue. Cost of revenue consequently declined from approximately 40.5% to 38.4% of revenue.

However, R&D spending increased faster than revenue and offset approximately 89 bps of gross-margin improvement.

This reflects a deliberate reinvestment of business-level operating leverage into:

- Frontier-model development;
- AI engineering talent;
- Technical infrastructure;
- Cloud product development;
- Consumer AI products;
- Security and data-analytics offerings.

Sequential Margin Bridge

Alphabet's operating margin declined from **36.1% in Q1 2026** to **34.0% in Q2 2026**.

Sequential Driver	Approximate Impact
Q1 2026 Operating Margin	36.1%
Lower gross margin	-80 bps

Sequential Driver	Approximate Impact
Lower R&D ratio	+29 bps
Higher sales and marketing ratio	–9 bps
Higher G&A ratio	–149 bps
Q2 2026 Operating Margin	34.0%

Q1 financial results and segment data are based on Alphabet's Q1 2026 release.

A significant portion of the sequential G&A increase was related to legal costs. Alphabet's MD&A states that Q2 G&A included a **\$1.5 billion charge related to a legal matter**.

Excluding only this identified charge:

Adjusted Q2 operating income = \$40.77B + \$1.50B = \$42.27B

Adjusted Q2 operating margin = \$42.27B ÷ \$119.80B = approximately 35.3%

This indicates that legal expenses explain a significant, but not complete, portion of the sequential margin decline. Even after adjusting for the identified charge, operating margin would have remained below Q1's 36.1%, reflecting higher infrastructure, depreciation and commercial investment.

3.4 Gross Margin and Cost of Revenue

Alphabet's gross margin increased from approximately **59.5% to 61.6%**, despite rapid growth in technical infrastructure expenses.

This improvement was driven primarily by:

- Revenue growing faster than total cost of revenue;
- A lower TAC rate;
- Higher Search revenue relative to Network advertising;
- Strong Cloud revenue growth;
- Operating leverage from existing infrastructure;
- A greater contribution from scalable digital and subscription revenue.

Cost of Revenue Breakdown

Cost Category	Q2 2025	Q2 2026	YoY Growth
Traffic Acquisition Costs	\$14.71B	\$16.18B	+10.0%
Other Cost of Revenue	\$24.33B	\$29.76B	+22.3%
Total Cost of Revenue	\$39.04B	\$45.94B	+17.7%

Alphabet identified depreciation, TPU-related inventory costs, YouTube content-acquisition costs and other technical-infrastructure operating costs as major drivers of the increase in other cost of revenue.

Traffic Acquisition Cost Leverage

Alphabet's TAC rate declined from **20.6% to 19.8%** on the relevant advertising revenue base.

The primary reason was a favorable revenue-mix shift:

Lower-growth Google Network revenue → Lower consolidated TAC burden

Higher-growth Google Search revenue → Better consolidated advertising economics

The TAC rate within Search itself was broadly stable, meaning that the improvement was largely mix-driven rather than the result of Alphabet negotiating materially lower distribution payments.

This distinction matters because mix-based TAC leverage may continue as long as Search grows faster than Network, but it does not necessarily indicate structural improvement in Alphabet's distribution agreements.

3.5 Depreciation: The Delayed AI Infrastructure Expense

Alphabet's depreciation expense increased from **\$5.00 billion to \$7.10 billion**, representing growth of approximately **42% year over year**.

Depreciation rose from approximately **5.2% to 5.9% of revenue**, creating an estimated 75-basis-point headwind to profitability before considering any offsetting revenue generated by the new infrastructure.

Depreciation appears in multiple parts of Alphabet's income statement:

- Cost of revenue;
- Research and development expense;
- Technical infrastructure usage allocations;
- Potentially other operational expense categories.

The economic pathway is:

Capital expenditure → Assets under construction → Assets placed into service → Depreciation expense

This timing relationship is critical.

Alphabet spent **\$80.6 billion** on capital expenditures during the first half of 2026, more than double the amount spent during the first half of 2025. However, depreciation increased by a much smaller amount because substantial infrastructure was not yet ready for use. Alphabet states that technical infrastructure may remain under construction or assembly for months or years before depreciation begins.

As a result:

Current CapEx growth is a leading indicator of future depreciation growth.

The full P&L effect of the 2026 infrastructure program has therefore not yet appeared.

Simplified Timing Example

Investment Stage	Cash-Flow Effect	Income-Statement Effect
Server purchased	CapEx increases	No immediate full expense
Data center under construction	CapEx increases	No depreciation until ready
Asset placed in service	No new purchase required	Depreciation begins

Investment Stage	Cash-Flow Effect	Income-Statement Effect
Compute capacity is used	Energy and network costs rise	Cost of revenue or R&D rises
Asset reaches end of useful life	Replacement CapEx may be required	Depreciation continues until fully depreciated

This creates an important margin risk:

Alphabet could experience strong revenue growth and high CapEx at the same time, followed by a later period in which revenue growth normalizes but depreciation continues increasing.

3.6 Google Services Profitability

Google Services generated:

- Revenue of **\$94.54 billion**;
- Operating income of **\$39.54 billion**;
- Operating margin of **41.8%**.

The margin increased from approximately 40.1% in Q2 2025, representing expansion of approximately 177 bps.

Google Services Metric	Q2 2025	Q2 2026	Change
Revenue	\$82.54B	\$94.54B	+14.5%
Operating Income	\$33.06B	\$39.54B	+19.6%
Operating Margin	40.1%	41.8%	+177 bps
Incremental Operating Margin	N/A	54.0%	Strong

The 54% incremental margin demonstrates the scalability of Search, advertising and consumer-subscription revenue.

The main positive drivers were:

- Search revenue growth;
- Higher advertising volume and pricing;
- Strong subscription growth;
- Lower TAC intensity due to business mix;
- Scale benefits across existing consumer platforms.

The principal offsets were:

- YouTube content-acquisition costs;
- AI inference and infrastructure usage;
- Marketing investment for Gemini and Search;
- Higher compensation;
- Legal expenses.

Alphabet stated that sales and marketing spending increased partly because of investments supporting the Gemini app and Search.

AI Search Margin Consideration

Google Services benefits from AI-enhanced Search revenue, but AI-powered queries also require greater compute resources than traditional search queries.

The relevant unit-economic relationship is:

Incremental AI advertising revenue – AI inference cost = Incremental AI Search contribution profit

Management stated that engineering and hardware optimization reduced the cost of AI Mode responses during the quarter. Nevertheless, Alphabet does not disclose:

- Cost per AI query;
- Revenue per AI query;
- AI Search gross margin;
- AI-related depreciation attributed to Search;
- The percentage of AI queries carrying advertising.

Therefore, the 41.8% Services operating margin provides evidence that Alphabet is currently absorbing AI Search costs successfully at the segment level, but it does not provide a complete measure of AI Search profitability.

3.7 Google Cloud Margin Expansion

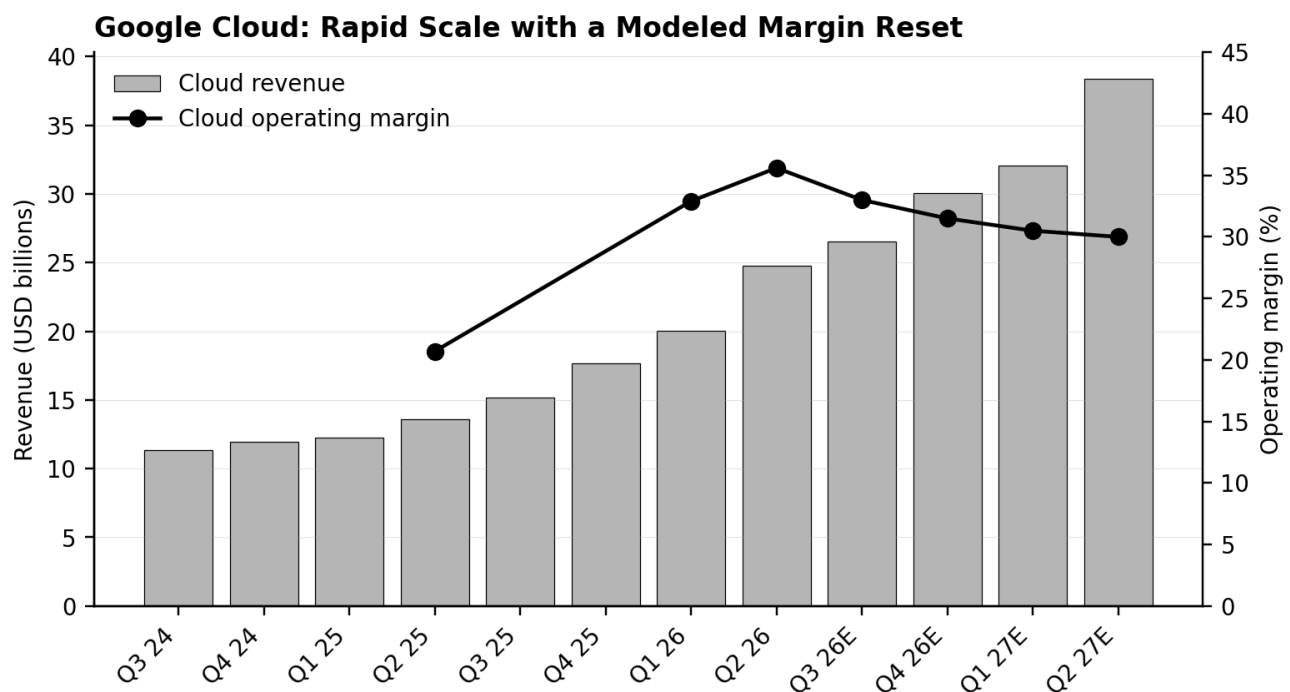


Figure 2. Google Cloud revenue and operating-margin progression. Source: company filings and analyst estimates.

Google Cloud produced the most significant profitability improvement in the quarter.

Google Cloud Metric	Q2 2025	Q1 2026	Q2 2026
Revenue	\$13.62B	\$20.03B	\$24.77B
Operating Income	\$2.83B	\$6.60B	\$8.81B

Google Cloud Metric	Q2 2025	Q1 2026	Q2 2026
Operating Margin	20.7%	32.9%	35.6%

Cloud revenue and operating-income figures are reported by Alphabet in its Q1 and Q2 releases and Form 10-Q.

Google Cloud operating margin expanded:

- Approximately 1,484 bps year over year;
- Approximately 265 bps sequentially.

The segment generated:

- **53.7% year-over-year incremental operating margin;**
- **46.8% sequential incremental operating margin.**

These incremental margins are calculated from the change in reported Cloud revenue and operating income.

Interpretation of Cloud Incremental Margin

The year-over-year calculation is:

Increase in Cloud operating income = \$8.81B – \$2.83B = \$5.99B

Increase in Cloud revenue = \$24.77B – \$13.62B = \$11.14B

Incremental Cloud margin = \$5.99B ÷ \$11.14B = approximately 53.7%

This is exceptionally strong operating leverage for a capital-intensive Cloud business.

It indicates that:

- Existing infrastructure was utilized more intensively;
- Revenue grew significantly faster than segment operating costs;
- High-value AI and platform services contributed meaningfully;
- Fixed engineering and commercial costs were spread over a larger revenue base;
- Cloud may have benefited from a favorable service and customer mix.

Alphabet states that the increase in Cloud operating income was driven by revenue growth, partially offset by technical-infrastructure usage costs and employee compensation.

The word “partially” is important: infrastructure and compensation costs increased, but they increased much more slowly than Cloud revenue.

3.8 What Is Driving Cloud Operating Leverage?

1. Revenue Growth Faster Than Infrastructure Cost Growth

Cloud revenue increased 82%, while total Alphabet cost of revenue increased only 18%.

Alphabet does not disclose Cloud-specific cost of revenue, but the segment's operating-income growth indicates that incremental Cloud revenue significantly exceeded incremental allocated infrastructure and labor costs.

This can occur when:

- Previously installed capacity gains utilization;

- Enterprise customers consume beyond contractual commitments;
- Software and platform revenue grows faster than physical infrastructure costs;
- Fixed development costs are distributed across a larger customer base;
- Existing sales and support infrastructure becomes more productive.

2. Higher Utilization of Owned Infrastructure

Cloud economics improve materially when owned data-center and server capacity moves from underutilized to highly utilized.

The simplified relationship is:

Higher utilization → More revenue per server → Lower depreciation cost per revenue dollar

This is likely an important contributor to Q2's margin expansion, although Alphabet does not disclose utilization rates.

3. Custom TPU Economics

Alphabet designs and uses its own TPUs across model training, inference and Cloud services.

Potential economic advantages include:

- Greater performance per dollar for selected workloads;
- Better power efficiency;
- Reduced dependence on external accelerator vendors;
- Hardware and software co-optimization;
- Greater control over deployment and capacity allocation.

Management stated that there are economic benefits from designing and manufacturing its own chips, but it does not disclose TPU-specific margins.

Custom silicon may improve long-term Cloud economics, but the benefit should not be assumed to apply equally to every workload. Google Cloud also uses GPUs, and external customers may require a mixture of TPU and GPU infrastructure.

4. Enterprise AI Solutions and Software Mix

Cloud margins are likely to be higher when customers purchase integrated software and platform solutions rather than raw compute alone.

Examples include:

- Gemini Enterprise;
- Vertex AI;
- Cybersecurity;
- Data analytics;
- Google Workspace;
- Agentic workflows.

Software, subscriptions and higher-level platform services generally have greater potential for differentiation and margin than undifferentiated compute capacity.

Alphabet's full-stack strategy allows it to monetize multiple layers:

Chip → Infrastructure → Model → Platform → Enterprise application

This is strategically important because the margin on the complete solution can be more attractive than the margin on infrastructure alone.

5. Customer Expansion and Contract Scale

Large enterprise contracts can improve operating leverage because sales, engineering and customer-acquisition costs do not necessarily increase proportionately with customer usage.

However, very large customers may also negotiate lower prices or require substantial dedicated infrastructure, creating a balance between scale efficiency and customer concentration.

3.9 Why the 35.6% Cloud Margin Is Not Fully Loaded

Alphabet's reported Google Cloud margin includes:

- Direct Cloud operating expenses;
- Direct infrastructure usage costs;
- Direct and allocated employee expenses;
- Allocated technical-infrastructure costs;
- Allocated office-facility costs;
- Inventory and product costs related to TPU system sales.

However, certain AI-focused expenses are not allocated to Google Cloud or Google Services.

Alphabet classifies shared AI research and development activities at the Alphabet level, including:

- Compensation for shared AI researchers and engineers;
- Technical-infrastructure usage for development of general AI models;
- Certain shared corporate and legal costs.

Therefore:

Google Cloud's 35.6% reported margin is a segment accounting margin, not a fully loaded standalone economic margin for Alphabet's enterprise AI ecosystem.

A portion of Gemini model-development cost supports Cloud products but remains within Alphabet-level activities.

It would be inappropriate to subtract the entire Alphabet-level loss from Cloud because shared AI R&D also supports:

- Search;
- YouTube;
- Gemini consumer products;
- Advertising;
- Other Alphabet products.

Nevertheless, investors should not interpret the Cloud margin as including all model-development expenses required to sustain its AI offerings.

3.10 Alphabet-Level AI R&D

Alphabet-level operating loss increased from **\$3.37 billion to \$5.79 billion**, an increase of approximately **\$2.42 billion**.

Alphabet states that these activities primarily reflect shared AI research and development expenses.

Alphabet-Level Activities	Q2 2025	Q2 2026	Change
Operating Loss	-\$3.37B	-\$5.79B	Loss increased by \$2.42B
Loss as % of Revenue	3.5%	4.8%	+134 bps

Before Alphabet-level activities, the change in segment operating performance was approximately:

- Google Services operating-income increase: +\$6.48B;
- Google Cloud operating-income increase: +\$5.99B;
- Other Bets deterioration: −\$0.55B.

Combined incremental segment profit before Alphabet-level costs was approximately **\$11.92 billion**.

The increase in Alphabet-level losses absorbed approximately:

$\$2.42\text{B} \div \$11.92\text{B} = 20.3\%$

of the incremental segment profit.

This is one of the most important findings in the margin analysis:

Google Services and Google Cloud are generating very strong operating leverage, but approximately one-fifth of their combined incremental segment profit was absorbed by higher shared AI and corporate expenses.

This spending may generate future revenue across multiple businesses, but it reduces current consolidated profitability.

3.11 AI Infrastructure Cost Architecture

AI infrastructure affects Alphabet’s financial statements through several distinct channels.

A. Internally Owned Servers and Data Centers

Financial Effect	Timing
Capital expenditure	When servers, land and equipment are purchased
Cash-flow reduction	Immediate
Depreciation	Begins when assets enter service
Energy and cooling cost	Incurred as infrastructure operates
Network-capacity cost	Incurred as data is transmitted
Maintenance and labor	Ongoing

Approximately 60% of Q2 technical-infrastructure investment was allocated to servers, while approximately 40% went to data centers and networking equipment.

Servers generally enter service faster than large data-center projects, meaning server-heavy CapEx can translate into depreciation sooner.

B. AI Model Training

Training frontier models requires:

- Accelerators;
- Networking;
- Energy;
- Data-center capacity;
- Engineering compensation;

- Software and data infrastructure.

Training expenses may appear within R&D or Alphabet-level activities, depending on their nature and allocation.

The training of larger frontier models can increase both:

- Current infrastructure usage costs;
- Future required capital capacity.

C. AI Inference and Product Serving

Once models are deployed, each customer or consumer interaction creates inference demand.

Inference costs include:

- Compute usage;
- Energy;
- Memory and networking;
- Data-center operations;
- Depreciation of serving infrastructure.

Cloud inference used by enterprise customers is more directly monetized. Consumer inference in Search, YouTube or the Gemini app may be monetized indirectly through advertising, subscriptions or ecosystem engagement.

D. TPU Systems Sold to Customers

Alphabet began recognizing revenue from TPU systems delivered to customer data centers in Q2.

The accounting sequence is:

Components purchased → Inventory built → Cash tied up in working capital → System delivered → Revenue and cost of goods recognized

Alphabet's inventory increased sharply during the first half of 2026, and the company identified TPU systems as an important component of inventory. Inventory costs associated with TPU system sales contributed to the increase in Q2 cost of revenue.

This business has different economics from standard Cloud consumption:

- Revenue may be recognized in large amounts at delivery;
- Hardware costs are recognized as cost of revenue;
- Working-capital requirements are higher;
- Gross margin may be lower than software or managed AI services;
- Revenue may be less recurring.

Management declined to disclose TPU-specific margins.

Therefore, future Cloud margin analysis should separate, as far as possible:

Recurring Cloud services and consumption

from

TPU system product sales

E. Third-Party Compute Capacity

Alphabet plans to increase its use of third-party capacity in Q3 as a temporary bridge while internally owned capacity is constructed.

Management explicitly stated that the higher cost of this capacity will pressure Google Cloud operating margins.

The economic comparison is:

Owned Internal Capacity	Third-Party Capacity
Requires substantial upfront CapEx	Lower initial CapEx
Depreciated over asset life	Expense or lease cost recognized more directly
Potentially lower long-term unit cost	Typically higher short-term unit cost
Greater operational control	Faster access to scarce capacity
Higher construction risk	Greater supplier dependence
Better long-term margins if well utilized	Near-term margin pressure

Management's rationale is that accepting higher short-term infrastructure costs may be worthwhile to secure large, multiyear customer relationships. Alphabet described these opportunities as potentially highly positive over the lifetime of the contract despite several months of elevated bridging costs.

This strategy may therefore produce:

Short-term Cloud margin compression → Customer retention and revenue capture → Improved long-term economics after internal capacity comes online

3.12 TPU Sales and Cloud Margin Risk

TPU systems create several uncertainties for the reported Cloud margin.

Potential Positive Effects

- Additional addressable market;
- Monetization of Google's proprietary chip design;
- Stronger customer commitment to the Google ecosystem;
- Potential hardware-software integration benefits;
- Expanded backlog and future revenue.

Potential Negative Effects

- Product revenue may carry lower margins than software;
- Inventory requirements reduce operating cash flow;
- Delivery timing may create quarterly revenue volatility;
- Supply-chain and component costs may fluctuate;
- Warranty and support obligations may extend beyond delivery;
- Large contracts may increase customer concentration.

Alphabet stated that TPU system sales were a relatively small contribution in Q2, would ramp during the remainder of 2026, and that the majority of revenue under existing agreements was expected in 2027.

Because TPU product margins are undisclosed, the current 35.6% Cloud margin should not automatically be extrapolated into 2027.

A reasonable analytical framework is:

Cloud service margin

plus

TPU product margin × TPU revenue mix

equals

Blended reported Cloud margin

If TPU product margins are lower than the current segment average, rapidly growing TPU sales could dilute the reported Cloud margin even while increasing total operating income.

3.13 Cloud Margin Outlook

Near-Term Headwinds

Third-Party Capacity

Alphabet expects third-party compute capacity to pressure Cloud operating margin beginning in Q3.

Wiz Integration

Management expects the integration of Wiz to create a near-term Cloud margin headwind during 2026.

Depreciation

Recent technical-infrastructure CapEx will move into depreciation as assets are placed into service.

Energy and Data-Center Operations

Management expects higher depreciation and data-center operating costs, including energy, to pressure the income statement.

AI and Cloud Hiring

Alphabet plans to continue hiring in AI and Cloud, increasing compensation expenses.

TPU Revenue Mix

A larger proportion of hardware-system revenue could dilute Cloud margins depending on product economics.

Potential Offsets

Continued Revenue Growth

Strong Cloud demand and backlog conversion could keep revenue growth ahead of infrastructure expense growth.

Internal Capacity Additions

As owned capacity enters service, Alphabet may reduce dependence on higher-cost third-party capacity.

Higher Infrastructure Utilization

Greater customer demand can increase revenue generated per unit of installed capacity.

Custom Silicon Efficiency

TPUs may provide greater cost control and performance efficiency for selected workloads.

Software and Solution Mix

Growth in Gemini Enterprise, cybersecurity, data analytics and Workspace could support higher blended margins.

Pricing

Management indicated that rising input costs are considered in the pricing and return framework for Cloud solutions.

Base-Case Interpretation

The 35.6% Q2 Cloud margin likely represents an unusually strong combination of:

- Rapid top-line acceleration;
- Existing-capacity utilization;
- Scale benefits;
- Limited initial TPU system revenue;
- Strong operational discipline.

Given explicit management commentary regarding third-party capacity and Wiz, a near-term moderation in Cloud margin would not necessarily indicate deterioration in the underlying business.

The more important question is whether:

Cloud operating income continues growing strongly even if the margin percentage temporarily declines.

3.14 Key Margin Indicators to Monitor

Indicator	Why It Matters
Google Cloud operating margin	Measures reported segment profitability
Cloud operating income growth	Shows absolute profit creation
Cloud incremental margin	Measures operating leverage on new revenue
TPU system revenue contribution	Identifies hardware mix
Other cost of revenue growth	Captures depreciation, TPU and infrastructure costs
Total depreciation expense	Measures delayed P&L impact of CapEx
Alphabet-level operating loss	Captures shared AI R&D burden
R&D as % of revenue	Indicates reinvestment intensity
Third-party capacity usage	Signals temporary Cloud margin pressure
Property and equipment not yet in service	Indicates future depreciation pipeline

Indicator	Why It Matters
Inventory balance	Indicates TPU system working-capital exposure
Energy and network costs	Measures infrastructure operating intensity
Cloud backlog conversion	Connects infrastructure spending to revenue
Revenue per dollar of CapEx	Proxy for capital productivity
Operating profit per dollar of CapEx	Proxy for AI investment returns

3.15 Overall Profitability Conclusion

Alphabet's Q2 2026 profitability was fundamentally strong, but the headline consolidated margin understates the operating leverage generated within its two core segments.

Google Services and Google Cloud each generated an incremental operating margin of approximately 54%. Cloud's reported operating margin expanded to 35.6%, demonstrating that enterprise AI demand is translating into substantial segment profit rather than revenue growth alone.

However, three qualifications are essential.

First, Alphabet's shared AI model-development costs are reported outside Google Cloud. The Cloud margin is therefore not a fully loaded measure of enterprise AI profitability.

Second, the P&L has not yet absorbed the full depreciation impact of the 2026 capital-expenditure program. Assets under construction will create future depreciation and operating costs as they enter service.

Third, Google Cloud will face near-term margin pressure from third-party capacity, Wiz integration and a potentially larger contribution from TPU hardware sales.

The quarter can be summarized as:

Strong Services operating leverage

plus

Exceptional Cloud margin expansion

minus

Higher shared AI R&D and future infrastructure costs

The central margin question is no longer whether Alphabet can generate profit from AI demand. Q2 demonstrates that it can.

The more important question is:

Can the growth in Search and Cloud operating profit remain ahead of the combined growth in depreciation, energy costs, third-party capacity, AI talent expenses and shared model-development spending?

At present, the answer appears positive, but Q2's 35.6% Cloud margin should be treated as a strong operating result rather than a guaranteed long-term margin level.

4. Earnings Quality

4.1 Executive Assessment

Alphabet's Q2 2026 operating performance was strong, but its reported net income and GAAP EPS substantially overstated the earnings generated by the company's recurring operations.

Alphabet reported:

Metric	Q2 2026
Operating Income	\$40.77B
Other Income, Net	\$97.98B
Income Before Taxes	\$138.75B
Net Income	\$112.19B
Net Income Available to Common Stockholders	\$112.11B
Diluted EPS	\$9.11

Approximately **71% of Alphabet's pre-tax income** was attributable to net equity-security gains rather than operating activities:

\$99.03B equity-security gains ÷ \$138.75B pre-tax income = approximately 71.4%

Alphabet stated that the equity gains increased Q2 net income by **\$77.1 billion** and diluted EPS by **\$6.26 per share**. Removing this contribution reduces reported EPS from \$9.11 to approximately **\$2.85**.

The principal conclusion is:

Alphabet's core operating earnings were materially stronger than the prior year, but reported GAAP EPS was dominated by noncash and largely unrealized investment gains.

The earnings-quality assessment should therefore separate:

1. Recurring operating income;
2. Non-operating investment gains;
3. Identifiable legal charges;
4. Cash conversion;
5. Stock-based compensation and dilution;
6. Acquisition-related accounting;
7. New financing costs.

4.2 Reported Earnings Versus Core Earnings

Alphabet's reported earnings can be divided into two broad components:

Earnings Component	Q2 2026	Quality Assessment
Operating Income	\$40.77B	High-quality recurring earnings
Net Equity-Security Gains	\$99.03B	Low-quality for forecasting purposes
Other OI&E Items, Net	Approximately -\$1.05B	Mixed
PriceRunner Legal Charge	\$2.08B pre-tax	Identifiable non-recurring charge
Income Tax Expense	\$26.56B	Significantly influenced by investment gains
Reported Net Income	\$112.19B	Not representative of recurring profitability

Alphabet's operating income increased 30% year over year, supported by revenue growth and operating leverage in Google Services and Google Cloud. This represents genuine improvement in recurring business performance. By contrast, the increase in net income of nearly 300% was primarily the result of investment revaluations rather than equivalent growth in operating profit.

A useful distinction is:

Operating income growth = evidence of stronger underlying business performance

Net income growth = heavily distorted by investment valuation changes

For forecasting and valuation, operating income, normalized net income and normalized EPS are substantially more informative than reported GAAP net income.

4.3 Equity Investment Gains

Alphabet recognized **\$99.03 billion** of net gains on equity securities in Q2 2026.

Composition of Equity-Security Gains

Component	Q2 2026
Unrealized Gain on Non-Marketable Equity Securities	\$77.35B
Unrealized Gain on Marketable and Other Equity Securities	\$21.40B
Realized Gain on Securities Sold	\$0.28B
Total Net Equity-Security Gain	\$99.03B

Approximately **99.7%** of the reported gain was unrealized:

\$98.75B unrealized gains ÷ \$99.03B total gains = approximately 99.7%

This means that almost none of the reported gain resulted from investments being sold and converted into cash during the quarter.

Alphabet attributed the increase primarily to unrealized gains associated with SpaceX and another private company.

Why the Gains Are Low Quality for Earnings Forecasting

The gains are legitimate under GAAP, but they should not be treated as recurring operating earnings for several reasons.

1. The gains were largely unrealized

The investments were revalued upward, but Alphabet did not receive an equivalent amount of cash.

2. The gains depend on external valuation events

Non-marketable securities are generally remeasured when observable transactions involving identical or similar securities provide new valuation information.

This means that earnings recognition can be:

- Irregular;
- Transaction-dependent;
- Highly concentrated in individual quarters;
- Disconnected from Alphabet's underlying operating performance.

3. The gains can reverse

Future financing rounds, public-market changes, impairments or weaker investor demand could result in downward revaluations.

Alphabet explicitly states that fluctuations in investment values could create significant volatility in future other income and expense.

4. Private-company valuations are less liquid

A carrying value based on an observable private transaction does not necessarily mean Alphabet could sell its entire investment at that valuation.

Large ownership positions may face:

- Liquidity discounts;
- Lock-up restrictions;
- Limited buyer availability;
- Tax consequences;
- Strategic restrictions;
- Market-price pressure if sold.

Investment Concentration

Alphabet's non-marketable securities increased from **\$68.69 billion** at December 31, 2025 to **\$131.46 billion** at June 30, 2026. The filing states that non-marketable equity securities measured under the measurement alternative primarily consisted of an investment in a private company.

This creates a meaningful concentration of reported balance-sheet value and potential earnings volatility in a small number of private investments.

Equity-Gain Assessment

Factor	Assessment
GAAP legitimacy	Valid
Cash realization	Very low
Recurrence	Low
Forecastability	Very low
Reversal risk	High
Relevance to operating valuation	Low
Relevance to sum-of-the-parts valuation	High

The investment gains should be excluded from Alphabet's operating EPS and earnings-growth analysis, but the underlying investment portfolio should still be recognized separately in a sum-of-the-parts valuation.

4.4 Normalized EPS Bridge

Alphabet does not provide a company-defined adjusted EPS measure. However, its earnings release disclosed the exact impact of equity-security gains on net income and EPS.

Reported to Normalized EPS

EPS Measure	Q2 2026
Reported GAAP Diluted EPS	\$9.11
Less: EPS Contribution from Equity-Security Gains	(\$6.26)
EPS Excluding Equity-Security Gains	\$2.85

The \$2.85 figure retains:

- Operating business earnings;
- Interest income and interest expense;
- Foreign-exchange effects;
- Debt-security gains and losses;
- Legal expenses;
- Acquisition-related amortization;
- Stock-based compensation;
- Other recurring and non-recurring expenses.

It should therefore be described as:

EPS excluding equity-investment gains, rather than a fully normalized core EPS measure.

Alphabet disclosed that the \$99.03 billion equity gain increased the tax provision by \$21.9 billion, net income by \$77.1 billion and diluted EPS by \$6.26. This allows the investment gain to be removed without relying on an estimated tax rate.

4.5 PriceRunner Legal Charge

Alphabet recognized approximately **\$2.08 billion** of Q2 legal charges related to the PriceRunner case:

- **\$1.50 billion** of principal damages was recorded in Google Services' general and administrative expenses;
- **\$581 million** of accrued interest and costs was recorded in other income and expense.

Alphabet appealed the decision.

Illustrative Legal-Adjusted EPS

Using Alphabet's reported Q2 effective tax rate of 19.1% as an approximation:

After-tax legal charge = \$2.08B × (1 – 19.1%) = approximately \$1.68B

Estimated EPS impact = \$1.68B ÷ 12.31B diluted shares = approximately \$0.14 per share

This produces an illustrative legal-adjusted earnings measure:

EPS Measure	Q2 2026
EPS Excluding Equity Gains	\$2.85
Add: Estimated After-Tax PriceRunner Impact	Approximately \$0.14
Illustrative Normalized EPS	Approximately \$2.99

The approximately \$2.99 figure is not reported by Alphabet. It is an analytical estimate and assumes that the legal charge would have been taxed at the consolidated effective tax rate.

Actual tax treatment may differ.

Treatment in Valuation

The PriceRunner charge should generally be excluded from forward recurring EPS because:

- It relates to a specific legal judgment;
- It is not part of the normal cost of delivering Alphabet's products;
- Its timing and amount are unusual.

However, legal and regulatory costs should not be removed without qualification. Alphabet faces repeated antitrust, competition and privacy proceedings. Therefore:

The individual PriceRunner charge is non-recurring, but elevated legal risk is a recurring feature of Alphabet's business model.

A reasonable forecast may exclude the full \$2.08 billion charge while retaining a normalized annual legal-cost assumption.

4.6 Other Income and Expense Excluding Equity Gains

Alphabet reported total other income of **\$97.98 billion**. Excluding the \$99.03 billion equity gain, other income and expense would have been approximately **negative \$1.05 billion**.

OI&E Component	Q2 2026
Interest Income	\$1.43B
Interest Expense	(\$1.28B)
Foreign-Exchange Loss	(\$0.16B)
Debt-Security Loss	(\$0.03B)
Equity-Security Gain	\$99.03B
Equity-Method Loss	(\$0.04B)
Other	(\$0.97B)
Total OI&E	\$97.98B

The “other” expense included approximately \$581 million of accrued PriceRunner interest and costs. Excluding that legal component, underlying OI&E excluding equity gains was approximately negative \$0.47 billion.

This indicates that Alphabet's recurring below-the-line contribution was no longer meaningfully positive in Q2.

The increase in interest expense from **\$261 million to \$1.28 billion** also reflects Alphabet's changing capital structure and increased debt financing.

Unlike the investment gains, higher interest expense is likely to remain a recurring cost.

4.7 Income-Tax Quality

Alphabet recorded:

Tax Metric	Q2 2025	Q2 2026
Income Before Taxes	\$33.93B	\$138.75B
Income-Tax Provision	\$5.74B	\$26.56B
Effective Tax Rate	16.9%	19.1%

The effective tax rate increased, but the absolute tax provision was heavily influenced by the investment gains. Alphabet disclosed that equity-security gains alone increased the tax provision by approximately **\$21.9 billion**.

The cash-flow statement also included a **\$20.62 billion deferred-tax adjustment** for Q2. This indicates that a substantial portion of the quarter's tax expense was noncash during the period, although Alphabet does not state that the entire deferred-tax adjustment was attributable exclusively to the investment gains.

Tax Interpretation

The reported tax provision should not be directly applied to recurring operating income without adjustment because:

- Investment gains may have different tax characteristics from operating income;
- Part of the provision was deferred rather than currently paid;
- Private securities may not create cash tax obligations until future transactions;
- Geographic income mix affects Alphabet's normalized tax rate.

For forward modeling, a normalized effective tax-rate assumption should be based on operating income and recurring below-the-line items rather than the Q2 GAAP net-income composition.

4.8 Cash Conversion

At first glance, Alphabet's reported earnings had weak cash conversion:

Metric	Q2 2026
Reported Net Income	\$112.19B
Operating Cash Flow	\$39.07B
Reported CFO / Net Income	34.8%

However, this comparison is misleading because reported net income included \$77.1 billion of after-tax equity gains that did not generate operating cash.

After removing the equity-gain contribution:

Net income excluding equity gains = \$112.19B – \$77.10B = approximately \$35.09B

Normalized cash conversion = \$39.07B CFO ÷ \$35.09B adjusted net income = approximately 111%

This indicates that recurring earnings converted into operating cash relatively well.

Q2 Operating Cash Flow Bridge

Cash-Flow Item	Q2 2026
Reported Net Income	\$112.19B
Depreciation Add-Back	+\$7.10B
Stock-Based Compensation Add-Back	+\$7.96B
Deferred Taxes	+\$20.62B
Securities Gains Removed	–\$99.00B
Other Noncash Adjustments	+\$1.90B
Working-Capital and Other Changes	Approximately –\$11.70B

Cash-Flow Item	Q2 2026
Operating Cash Flow	\$39.07B

The approximately \$11.7 billion working-capital and other cash-flow drag was primarily associated with:

- Higher accounts receivable;
- Higher inventory;
- Growth in other assets;
- Partially offset by increases in accounts payable and accrued liabilities.

Inventory alone used approximately **\$6.70 billion** of cash during Q2, reflecting, among other factors, the growing scale of TPU systems and infrastructure-related inventory.

Cash-Conversion Conclusion

Alphabet’s operating cash-generation quality remained strong after removing the unrealized investment gains.

The correct interpretation is:

Reported net income had weak cash conversion because the numerator was distorted, while normalized operating earnings had strong cash conversion.

4.9 Free Cash Flow Quality

Alphabet generated **\$39.07 billion** of operating cash flow but spent **\$44.92 billion** on property and equipment, resulting in negative free cash flow of approximately **\$5.86 billion**.

Metric	Q2 2026
Operating Cash Flow	\$39.07B
Capital Expenditures	(\$44.92B)
Free Cash Flow	(\$5.86B)
Free Cash Flow Margin	−4.9%

This does not necessarily indicate poor operating-earnings quality. Instead, it reflects Alphabet’s rapidly increasing capital intensity.

The distinction is:

Strong operating cash conversion

minus

Exceptionally high AI infrastructure investment

equals

Negative free cash flow

Therefore, Alphabet's Q2 earnings quality was stronger than its free-cash-flow result alone would suggest. Nevertheless, shareholder returns ultimately depend on whether the infrastructure spending generates sufficient future revenue and operating profit.

Free Cash Flow After Stock-Based Compensation

Stock-based compensation is already included as an expense in GAAP net income but is added back because it is noncash.

An illustrative shareholder-oriented cash-flow measure is:

Reported FCF – Stock-Based Compensation

Negative \$5.86B – \$7.96B = approximately negative \$13.81B

This is not a standard Alphabet-reported measure. It highlights that stock-based compensation has an economic cost even when it does not reduce current operating cash flow.

4.10 Stock-Based Compensation

Alphabet recorded **\$7.96 billion** of stock-based compensation in Q2 2026, compared with approximately \$6.00 billion in Q2 2025.

SBC Metric	Q2 2026
Stock-Based Compensation	\$7.96B
SBC as % of Revenue	6.6%
SBC as % of Operating Income	19.5%
YoY Growth	Approximately 32.7%
Unrecognized Future Compensation Cost	\$59.1B
Recognition Period	2.7 years weighted average

Alphabet had \$59.1 billion of unrecognized compensation cost related to unvested RSUs and performance-stock units at June 30, 2026.

SBC Earnings-Quality Implications

Stock-based compensation is:

- A genuine expense under GAAP;
- Noncash in the current period;
- A source of employee incentives;
- A potential source of shareholder dilution;
- A future cash requirement when the company pays tax withholding obligations associated with vesting.

Alphabet made **\$6.57 billion** of net payments related to stock-based award activities during Q2, primarily reflecting tax withholding and related activity.

This means that SBC should not be dismissed as a purely accounting expense.

Diluted Share Count

Alphabet's diluted weighted-average share count increased from **12.20 billion to 12.31 billion**, an increase of approximately 0.9% year over year.

Alphabet conducted no share repurchases during Q2 2026, while issuing common stock and mandatory convertible preferred securities to fund infrastructure and other corporate purposes.

The absence of repurchases means that share-based compensation and new equity issuance were not offset by buybacks during the quarter.

SBC Assessment

Factor	Assessment
Accounting treatment	Appropriate
Current cash impact	Limited in income statement
Economic cost	Material
Dilution risk	Increasing
Forecast visibility	High
Earnings-quality impact	Negative

4.11 New Equity and Preferred-Stock Dilution

Alphabet issued approximately **\$30.5 billion of common stock** and **\$19.0 billion of mandatory convertible preferred stock** in Q2.

The preferred stock carries a **6.25% annual dividend rate** and is scheduled to convert into common shares by approximately May 2029 unless converted earlier.

Preferred Dividend Burden

The annual preferred dividend based on the \$19 billion liquidation preference is approximately:

$$\text{\$19.0B} \times 6.25\% = \text{approximately } \text{\$1.19B annually}$$

Equivalent full-quarter preferred dividends would be approximately:

$$\text{\$1.19B} \div 4 = \text{approximately } \text{\$297M per quarter}$$

Only \$86 million of preferred dividends affected the Q2 common-stock earnings numerator because the securities were issued in June. Future quarters should reflect a larger preferred-dividend burden.

At the current diluted share count, a full-quarter preferred dividend would reduce common EPS by approximately:

$$\text{\$297M} \div 12.31\text{B shares} = \text{approximately } \text{\$0.02 per quarter}$$

This does not include the eventual common-stock dilution from conversion.

Common-Share Issuance

Because the common shares were issued partway through Q2, Q2 weighted-average diluted shares do not capture their full-quarter impact.

Therefore:

Future EPS growth may be lower than net-income growth because the share denominator will reflect a full period of the newly issued common shares.

This financing activity is important to earnings quality because it means some of Alphabet's AI-investment program is being funded through shareholder dilution rather than solely through internally generated cash.

4.12 Acquisition-Related Earnings Quality

Alphabet completed the acquisition of Wiz for approximately **\$29.5 billion** in March 2026.

The preliminary purchase-price allocation included:

- \$8.3 billion of identifiable intangible assets;
- \$22.7 billion of goodwill;
- Approximately \$1.5 billion of net liabilities assumed.

Wiz's financial performance has been included in the Google Cloud segment since the acquisition closed.

Organic Versus Acquired Growth

Alphabet did not separately disclose:

- Wiz revenue;
- Wiz operating income;
- Wiz's contribution to Cloud growth;
- Organic Google Cloud revenue growth;
- Acquisition-related integration costs.

Therefore, Google Cloud's reported 82% growth cannot be fully separated into organic and acquired components.

Management stated that core GCP growth accelerated even excluding TPU system sales, but the filing does not provide an equivalent quantitative organic-growth bridge excluding Wiz.

This reduces revenue and earnings transparency.

Intangible-Asset Amortization

Alphabet's amortization expense increased from **\$124 million to \$367 million** year over year.

Expected future amortization associated with definite-lived intangible assets included:

- \$747 million during the remainder of 2026;
- \$1.30 billion in 2027;
- More than \$1 billion annually from 2028 through 2030.

The increase reflects, in part, acquired technologies and customer relationships.

Amortization is noncash after the acquisition date, but it represents the accounting consumption of assets acquired with real shareholder capital.

It may be excluded in some non-GAAP analyses, but investors should not automatically assume that all acquisition-related amortization is economically irrelevant.

Goodwill Risk

Goodwill is not amortized but must be tested for impairment.

Alphabet's Google Cloud goodwill increased substantially following the Wiz acquisition. If expected revenue growth, customer retention or acquisition synergies fail to materialize, future impairment charges could reduce reported earnings.

Acquisition-Accounting Assessment

Factor	Assessment
Organic-growth transparency	Limited
Current amortization burden	Increasing
Cash cost	Already incurred at acquisition
Goodwill risk	Long-term
Recurrence of acquisitions	Possible
Earnings-quality impact	Mixed

4.13 Earnings Quality Scorecard

Earnings-Quality Dimension	Rating	Rationale
Core Revenue Growth	Strong	Broad growth led by Search and Cloud
Operating Income Growth	Strong	Operating income increased 30%
Operating Margin Quality	Strong	Expansion supported by real business leverage
Reported Net Income Quality	Very Weak	Approximately 71% of pre-tax income came from equity gains
GAAP EPS Comparability	Very Weak	\$6.26 of \$9.11 EPS came from investment gains
Normalized EPS	Strong	Approximately \$2.85 before legal normalization
Legal-Adjusted EPS	Approximately \$2.99	Illustrative, not company-reported

Earnings-Quality Dimension	Rating	Rationale
Cash Conversion	Strong After Normalization	CFO exceeded net income excluding equity gains
Free Cash Flow	Weak	High CapEx drove FCF negative
Stock-Based Compensation	Weak	SBC grew faster than revenue and creates dilution
Share-Count Quality	Weakening	New common and preferred issuance
Tax Quality	Mixed	Large deferred and investment-related tax effects
Acquisition Transparency	Mixed	Wiz contribution not separately disclosed
Forecastability	Moderate	Core operations predictable; GAAP net income highly volatile

4.14 Overall Earnings Quality Conclusion

Alphabet's Q2 2026 reported earnings should not be interpreted using GAAP EPS alone.

The quarter contained three distinct earnings layers.

Layer 1: High-Quality Operating Earnings

Alphabet generated \$40.77 billion of operating income, representing 30% year-over-year growth. Search remained resilient, Google Cloud delivered exceptional operating leverage and consolidated operating margin expanded.

This is the strongest component of the earnings report.

Layer 2: Low-Quality Investment Gains

Net equity-security gains of \$99.03 billion contributed \$77.1 billion to net income and \$6.26 to diluted EPS.

These gains were almost entirely unrealized, difficult to forecast and potentially reversible.

They should be removed from operating valuation multiples.

Layer 3: Capital and Shareholder Costs

Normalized operating earnings converted well into operating cash flow, but shareholder-level earnings quality was weakened by:

- Negative free cash flow;
- Rapidly rising capital expenditures;
- Material stock-based compensation;
- Common-share issuance;

- Preferred-stock dividends and future conversion;
- Higher interest expense;
- Increasing acquisition-related amortization.

The quarter can be summarized as:

Strong recurring operating earnings

plus

Large noncash investment gains

minus

Higher capital intensity and dilution

The most useful EPS figures are:

EPS Definition	Q2 2026
Reported GAAP EPS	\$9.11
EPS Excluding Equity Gains	\$2.85
Illustrative EPS Excluding Equity Gains and PriceRunner Charge	Approximately \$2.99

For valuation purposes, the appropriate starting point is approximately **\$2.85 to \$2.99 per share**, not \$9.11.

The investment portfolio should then be valued separately as a non-operating asset.

The central conclusion is:

Alphabet’s underlying earnings quality was strong at the operating level, but weak at the reported net-income level. Investors should focus on normalized operating EPS, cash conversion and future dilution rather than the headline GAAP EPS growth rate.

5. AI Monetization

5.1 Executive Assessment

Alphabet is monetizing artificial intelligence through three primary pathways:

1. **Consumer AI**, primarily through Search advertising, consumer subscriptions and AI-enhanced YouTube engagement;
2. **Enterprise AI**, through Google Cloud consumption, Gemini Enterprise, Vertex AI, Workspace subscriptions and cybersecurity solutions;
3. **AI Infrastructure**, through TPU- and GPU-based cloud capacity, model training and inference services, and direct TPU system sales.

Alphabet does not report a standalone “AI revenue” line. AI-related revenue is distributed across Google Search, Subscriptions, Platforms and Devices, Google Cloud Platform, Google Workspace and TPU product sales. The Form 10-Q confirms that Google Cloud includes enterprise AI infrastructure, Vertex AI, Gemini Enterprise, Gemini for Workspace and TPU system sales, while Google One subscriptions include access to Alphabet’s most capable Gemini models.

As a result, AI monetization should not be measured using a single reported revenue number. It should be evaluated through a sequence:

AI adoption → Increased usage → Monetized transactions → Recognized revenue → Contribution profit

User growth, query growth and token consumption are important leading indicators, but they do not independently demonstrate profitable monetization.

The principal conclusion is:

Alphabet has moved beyond the experimental stage of AI monetization. Search advertising and enterprise Cloud are already generating measurable commercial benefits, while consumer Gemini subscriptions, YouTube AI features and direct TPU system sales remain earlier-stage opportunities.

5.2 AI Monetization Framework

Monetization Path	Primary Products	Revenue Mechanism	Current Maturity
Consumer Search AI	AI Overviews, AI Mode	Search advertising and commerce	Scaling
Consumer Gemini	Gemini App, Google AI plans	Paid subscriptions through Google One	Early commercial scale
YouTube AI	Ask YouTube, recommendations, creator tools	Advertising, subscriptions and commerce	Early indirect monetization
Enterprise Models	Gemini APIs, Vertex AI	Token- and consumption-based fees	Scaling rapidly
Enterprise Applications	Gemini Enterprise, Workspace AI	Subscription and consumption revenue	Scaling
Security AI	Google Cloud Security, Wiz, CodeMender	Cloud subscriptions and platform usage	Scaling
AI Infrastructure Services	TPU and GPU Cloud capacity	Consumption-based infrastructure fees	Large-scale commercial deployment
TPU System Sales	On-premises TPU systems	Hardware, software, installation and support	Early revenue recognition

The economic quality of these revenue streams differs materially.

Search advertising and enterprise subscriptions can potentially produce highly recurring revenue. Infrastructure consumption is recurring but highly capital-intensive. TPU system sales may create large contract values but have more hardware, inventory and delivery exposure.

5.3 Consumer AI

Alphabet’s Consumer AI strategy monetizes AI across three primary surfaces:

- Search;
- The Gemini App and Google One;
- YouTube.

These products have different economic models.

Consumer Product	Primary Monetization
AI Overviews and AI Mode	Advertising and commerce
Gemini App	Subscription revenue and future ecosystem monetization
YouTube AI features	Advertising, subscriptions, commerce and engagement
AI-powered advertiser tools	Higher advertiser conversion and spending

5.3.1 AI Overviews and AI Mode

Search is currently Alphabet’s most commercially mature AI product.

AI Overviews and AI Mode support monetization through:

- Higher total query volume;
- More detailed commercial queries;
- Improved understanding of user intent;
- More relevant advertising;
- New conversational advertising formats;
- Agentic commerce and transaction completion.

AI Mode exceeded **one billion monthly active users** after its global expansion. Management stated that AI Mode and AI Overviews are generating incremental Search queries, rather than merely replacing traditional searches. Alphabet also reported that AI-powered Search features are sending billions of clicks to websites each week.

This supports a positive adoption thesis:

More AI usage → More Search queries → More commercial opportunities

However, the more important monetization question is whether these incremental queries generate sufficient advertising revenue to cover their higher inference costs.

AI Search Advertising

Alphabet stated that monetization on queries displaying AI Overviews remained encouraging, including as the product expanded into more commercial queries. Users of AI Overviews and AI Mode are asking longer and more detailed questions, which can improve the matching of advertisements to commercial intent.

Alphabet is monetizing AI Search through three layers:

1. Improved Ad Quality

Gemini improves Alphabet’s understanding of complex queries that were previously difficult to monetize.

Management reported that Gemini contributed to a **20% improvement in the relevance of Shopping advertisements** for selected queries.

Better relevance can improve:

- Click-through rates;
- Conversion rates;
- Advertiser return on advertising spending;
- Auction participation;
- Cost per click.

2. AI-Powered Advertiser Tools

AI Max and Performance Max automate targeting, creative development and bidding.

Approximately **500,000 advertisers** had adopted AI Max. Advertisers using AI Max or Performance Max generated, on average, approximately **15% more conversions or conversion value at a similar return on advertising spending**.

The monetization mechanism is indirect but economically important:

Improved advertiser results → Greater advertising budgets → Higher Google advertising revenue

3. New AI Advertising Formats

Alphabet is testing several formats within AI Mode, including:

- Contextual sitelinks;
- Direct Offers;
- Sponsored links embedded within clearly identified responses;
- Commerce and travel-related offers;
- Agentic purchasing experiences.

Management described these formats as early-stage, indicating that AI Mode advertising is not yet fully mature.

AI Search Cost Efficiency

Alphabet stated that the cost of generating an AI Mode response fell to its lowest level since launch, despite the inclusion of more advanced capabilities. The reduction was attributed to engineering and hardware optimization.

This is essential because the relevant Search AI economics are:

AI Search contribution profit = Incremental advertising revenue – Inference cost – TAC – Allocated model and infrastructure costs

Alphabet does not disclose:

- Revenue per AI query;
- Cost per AI query;
- AI query advertising click-through rates;
- AI Mode advertising revenue;
- The percentage of AI queries containing ads;
- The gross margin of AI-generated searches.

Consequently, the company has demonstrated adoption and early monetization, but not full AI Search unit economics.

Consumer Search AI Assessment

Indicator	Current Assessment
AI Mode monthly users	More than 1 billion

Indicator	Current Assessment
Incremental query contribution	Positive
Commercial-query adoption	Expanding
Advertising monetization	Active and improving
New advertising formats	Early-stage testing
Cost per response	Declining
Revenue disclosure	Limited
Profitability disclosure	Not provided

AI Search is already commercially relevant, but Alphabet has not yet disclosed enough information to calculate whether an average AI query is more or less profitable than a traditional Search query.

5.3.2 Gemini App and Consumer Subscriptions

The Gemini App reached approximately **950 million monthly active users**, while daily active users tripled over the preceding year. Alphabet also reported a 40% increase in daily active users creating video content following the launch of its Omni product.

The user scale is substantial, but a monthly active user is not necessarily a paid customer.

Gemini App monetization currently occurs primarily through Google One AI subscription plans. Alphabet's Subscriptions, Platforms and Devices category includes Google One, which provides access to its most capable Gemini models.

Management stated that Google One revenue growth was supported by demand for AI plans. However, Alphabet did not separately disclose:

- The number of Gemini or Google AI paying subscribers;
- Subscription revenue attributable to Gemini;
- Average revenue per paid AI subscriber;
- Free-to-paid conversion;
- Subscriber churn;
- Inference cost per subscriber;
- Subscription gross margin.

The monetization funnel is:

Gemini App users → Free product engagement → Paid AI-plan conversion → Recurring subscription revenue

The current evidence indicates that direct consumer AI monetization has begun, but its financial scale cannot yet be isolated from the broader Google One and subscription category.

Consumer AI Subscription Economics

Subscription profitability depends on both pricing and usage.

A simplified model is:

Subscription contribution per user = Monthly subscription fee – Inference and storage costs – Customer acquisition and support costs

Heavy users may generate higher engagement and retention but also incur substantially higher model-serving costs. Subscription revenue is therefore not automatically equivalent to high-margin software revenue.

Potential Additional Monetization Paths

Over time, the Gemini App may support:

- Higher-priced premium plans;
- Agentic commerce;
- Travel and local-service transactions;
- Developer and creator tools;
- Integration with Google Workspace;
- Cross-selling into Cloud services;
- Advertising or sponsored commercial experiences.

Alphabet has not indicated that conventional advertising is currently the primary monetization model for the standalone Gemini App. The current approach appears to prioritize user adoption, product quality and paid subscriptions.

Gemini Consumer Assessment

Indicator	Current Assessment
Monthly active users	Approximately 950 million
Daily-user growth	Strong
Paid subscription demand	Positive
Paid subscriber disclosure	Not provided
ARPU disclosure	Not provided
Subscription gross margin	Not provided
Monetization maturity	Early to intermediate
Strategic value	High

The Gemini App has achieved broad consumer distribution, but Alphabet has not yet demonstrated the size or economics of the paid-user base. Consumer Gemini should currently be treated as a large monetization option rather than a separately measurable earnings driver.

5.3.3 YouTube AI Monetization

Alphabet is integrating Gemini into YouTube through:

- Ask YouTube;
- AI-powered search and discovery;
- Recommendation improvements;
- Creator production tools;
- Automated translation and dubbing;
- AI-generated advertising and sponsorship tools;
- Shopping and connected-TV commerce.

More than **140 million users** engaged with Ask YouTube on the video watch page during June 2026. Ask YouTube allows users to ask questions, receive summaries and navigate directly to relevant moments within a video.

The immediate monetization pathway is primarily indirect:

Better discovery and engagement → More viewing time → More advertising and subscription opportunities

AI can also increase monetization by:

- Improving advertisement targeting;
- Expanding creator content across languages;
- Reducing creator production costs;
- Increasing the number of monetizable videos;
- Supporting shoppable and performance advertising;
- Improving content recommendations.

YouTube subscriptions were growing faster than YouTube advertising, particularly YouTube Music and Premium.

However, Alphabet does not disclose:

- Revenue attributable to Ask YouTube;
- Incremental watch time generated by AI features;
- AI-driven YouTube ad yield;
- Revenue generated by AI creator tools;
- Inference costs associated with Ask YouTube;
- Subscriber conversion caused by AI features.

YouTube AI Assessment

Indicator	Current Assessment
Ask YouTube user engagement	Strong early adoption
Direct AI revenue	Not separately disclosed
Advertising benefit	Primarily indirect
Subscription benefit	Potentially positive
Creator productivity benefit	Positive but unquantified
Monetization maturity	Early

YouTube AI features are currently better understood as engagement and monetization enhancers than as a standalone revenue business.

5.4 Enterprise AI

Enterprise AI is the clearest direct AI revenue pathway outside Search.

Alphabet monetizes enterprise AI through:

- Gemini Enterprise;
- Vertex AI;
- Gemini model APIs;
- Google Workspace with Gemini;
- Data analytics;
- Cybersecurity;
- AI agents;

- AI-related Google Cloud infrastructure consumption.

The Form 10-Q states that GCP generates consumption and subscription fees from AI infrastructure, Vertex AI and Gemini Enterprise, while Workspace generates subscription revenue from productivity products containing integrated Gemini functionality.

5.4.1 Enterprise Adoption and Usage

Management disclosed several indicators of enterprise AI adoption:

- More than nine million developers build monthly with Alphabet's models and developer products;
- Model APIs process approximately **22 billion tokens per minute**, up from 16 billion in the previous quarter;
- Nearly 90% of the Fortune 100 use Gemini Enterprise;
- Nearly 500 Cloud customers each processed more than one trillion tokens during the preceding year;
- More than 2,000 enterprises consumed over 100 billion tokens during the preceding 12 months.

Alphabet specifically described the enterprise token statistics as growth in **paid token usage**, making them more meaningful than a broad measure that includes free consumer tokens.

Nevertheless, token volume alone does not determine revenue because revenue depends on:

- Model selected;
- Input versus output tokens;
- Contracted pricing;
- Volume discounts;
- Caching;
- Batch processing;
- Customer credits;
- TPU or GPU infrastructure used;
- Cost per token.

The enterprise model API economics can be simplified as:

Model API revenue = Paid tokens × Effective price per token

Model API contribution = Revenue – Compute cost – Support and sales expense

Alphabet does not disclose average revenue per token or contribution margin per token.

5.4.2 Gemini Enterprise

Gemini Enterprise is Alphabet's enterprise platform for:

- Developing AI agents;
- Connecting AI to internal enterprise systems;
- Automating business processes;
- Managing governance and security;
- Controlling AI costs;
- Integrating data, models and workflows.

Nearly 90% of the Fortune 100 use Gemini Enterprise, according to management. The Agent Development Kit had reached nearly 70 million cumulative downloads by Q2.

The primary revenue mechanisms include:

- Enterprise subscriptions;
- Model and token consumption;
- Infrastructure usage;
- Data and analytics services;

- Integration and platform services.

Gemini Enterprise has strategic value because it allows Alphabet to monetize multiple layers of the technology stack within the same customer relationship:

Enterprise agent → Gemini model usage → Vertex AI → Data services → Cloud infrastructure

This creates the potential for higher customer lifetime value than selling model access alone.

Gemini Enterprise Risks

Key risks include:

- Enterprises conducting pilots without moving into production;
- Lower-priced competing models;
- Falling token prices;
- High integration and support costs;
- Customers using multiple model providers;
- Difficulty proving measurable return on investment;
- Increased demand for data sovereignty and private infrastructure.

5.4.3 Vertex AI and Model APIs

Vertex AI allows customers to build, deploy and manage models and AI applications.

Revenue is generated through:

- Model inference;
- Model training;
- Data processing;
- Storage;
- Agent deployment;
- Supporting infrastructure;
- Enterprise platform subscriptions.

Vertex AI also enables Alphabet to offer both Gemini and third-party models. This reduces dependence on the commercial performance of a single model family and strengthens Google Cloud's role as an AI platform.

Alphabet stated that both TPUs and GPUs are used to serve Gemini Enterprise, Vertex AI and agentic workloads.

Model Commoditization Risk

Model capabilities may become increasingly competitive, reducing price per token.

Alphabet's defense is its full-stack offering:

- Proprietary models;
- Proprietary TPUs;
- NVIDIA GPU availability;
- Data services;
- Security;
- Workspace;
- Enterprise relationships;
- Agent-management tools.

The value proposition is therefore not limited to selling the Gemini model itself.

Gemini can serve as an ingredient within a broader enterprise solution rather than the entire commercial product.

Management emphasized this point when discussing cybersecurity, data analytics and end-to-end enterprise workflows.

5.4.4 Google Workspace AI

Google Workspace integrates Gemini into products such as:

- Gmail;
- Docs;
- Drive;
- Meet;
- Calendar;
- Other productivity and collaboration tools.

Workspace monetization is primarily subscription-based, which provides greater predictability than pure consumption revenue.

AI can increase Workspace revenue through:

- Higher revenue per seat;
- Premium AI subscription tiers;
- Greater customer retention;
- New enterprise customers;
- Upselling existing customers;
- Higher-value workflows and agents.

However, Alphabet does not separately disclose:

- Workspace AI revenue;
- Gemini revenue per Workspace seat;
- AI-related seat growth;
- Workspace AI customer retention;
- AI gross margin.

The core metric to monitor is not simply the number of Workspace users with access to Gemini. It is:

Incremental Workspace revenue per seat relative to incremental AI inference cost per seat

5.4.5 Security AI

Alphabet's AI security strategy combines:

- Google threat intelligence;
- Google Cloud security products;
- Wiz;
- AI-powered scanning;
- Automated code remediation;
- Security operations;
- Gemini models;
- CodeMender.

Management stated that approximately 90% of the Fortune 100 use Google Cloud Security, nearly 90% of Wiz customers use AI-powered security features, and the number of AI workloads scanned and protected increased by more than 45% sequentially.

Security AI can be monetized through:

- Platform subscriptions;
- Usage-based security processing;
- Cloud infrastructure;
- Enterprise contracts;
- Cross-selling between Wiz and Google Cloud;
- Higher customer retention.

Security may be a relatively attractive AI monetization category because customer willingness to pay is supported by:

- High cost of cyber incidents;
- Regulatory requirements;
- Increasing AI-related attack risks;
- Need for continuous monitoring;
- Deep integration with enterprise infrastructure.

Enterprise AI Assessment

Indicator	Current Assessment
Developer adoption	Strong
Paid token growth	Very strong
Large-enterprise penetration	High
Gemini Enterprise adoption	Strong
Workspace monetization	Active but undisclosed
Security AI adoption	Strong
Revenue disclosure	Aggregated within Cloud
Margin disclosure	Not separately provided
Monetization maturity	Scaling

Enterprise AI is already generating direct revenue through subscriptions, token usage and infrastructure consumption. The principal analytical limitation is that Alphabet does not separate AI revenue from the broader Google Cloud segment.

5.5 AI Infrastructure

AI infrastructure is both a revenue engine and the principal source of Alphabet's rising capital intensity.

Alphabet monetizes infrastructure through:

1. TPU-based Cloud consumption;
2. GPU-based Cloud consumption;
3. Model training and inference;
4. Direct TPU system sales;
5. Networking, storage and related platform services;
6. Large, long-duration Cloud contracts.

5.5.1 TPU Cloud Services

Alphabet uses TPUs for:

- Internal model training;

- Internal model inference;
- Search and consumer products;
- Gemini Enterprise;
- Vertex AI;
- Customer infrastructure workloads.

TPUs can support better economics through:

- Hardware and software co-design;
- Greater control over the supply chain;
- Potentially lower cost for selected workloads;
- Greater energy efficiency;
- Reduced dependence on third-party chip providers;
- Optimized model-serving performance.

Alphabet also offers TPU systems directly to customers. The company began recognizing TPU system revenue in Q2 2026.

The economic model differs depending on whether the TPU is consumed as a Cloud service or sold as a system.

TPU Model	Revenue Type	Economic Characteristics
TPU Cloud consumption	Usage-based service revenue	Recurring, capital-intensive
TPU system sale	Product and service revenue	Delivery-based, inventory-intensive
Internal TPU usage	Indirect product monetization	Supports Search, Gemini and Cloud margins

5.5.2 GPU Cloud Services

Google Cloud also offers NVIDIA accelerators and uses GPUs to support enterprise customers and serve AI models.

GPU revenue is generally generated through:

- Cloud compute consumption;
- Reserved capacity;
- Long-term infrastructure contracts;
- Training and inference usage;
- Supporting storage and networking.

Management stated that Google Cloud provides customers with both TPU- and GPU-based infrastructure and uses both to serve Vertex AI, Gemini Enterprise and agentic workloads.

The GPU pathway expands customer choice but can create greater exposure to:

- External chip costs;
- Supplier availability;
- Lower hardware differentiation;
- Competitive pricing across hyperscalers.

5.5.3 AI Training and Inference Demand

AI infrastructure demand is divided into two primary workloads.

Training

Training requires large amounts of compute over concentrated periods.

Training demand is affected by:

- Model size;
- Training dataset size;
- Number of model experiments;
- Frequency of model releases;
- Use of reinforcement learning and post-training;
- Development of multimodal models.

Inference

Inference occurs when users or enterprise applications interact with deployed models.

Inference demand is affected by:

- Number of users;
- Number of queries;
- Tokens per query;
- Model complexity;
- Agentic workflows;
- Video, image and audio generation;
- Enterprise adoption.

Inference may eventually represent a more recurring revenue source because it grows with ongoing application usage.

The relevant infrastructure revenue relationship is:

AI infrastructure revenue = Training consumption + Inference consumption + Storage + Networking + Platform services

The relevant profitability relationship is:

AI infrastructure contribution = Infrastructure revenue – Depreciation – Energy – Networking – Hardware – Third-party capacity costs

5.5.4 TPU System Sales

Alphabet's TPU system products include:

- Hardware;
- Software;
- Installation;
- Support;
- Extended warranties.

Hardware and software revenue is generally recognized when control transfers to the customer, while installation, support and warranty revenue is recognized as the services are performed or over the service period.

Alphabet began recognizing TPU system revenue in Q2, but management described the Q2 contribution as relatively small. Revenue is expected to ramp through the remainder of 2026, with the substantial majority under current agreements expected to be recognized in 2027.

TPU Sales Monetization Benefits

- Expands Alphabet's addressable infrastructure market;
- Allows customers to operate TPUs in their own or third-party data centers;
- Creates hardware and support revenue;

- Strengthens customer commitment to Google's software ecosystem;
- Monetizes Alphabet's proprietary silicon externally.

TPU Sales Risks

- Lower margins than Cloud software or consumption;
- Large inventory requirements;
- Working-capital usage;
- Delivery and installation risk;
- Warranty obligations;
- Potential quarterly revenue volatility;
- Concentration in a limited number of contracts.

The Form 10-Q states that Alphabet has signed only a limited number of TPU system agreements for specialized, high-scale workloads.

Alphabet does not separately disclose TPU revenue or TPU margin.

5.5.5 Cloud Backlog

Alphabet reported total revenue backlog of **\$519.5 billion**, of which **\$513.9 billion** related to Google Cloud. Just over 50% of the backlog is expected to be recognized over the following 24 months.

Cloud backlog increased by more than \$50 billion sequentially, driven by enterprise AI demand. Management stated that most of the backlog relates to conventional GCP contracts, although TPU system contracts are included.

Backlog is a strong forward indicator, but it is not equivalent to AI revenue because it includes:

- Non-AI Cloud services;
- Long-duration customer commitments;
- TPU contracts;
- Infrastructure consumption;
- Services to be provided over multiple years.

Key backlog metrics should include:

- Total Cloud backlog growth;
- AI-related share of backlog;
- Backlog conversion;
- Contract duration;
- Customer concentration;
- Revenue recognized per dollar of backlog;
- Margin on backlog conversion.

A useful analytical measure is:

Backlog conversion rate = Recognized Cloud revenue ÷ Beginning Cloud backlog

5.5.6 Third-Party Data Center Capacity

Alphabet remains supply-constrained and plans to expand its use of third-party capacity as a temporary bridge while internal infrastructure is constructed.

Management expects this strategy to create modest near-term pressure on Google Cloud's operating margin because third-party capacity is more expensive than internally owned infrastructure.

The economic trade-off is:

Higher short-term capacity cost → Ability to serve customers immediately → Greater long-term contract value

Management described certain arrangements as cases in which Alphabet may accept several months of higher costs to support multiyear customer opportunities with attractive lifetime economics.

Alphabet’s filing also identifies financial guarantees and backstop commitments that support the construction of third-party data centers and power infrastructure. These agreements may expose Alphabet to additional liabilities, excess capacity or counterparty risk.

Third-Party Capacity Assessment

Factor	Impact
Near-term revenue capacity	Positive
Customer acquisition and retention	Positive
Near-term Cloud margin	Negative
Upfront CapEx requirement	Lower
Long-term unit economics	Potentially weaker
Supply-chain flexibility	Positive
Contractual risk	Higher

5.6 AI Monetization Logic Table

AI Investment	Corresponding Revenue Source	Current Progress	Future Monitoring Indicators
TPU servers and owned data centers	Cloud AI infrastructure consumption	Cloud AI infrastructure growing rapidly	Utilization, compute revenue, operating margin and depreciation per revenue dollar
TPU system development	Direct TPU product sales	Initial revenue recognized in Q2; larger contribution expected in 2027	Orders, revenue recognition, inventory, product margin and customer concentration
GPU infrastructure	Cloud training and inference	Supports broad customer demand	GPU utilization, pricing, supply availability and gross margin
Gemini model development	APIs, Vertex AI and enterprise AI	Paid token usage growing rapidly	Paid tokens, revenue per token, cost per token and customer retention

AI Investment	Corresponding Revenue Source	Current Progress	Future Monitoring Indicators
Gemini Enterprise	Enterprise subscriptions and Cloud consumption	Nearly 90% of Fortune 100 using the product	Active enterprise customers, agents deployed, seat revenue and production usage
AI Search	Search advertising and commerce	AI Mode above one billion monthly users; monetization active	Incremental queries, ad load, click-through rate, revenue per query and inference cost
Consumer Gemini	Google One AI subscriptions	Gemini App at approximately 950 million monthly users; AI-plan demand positive	Paid subscribers, conversion rate, ARPU, churn and inference cost per subscriber
YouTube AI	Advertising, subscriptions and commerce	Ask YouTube reached more than 140 million users in June	Incremental watch time, ad yield, subscriber conversion and shopping revenue
Security AI	Google Cloud Security and Wiz	High Fortune 100 adoption and growing protected workloads	Security ARR, AI workload volume, cross-selling and retention
AI talent	Product development and internal productivity	R&D and compensation spending increased	Revenue per employee, development speed, model cost efficiency and product adoption
Third-party capacity	Immediate Cloud infrastructure revenue	Use expected to expand as a capacity bridge	Capacity cost, contract lifetime value and Cloud margin impact
Networking and energy infrastructure	Supports all AI products	Capacity expansion accelerating	Energy cost, network cost, uptime and revenue per installed capacity unit

5.7 Monetization Maturity by Business

AI Business	Usage Scale	Direct Revenue Evidence	Profitability Evidence	Maturity
AI Search	Very high	Search advertising growth and active AI ad formats	Services margin remains strong, but AI unit economics undisclosed	Scaling
Gemini Consumer	Very high	Demand for Google One AI plans	No product-level profit disclosure	Early commercial
YouTube AI	High	Primarily indirect through advertising and subscriptions	Not separately disclosed	Early
Gemini Enterprise	High and expanding	Cloud subscriptions, usage and paid tokens	Reflected within strong Cloud margin	Scaling
Vertex AI and APIs	Rapid token growth	Direct usage-based revenue	Product margin not disclosed	Scaling
Workspace AI	Large installed base	Subscription revenue	Incremental margin not disclosed	Scaling
Security AI	Strong enterprise adoption	Cloud and security contracts	Product margin not disclosed	Scaling
TPU Cloud Infrastructure	High demand and supply-constrained	Consumption revenue	Reflected within Cloud margin	Large-scale
TPU System Sales	Early	Initial Q2 revenue	Margin undisclosed	Emerging

5.8 Key Analytical Limitations

Alphabet provides considerable AI usage data but limited AI financial disclosure.

The most important missing metrics are:

- Total AI-related revenue;
- Consumer Gemini subscription revenue;
- Paid Gemini subscribers;
- AI Search revenue;
- Revenue and cost per AI query;
- Revenue and cost per paid token;
- Vertex AI revenue;
- Gemini Enterprise revenue;
- AI-related Workspace ARPU;

- TPU system revenue and margin;
- TPU versus GPU infrastructure economics;
- AI share of Cloud backlog;
- AI infrastructure utilization;
- Fully loaded AI operating profit.

This creates a risk of confusing rapid product adoption with economic value creation.

The appropriate analytical rule is:

Usage growth should be treated as evidence of demand, not as proof of revenue or profitability.

5.9 AI Monetization Scorecard

Dimension	Rating	Rationale
Consumer AI adoption	Very Strong	AI Mode and Gemini App have reached large-scale usage
Search AI monetization	Strong	Ads are active and commercial query coverage is expanding
Consumer subscription monetization	Positive but Early	AI plans are contributing to Google One growth, but financial disclosure is limited
YouTube AI monetization	Early	Engagement is meaningful, but direct revenue is undisclosed
Enterprise AI adoption	Very Strong	High Fortune 100 adoption and rapid paid-token growth
Enterprise AI revenue	Strong	Directly supports Cloud subscriptions and consumption
AI infrastructure revenue	Very Strong	Major contributor to Cloud growth
TPU system monetization	Emerging	Revenue recognition has begun, with larger 2027 contribution expected
Revenue visibility	Strong	Cloud backlog exceeds \$500 billion
Product-level disclosure	Weak	AI revenue and margins are not separately reported
AI profitability evidence	Moderate to Strong	Cloud margin is strong, but fully loaded AI economics remain unclear
Capital efficiency	Uncertain	Revenue growth is strong, but CapEx and operating costs are increasing rapidly

5.10 Overall AI Monetization Conclusion

Alphabet's AI monetization strategy is broader than that of a standalone model developer because the company can monetize the same AI investment across multiple products and customer groups.

A Gemini model can simultaneously support:

- Search advertising;
- Consumer subscriptions;
- YouTube engagement;
- Workspace productivity;
- Cloud APIs;
- Enterprise agents;
- Cybersecurity;
- Infrastructure consumption.

This full-stack model creates multiple opportunities to recover the cost of model development and technical infrastructure.

The current monetization position can be summarized as:

Consumer AI adoption at massive scale

plus

Active Search advertising monetization

plus

Rapidly scaling enterprise AI and infrastructure revenue

plus

Emerging consumer subscriptions and TPU system sales

minus

Limited product-level disclosure and high infrastructure costs

Consumer AI has achieved strong distribution, but direct monetization outside Search remains relatively early. Enterprise AI is further advanced because token usage, subscriptions and infrastructure consumption create identifiable customer payments. AI infrastructure is already a major revenue engine, but it also requires substantial capital expenditure and exposes Alphabet to depreciation, energy, hardware and third-party capacity costs.

The central question is not whether Alphabet can monetize AI. The evidence indicates that it already is.

The more important investment question is:

Can Alphabet increase AI revenue faster than the combined growth in inference costs, depreciation, energy, model-development spending, customer incentives and third-party infrastructure costs?

The strongest current evidence comes from Google Cloud, where AI solutions and AI infrastructure contributed to 82% revenue growth and a 35.6% reported operating margin. However, because Alphabet does not separately disclose AI revenue or fully loaded AI costs, investors cannot yet calculate a precise consolidated AI return on invested capital.

Future evidence of successful monetization should include:

- Continued Search growth without deterioration in Services margins;
- Rising paid Gemini subscription penetration;
- Paid-token revenue growing faster than token volume;
- Stable or improving Cloud margins after third-party capacity and TPU sales;
- Strong backlog conversion;
- Lower inference cost per query and token;

- Improving free cash flow as installed infrastructure begins generating revenue.

6. CapEx, Cash Flow and AI ROIC

6.1 Executive Assessment

Alphabet’s AI investment cycle is producing visible revenue and operating-profit growth, particularly within Google Cloud. However, the company has not yet demonstrated that these investments are generating attractive returns at the consolidated free-cash-flow level.

The quarter showed three simultaneous trends:

- Google Cloud revenue and operating income accelerated sharply;
- Capital expenditures increased substantially faster than operating cash flow;
- Free cash flow turned negative as Alphabet continued building servers, data centers, networking infrastructure and power capacity.

Alphabet generated **\$39.1 billion of operating cash flow** in Q2 2026 but spent **\$44.9 billion on capital expenditures**, resulting in **negative free cash flow of \$5.9 billion**. The company raised its full-year 2026 CapEx guidance to **\$195–\$205 billion** and stated that CapEx is expected to increase significantly again in 2027.

The current investment conclusion is:

Alphabet has demonstrated positive AI monetization and strong operating leverage, but it has not yet demonstrated mature AI free-cash-flow returns.

The evidence is strongest at the Google Cloud operating-income level and weakest at the consolidated free-cash-flow level.

6.2 Capital Expenditure Scorecard

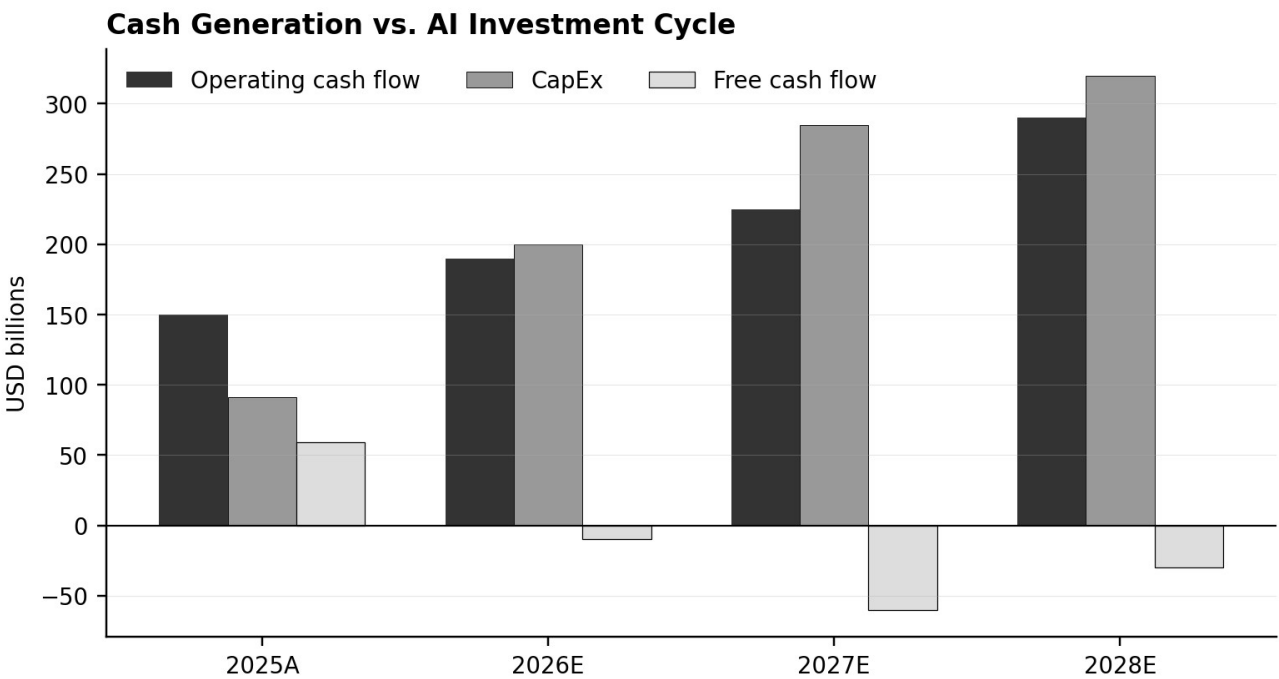


Figure 3. Operating cash flow, capital expenditures and free cash flow. Source: company filings and analyst estimates.

Quarterly Capital Intensity

Metric	Q2 2025	Q2 2026	YoY Change
Revenue	\$96.43B	\$119.80B	+24.2%
Operating Cash Flow	\$27.75B	\$39.07B	+40.8%
Capital Expenditures	\$22.45B	\$44.92B	+100.1%
Free Cash Flow	\$5.30B	-\$5.86B	-\$11.16B
CapEx as % of Revenue	23.3%	37.5%	+1,423 bps
Operating Cash Flow as % of Revenue	28.8%	32.6%	+381 bps
Free Cash Flow Margin	5.5%	-4.9%	-1,039 bps
Operating Cash Flow / CapEx	1.24x	0.87x	Deteriorated
Revenue / CapEx	4.30x	2.67x	Deteriorated

Q2 2025 CapEx was approximately \$22.4 billion and free cash flow was \$5.3 billion. Q2 2026 CapEx increased to \$44.9 billion while free cash flow declined to negative \$5.9 billion.

The most important relationship is:

Operating cash flow increased 41%, but CapEx increased 100%.

This explains why free cash flow deteriorated despite strong operating performance.

The decline in Revenue / CapEx from 4.30x to 2.67x does not prove that capital productivity has permanently weakened. Current-period revenue is generated partly by infrastructure purchased in previous periods, while current CapEx will support revenue over several future years.

Nevertheless, the ratio shows that Alphabet is currently investing capital much faster than it is generating incremental revenue.

First-Half Capital Intensity

Metric	H1 2025	H1 2026	YoY Change
Revenue	\$186.66B	\$229.69B	+23.1%
Operating Cash Flow	\$63.90B	\$84.86B	+32.8%
Capital Expenditures	\$39.64B	\$80.60B	+103.3%
Free Cash Flow	\$24.25B	\$4.26B	-82.4%
CapEx as % of Revenue	21.2%	35.1%	+1,387 bps
Free Cash Flow Margin	13.0%	1.9%	-1,108 bps

Metric	H1 2025	H1 2026	YoY Change
Operating Cash Flow / CapEx	1.61x	1.05x	Deteriorated

Alphabet's first-half cash-flow statement shows that purchases of property and equipment more than doubled to \$80.6 billion, while operating cash flow increased to \$84.9 billion. As a result, nearly all first-half operating cash flow was absorbed by CapEx.

The first-half relationship can be expressed as:

\$84.86B operating cash flow – \$80.60B CapEx = \$4.26B free cash flow

This represents a major reduction in internally generated cash available for:

- Share repurchases;
- Dividends;
- Acquisitions;
- Debt repayment;
- Strategic investments;
- Balance-sheet accumulation.

6.3 Composition of Capital Expenditures

Alphabet stated that the vast majority of Q2 CapEx was invested in technical infrastructure supporting AI.

Approximately:

- **60%** of technical-infrastructure investment was allocated to servers;
- **40%** was allocated to data centers and networking equipment.

The capital-expenditure pathway is therefore:

Servers and accelerators → Immediate compute capacity

Data centers, networking and power → Long-duration capacity foundation

These categories have different investment and return timelines.

Investment Category	Typical Deployment Period	Primary Return Source
Servers, TPUs and GPUs	Relatively short	Cloud consumption, Search inference and model training
Networking equipment	Short to intermediate	Higher cluster utilization and data throughput
Data-center buildings	Multi-year	Long-duration compute capacity
Land and power infrastructure	Multi-year	Future expansion capacity

Investment Category	Typical Deployment Period	Primary Return Source
Third-party infrastructure support	Contract-dependent	Faster access to customer capacity
TPU inventory for sale	Delivery-dependent	Product revenue and support revenue

Server investment may begin generating revenue relatively quickly after installation. Data-center, power and construction projects generally have much longer payback periods.

6.4 Infrastructure Deployment Pipeline

Alphabet's balance sheet shows a significant increase in both operating infrastructure and infrastructure that is not yet ready for use.

Property and Equipment Metric	Dec. 31, 2025	June 30, 2026	Change
Technical Infrastructure in Service	\$203.68B	\$247.18B	+21.4%
Total Property and Equipment in Service	\$266.49B	\$304.31B	+14.2%
Assets Not Yet in Service	\$78.59B	\$122.81B	+56.3%
Net Property and Equipment	\$246.60B	\$321.21B	+30.3%
Accrued Purchases of Property and Equipment	\$8.88B	\$16.20B	+82.4%

Assets not yet in service consist primarily of technical infrastructure under construction or assembly. Alphabet states that these assets may take months or years to become operational, particularly for multi-phase data-center projects.

The increase in assets not yet in service has two major implications.

Positive Implication

The construction pipeline provides substantial future capacity that can support:

- Google Cloud revenue;
- Gemini training and inference;
- AI Search;
- YouTube AI features;
- Enterprise AI agents;
- TPU and GPU workloads.

Negative Implication

The assets have already required capital but are not yet generating their full revenue potential.

Once placed into service, they will also begin producing:

- Depreciation expense;

- Energy costs;
- Maintenance costs;
- Network expenses;
- Staffing and operational costs.

The investment sequence is:

Cash spent today → Asset under construction → Asset placed into service → Revenue generation and depreciation begin

Alphabet is currently carrying a large amount of capital between the first and third stages of this sequence.

6.5 The Depreciation Lag

Depreciation on property and equipment increased from **\$9.5 billion in H1 2025 to \$13.6 billion in H1 2026**, representing growth of approximately 43%. However, CapEx increased more than 100% over the same period.

This difference reflects the time lag between purchasing infrastructure and placing it into service.

The key relationship is:

CapEx growth today → Depreciation growth in future periods

Because assets not yet in service reached \$122.8 billion, the income statement has not yet absorbed the full depreciation burden associated with Alphabet’s current infrastructure program.

This creates a future earnings risk:

Revenue growth could eventually normalize while depreciation continues accelerating as previously purchased infrastructure enters service.

Management also expects higher data-center operating costs, including energy expenses, to accompany the increase in depreciation.

A complete return analysis must therefore consider both:

- The immediate cash outflow from CapEx;
- The later income-statement cost from depreciation and operations.

6.6 Operating Cash Flow Bridge

Alphabet’s operating cash generation remained strong, but working-capital requirements increased.

H1 2026 Operating Cash Flow Components

Cash-Flow Component	H1 2026
Reported Net Income	\$174.77B
Depreciation Add-Back	+\$13.59B
Stock-Based Compensation Add-Back	+\$14.71B

Cash-Flow Component	H1 2026
Deferred-Tax Adjustment	+\$27.54B
Securities Gains Removed	-\$135.80B
Accounts Receivable Change	-\$6.90B
Inventory Change	-\$7.74B
Other Assets Change	-\$9.95B
Accounts Payable Change	+\$2.09B
Other Adjustments	Net positive
Operating Cash Flow	\$84.86B

Alphabet's cash-flow statement shows that growth in accounts receivable, inventory and other assets created a significant use of operating cash during the first half.

Accounts Receivable

Accounts receivable used approximately \$6.9 billion of cash during H1.

This may reflect:

- Rapid revenue growth;
- Large enterprise Cloud contracts;
- Customer billing timing;
- Changes in payment terms;
- Higher infrastructure and system sales.

Higher receivables are not necessarily negative if they result from strong revenue growth, but persistent growth faster than revenue could weaken cash conversion.

Inventory

Inventory used approximately \$7.7 billion of cash during H1. Alphabet's inventory increased from **\$2.44 billion to \$9.99 billion** between December 2025 and June 2026.

Alphabet states that inventory primarily includes:

- TPU systems for sale to enterprise customers;
- Pixel and other consumer devices;
- Raw materials;
- Work in process;
- Finished goods.

The increase indicates that the TPU system business requires Alphabet to invest cash before customer delivery.

The TPU cash-flow sequence is:

Components purchased → Inventory produced → Cash absorbed → System delivered → Revenue recognized → Customer cash collected

Therefore, TPU system sales can create a temporary operating-cash-flow headwind before producing reported revenue.

Management explicitly stated that Alphabet is building inventory ahead of TPU system deliveries, with most existing contracted revenue expected to be recognized in 2027.

6.7 Free Cash Flow Interpretation

Alphabet defines free cash flow as:

Free cash flow = Operating cash flow – Capital expenditures

Q2 free cash flow was negative \$5.9 billion, while trailing-12-month free cash flow remained positive at \$53.3 billion.

Negative quarterly free cash flow does not mean Alphabet's existing business is unprofitable. It means current infrastructure investment exceeded current operating cash generation.

The appropriate interpretation is:

Strong operating cash generation

minus

Exceptionally high growth investment

equals

Weak current free cash flow

However, investors should not assume that all CapEx is optional growth investment.

Alphabet does not separately disclose:

- Maintenance CapEx;
- Replacement CapEx;
- Growth CapEx;
- AI-specific CapEx;
- Search-specific infrastructure investment;
- Cloud-specific CapEx.

A shareholder-oriented owner-earnings measure would ideally be:

Owner earnings = Operating cash flow – Maintenance CapEx

Reported free cash flow subtracts both maintenance and growth CapEx. This may understate current economic earnings during a rapid expansion period, but it may also overstate future returns if a large portion of CapEx must be repeated continuously to remain competitive.

6.8 Full-Year CapEx Outlook

Alphabet raised its 2026 CapEx guidance from **\$180–\$190 billion to \$195–\$205 billion**. The increase was attributed primarily to accelerated capacity delivery in response to strong demand. Management also expects CapEx to increase significantly again in 2027.

Alphabet spent \$80.6 billion during the first half.

The updated guidance implies second-half CapEx of:

FY2026 CapEx Scenario	FY Guidance	Implied H2 CapEx	Average Per Quarter
Low End	\$195.0B	\$114.4B	\$57.2B
Midpoint	\$200.0B	\$119.4B	\$59.7B
High End	\$205.0B	\$124.4B	\$62.2B

This is an analytical calculation based on reported first-half CapEx and management's guidance.

At the midpoint, average quarterly CapEx during the second half would be approximately **33% higher than Q2's already elevated \$44.9 billion**.

This suggests that free-cash-flow pressure is likely to continue unless operating cash flow accelerates substantially.

6.9 Contractual Infrastructure Commitments

Reported CapEx does not capture Alphabet's entire future infrastructure exposure.

Commitment or Exposure	June 30, 2026
Fixed or Guaranteed Long-Term Contract Commitments	\$707.0B
Total Material Purchase and Contractual Obligations	\$811.0B
Short-Term Portion of Material Obligations	\$200.7B
Data-Center Leases Not Yet Commenced	\$85.2B
Short-Term Lease Beginning in Q3	\$5.8B
Data-Center Credit-Derivative Backstops	\$43.8B
Financial Guarantees	\$7.6B
Estimated Future Infrastructure Backstops	\$24.1B

These amounts should **not be added together**, because the categories may overlap and include different accounting definitions. They also include certain obligations that are not exclusively AI-related.

However, they demonstrate that Alphabet's economic infrastructure exposure is substantially larger than current-period cash CapEx alone.

The \$707 billion of fixed or guaranteed commitments primarily relates to long-term supply agreements, with additional commitments for energy and content. Some energy contracts extend through 2054 and contain minimum-purchase or take-or-pay provisions.

Alphabet also has significant data-center lease and backstop arrangements designed to support third-party construction and power infrastructure. In a counterparty default, Alphabet may be required to assume leases, use the underlying facilities or make termination payments.

These arrangements improve access to scarce capacity but increase:

- Fixed-cost exposure;
- Counterparty risk;
- Excess-capacity risk;
- Long-term contractual obligations;
- Complexity of calculating invested capital.

6.10 Third-Party Capacity Economics

Alphabet plans to increase its use of third-party compute capacity as a temporary bridge while internally owned capacity is constructed.

Management expects this strategy to create modest near-term pressure on Google Cloud's operating margin because externally sourced capacity is more expensive.

The trade-off is:

Internally Owned Capacity	Third-Party Capacity
High upfront CapEx	Lower immediate CapEx
Lower potential long-run unit cost	Higher near-term operating cost
Construction and deployment delay	Faster availability
Greater operational control	Greater counterparty dependence
Utilization risk remains with Alphabet	Contract structure may transfer some risk
Higher long-term margin potential	Lower near-term margin potential

The strategic rationale is that Alphabet may accept higher short-term costs to secure:

- New enterprise customers;
- Multiyear Cloud contracts;
- Greater customer lifetime value;
- Higher backlog;
- Market share.

A proper return test should compare:

Lifetime customer operating profit

against

Third-party capacity cost + customer-acquisition cost + allocated infrastructure cost

A temporary decline in Cloud margin may be rational if the customer relationship generates sufficiently attractive lifetime economics.

6.11 Funding the AI Investment Cycle

Alphabet's AI infrastructure program is no longer being funded exclusively through internally generated free cash flow.

During H1 2026, Alphabet received:

- \$30.5 billion from common-stock issuance;
- \$19.1 billion from mandatory convertible preferred stock;
- \$56.2 billion of gross debt proceeds;
- Additional proceeds from sales of interests in consolidated entities.

Alphabet conducted no share repurchases during the first half.

The company ended Q2 with:

- \$242.5 billion of cash and marketable securities;
- \$98.2 billion of long-term debt;
- A \$40 billion at-the-market equity program that remained unused.

Alphabet therefore retains substantial liquidity. The financing activity does not indicate an immediate inability to fund operations.

Instead, it indicates that management is:

- Preserving strategic liquidity;
- Matching long-duration infrastructure with long-term financing;
- Expanding capacity faster than current free cash flow permits;
- Sharing part of the investment burden with debt and equity investors.

The cost of capital must still be reflected in AI returns.

Higher financing creates:

- Interest expense;
- Preferred dividends;
- Common-stock dilution;
- Future preferred-stock conversion;
- A higher required return on investment.

The relevant hurdle is no longer simply whether an AI project produces positive operating profit.

It must produce a return above Alphabet's blended cost of debt, preferred equity and common equity.

6.12 Defining AI ROIC

A true AI return on invested capital would be calculated as:

AI ROIC = AI-related after-tax operating profit ÷ Average AI-related invested capital

AI-related after-tax operating profit would ideally include:

- Google Cloud AI infrastructure profit;
- Gemini API and enterprise AI profit;
- Incremental Search advertising profit caused by AI;
- Consumer AI subscription profit;
- YouTube AI contribution;
- TPU system gross profit;

minus:

- AI model-development expenses;
- Shared AI R&D;
- Depreciation;
- Energy and networking costs;
- Third-party capacity costs;
- AI sales and marketing;
- Allocated support expenses.

AI-related invested capital would ideally include:

- Net AI servers, TPUs and GPUs;
- AI-related data centers and networking;
- Assets under construction;
- TPU inventory and working capital;
- Relevant lease liabilities;
- Third-party infrastructure commitments;
- AI acquisitions and goodwill;
- Capitalized or economically accumulated model-development spending.

Alphabet does not disclose these data separately.

Consequently:

A precise Alphabet AI ROIC cannot be calculated from the public financial statements.

Any single numerical AI ROIC estimate would require material assumptions about revenue attribution, cost allocation, asset allocation and project timing.

6.13 Accounting ROIC Versus Economic AI ROIC

Alphabet's AI investments create both direct and defensive returns.

Direct Returns

These include:

- Cloud infrastructure revenue;
- Gemini and Vertex AI consumption;
- Enterprise subscriptions;
- Consumer AI subscriptions;
- TPU system sales;
- Incremental Search and YouTube advertising revenue.

Defensive Returns

AI infrastructure also helps Alphabet:

- Preserve Search market share;
- Prevent user migration to competing assistants;
- Maintain advertiser relevance;
- Protect the Android and Google ecosystem;
- Retain Cloud customers;
- Maintain technological leadership.

Defensive returns are economically real but difficult to measure because they represent revenue or market share that Alphabet might otherwise have lost.

Therefore, direct AI revenue may understate the strategic return on investment.

Conversely, management may characterize necessary defensive spending as growth investment even when it produces limited incremental revenue.

A complete assessment must ask:

Is AI investment creating new profits, protecting existing profits, or merely increasing the cost of remaining competitive?

6.14 Current AI Return Proxies

Although true AI ROIC cannot be calculated, several operating proxies can be used.

Cloud Revenue and Profit Growth

Google Cloud Metric	Q2 2025	Q2 2026	Change
Revenue	\$13.62B	\$24.77B	+\$11.14B
Operating Income	\$2.83B	\$8.81B	+\$5.99B
Operating Margin	20.7%	35.6%	+1,484 bps
Cloud Backlog	Approximately \$106B	\$513.9B	Substantial increase

Google Cloud revenue increased 82%, while operating income more than tripled. Cloud backlog reached \$513.9 billion, with slightly more than half expected to be recognized over the following 24 months.

The backlog comparison should be treated cautiously because Alphabet changed its backlog reporting in Q1 2026 to include contracts with an original expected term of one year or less.

Current-Period Capital-Coverage Proxies

Proxy	Calculation	Result
Incremental Q2 CapEx	\$44.92B – \$22.45B	\$22.48B
Incremental Cloud Operating Income	\$8.81B – \$2.83B	\$5.99B
Incremental Consolidated Operating Income	\$40.77B – \$31.27B	\$9.50B
Cloud Incremental Operating Income / Incremental CapEx	\$5.99B ÷ \$22.48B	26.6%
Consolidated Incremental Operating Income / Incremental CapEx	\$9.50B ÷ \$22.48B	42.3%
Cloud Incremental Revenue / Incremental CapEx	\$11.14B ÷ \$22.48B	49.6%

These ratios are calculated from reported Q2 figures.

They are **not ROIC measures** and should not be annualized.

The reasons are:

- Current profit is supported by assets purchased in previous periods;
- Current CapEx may not generate revenue until future periods;
- CapEx supports Search, YouTube, Gemini, Cloud and internal R&D;
- Cloud operating income includes both AI and non-AI products;
- Shared AI R&D is partly reported outside Google Cloud;
- The calculations do not use average invested capital.

However, the ratios provide an important directional signal:

Operating profit is growing meaningfully alongside the investment program, but not yet rapidly enough to prevent free-cash-flow compression.

6.15 AI Return by Monetization Path

AI Investment Area	Return Mechanism	Current Evidence	Required Confirmation
Cloud TPUs and GPUs	Infrastructure-consumption profit	Cloud revenue and operating income accelerating	Stable margin after depreciation and third-party capacity
Data centers and networking	Higher compute availability	Large backlog and supply constraints	High utilization and backlog conversion
Gemini models	API, enterprise and subscription revenue	Paid token and enterprise usage growth	Revenue per token and contribution margin
AI Search infrastructure	Advertising revenue and franchise protection	Search revenue and paid clicks growing	Revenue per AI query relative to inference cost
Consumer Gemini	Subscription revenue	Demand for Google One AI plans	Paid subscribers, ARPU and churn
TPU system inventory	Product gross profit	Initial revenue recognized	Product margin and working-capital recovery
Third-party capacity	Multiyear Cloud customer value	Enables demand capture	Contract lifetime profit exceeds bridging costs
Shared AI R&D	Future products and model quality	Broad product adoption	Revenue and productivity gains exceed recurring R&D cost
Energy infrastructure	Enables long-term compute growth	Capacity commitments secured	Cost per unit of compute and utilization

The strongest measurable return currently comes from Google Cloud.

The weakest areas of financial disclosure are:

- Consumer Gemini;
 - AI Search unit economics;
 - TPU product margins;
 - Fully loaded Gemini model profitability;
 - Return on shared AI R&D.
-

6.16 AI Investment Lifecycle

Alphabet appears to be moving through several stages simultaneously.

Stage 1: Capacity Construction

Characteristics:

- Rapid CapEx growth;
- Large assets not yet in service;
- Negative or weak free cash flow;
- Increased debt and equity financing.

Alphabet remains heavily exposed to this stage.

Stage 2: Deployment

Characteristics:

- Servers enter service;
- Depreciation increases;
- Energy and operational expenses rise;
- Revenue capacity begins expanding.

Alphabet is also increasingly entering this stage.

Stage 3: Utilization and Monetization

Characteristics:

- Cloud consumption increases;
- AI Search query volume increases;
- Enterprise customers use contracted capacity;
- Operating margin expands if revenue grows faster than costs.

Google Cloud's Q2 results provide evidence that part of Alphabet's infrastructure has reached this stage.

Stage 4: Mature Cash Return

Characteristics:

- CapEx growth moderates;
- Installed capacity reaches high utilization;
- Operating cash flow grows faster than CapEx;
- Free cash flow recovers;
- Capital can again be returned through buybacks or debt reduction.

Alphabet has not yet reached this stage at the consolidated level.

The current position can be summarized as:

Alphabet is monetizing previously installed capacity while simultaneously conducting a substantially larger new construction cycle.

6.17 AI ROIC Scenario Framework

Bull Case

Conditions:

- Cloud revenue remains very strong;
- Backlog converts rapidly;
- Internal capacity replaces expensive third-party capacity;
- TPU and GPU utilization remains high;
- Cloud margin remains above 30%;
- AI Search monetization improves;
- Cost per query and token declines;
- CapEx growth eventually moderates;
- Free cash flow recovers strongly.

Interpretation:

Alphabet's current investment cycle generates attractive long-term returns despite weak near-term free cash flow.

Base Case

Conditions:

- Cloud growth gradually normalizes;
- Cloud margin declines moderately from Q2's exceptional level;
- Depreciation and energy costs increase;
- Search and subscriptions continue growing;
- CapEx remains elevated through 2027;
- Free cash flow recovers slowly after the main build-out phase.

Interpretation:

AI investment creates positive economic value, but the payback period is longer and the near-term equity return is reduced by financing and dilution.

Bear Case

Conditions:

- Cloud demand slows before capacity is fully utilized;
- Customers renegotiate or delay consumption;
- Backlog conversion disappoints;
- Third-party capacity remains expensive;
- TPU hardware carries lower margins;
- AI token and compute prices decline;
- Search AI costs increase faster than advertising revenue;
- Depreciation continues rising after revenue growth slows;
- Alphabet is left with excess or inflexible capacity.

Interpretation:

The infrastructure cycle produces low returns and prolonged free-cash-flow pressure.

6.18 Key Metrics Required to Validate AI ROIC

Metric	Positive Signal	Warning Signal
Cloud Revenue Growth	Remains above infrastructure-cost growth	Decelerates below depreciation growth
Cloud Operating Income	Continues growing rapidly	Stagnates despite high CapEx
Cloud Operating Margin	Remains resilient after capacity costs	Sustained major compression
Backlog Conversion	Revenue recognition accelerates	Backlog grows without conversion
CapEx / Revenue	Peaks and begins declining	Continues increasing indefinitely
Operating Cash Flow / CapEx	Returns above 1.0x	Remains below 1.0x
Free Cash Flow	Recovers after capacity deployment	Remains negative or structurally weak
Assets Not Yet in Service	Converts into productive capacity	Continues accumulating without revenue
Depreciation / Revenue	Stabilizes as utilization rises	Increases faster than revenue
Inventory	Converts into TPU sales and cash	Continues growing without delivery
Revenue per Installed Capacity	Improves	Declines
AI Cost per Query or Token	Falls	Remains high or increases
Third-Party Capacity Use	Declines as internal capacity opens	Becomes permanent and margin-dilutive
Share Count and Debt	Stabilize	Continue rising to fund CapEx
Share Repurchases	Resume from internally generated cash	Remain suspended due to cash needs

6.19 CapEx and AI Return Scorecard

Dimension	Rating	Rationale
AI Demand Evidence	Very Strong	Cloud demand, AI usage and backlog remain high
Cloud Revenue Return	Very Strong	Cloud revenue increased 82%
Cloud Operating-Profit Return	Very Strong	Operating income more than tripled
Search AI Return	Positive but Unquantified	Search growth remains strong, but AI economics are undisclosed
Operating Cash Generation	Strong	Q2 OCF increased 41%
Current Free Cash Flow	Very Weak	Q2 FCF was negative \$5.9B
Capital Intensity	Very High	CapEx represented 37.5% of revenue
Future Depreciation Risk	High	Assets not yet in service reached \$122.8B
Infrastructure Commitments	Very High	Significant leases, supply agreements and backstops
Funding Capacity	Strong	Large liquidity and access to debt and equity
Shareholder Dilution Risk	Increasing	Common and preferred equity issued
AI ROIC Visibility	Low	AI revenue, costs and invested capital not separately disclosed
Current AI ROIC Direction	Positive but Unproven	Strong operating returns but weak free-cash-flow returns

6.20 Overall CapEx, Cash Flow and AI ROIC Conclusion

Alphabet's AI investment program is producing strong commercial results.

Google Cloud revenue increased 82%, Cloud operating income more than tripled and Cloud margin expanded to 35.6%. Search revenue also remained strong, suggesting that AI investment is supporting both enterprise growth and the resilience of Alphabet's core advertising franchise.

However, the cash-flow evidence is less favorable.

CapEx doubled year over year, absorbed more than Alphabet's quarterly operating cash flow and caused free cash flow to turn negative. Assets not yet in service increased to \$122.8 billion, meaning substantial additional depreciation and operating costs have not yet reached the income statement.

Alphabet has also assumed significant supply, lease, energy and third-party infrastructure commitments. The economic capital at risk is therefore larger than reported CapEx alone.

The quarter can be summarized as:

Strong AI demand and operating-profit growth

plus

Large future contracted revenue

minus

Negative current free cash flow, rising depreciation and expanding capital commitments

The evidence supports the conclusion that AI is generating operating returns.

It does not yet support the stronger conclusion that Alphabet's current AI investment cycle is generating attractive consolidated free-cash-flow returns.

The central test over the next several years is:

Can incremental Search, Cloud, subscription and TPU operating profit grow faster than depreciation, energy costs, third-party capacity expenses, model-development spending, financing costs and the capital required to replace aging infrastructure?

A successful AI investment cycle should eventually produce all four outcomes:

1. Strong backlog conversion;
2. Stable or improving operating profit;
3. A declining CapEx-to-revenue ratio;
4. Recovery in free cash flow.

Until those conditions emerge, Alphabet's AI ROIC should be classified as:

Directionally positive at the operating level, but not yet proven at the consolidated cash-return level.

7. Balance Sheet and Financing

7.1 Executive Assessment

Alphabet remains in a strong liquidity position, but its capital structure has changed materially.

Historically, Alphabet could be analyzed as a highly cash-generative technology company with substantial net cash, limited debt and enough free cash flow to fund capital expenditures, acquisitions, dividends and share repurchases internally.

That description is no longer complete.

During the first half of 2026:

- Capital expenditures reached **\$80.6 billion**;
- Acquisition and intangible-asset payments reached **\$33.7 billion**;

- Net purchases of marketable and non-marketable securities absorbed approximately **\$30.2 billion**;
- Operating cash flow was **\$84.9 billion**;
- Free cash flow was only **\$4.3 billion**;
- Alphabet raised approximately **\$105.8 billion** through common equity, mandatory convertible preferred stock and debt issuance;
- Share repurchases were suspended.

Alphabet is not financing because it has exhausted its cash or faces an immediate solvency problem. The company held **\$242.5 billion** of cash, cash equivalents and short-term marketable securities at June 30, 2026, and retained access to unused commercial paper, credit facilities and an at-the-market equity program.

Instead, Alphabet is financing because:

1. The scale of AI infrastructure investment has grown faster than current free cash flow;
2. Long-duration data-center assets are better matched with long-duration financing;
3. Management wants to preserve liquidity for acquisitions, contractual commitments, legal obligations and strategic flexibility;
4. A meaningful portion of reported liquidity consists of equity securities whose values may be volatile;
5. External financing allows Alphabet to avoid relying on the forced liquidation of investments;
6. The company is attempting to spread the cost of the AI build-out across operating cash flow, debt investors and equity investors.

The central conclusion is:

Alphabet does not currently face a shortage of liquidity, but its AI investment program is sufficiently large that the company is actively optimizing its capital structure rather than relying exclusively on internally generated cash.

7.2 Balance Sheet Snapshot

Balance Sheet Metric	Dec. 31, 2025	June 30, 2026	Change
Cash and Cash Equivalents	\$30.71B	\$55.91B	+82.1%
Marketable Securities	\$96.14B	\$186.56B	+94.1%
Total Cash, Cash Equivalents and Marketable Securities	\$126.84B	\$242.47B	+91.1%
Non-Marketable Securities	\$68.69B	\$131.46B	+91.4%
Accounts Receivable	\$62.89B	\$69.18B	+10.0%
Inventory	\$2.44B	\$9.99B	+309.6%
Property and Equipment, Net	\$246.60B	\$321.21B	+30.3%
Goodwill	\$33.38B	\$57.83B	+73.2%
Intangible Assets	\$1.28B	\$9.11B	+609.7%
Total Assets	\$595.28B	\$921.98B	+54.9%
Current Liabilities	\$102.75B	\$126.11B	+22.7%

Balance Sheet Metric	Dec. 31, 2025	June 30, 2026	Change
Long-Term Debt	\$46.55B	\$98.17B	+110.9%
Total Liabilities	\$180.02B	\$281.50B	+56.4%
Stockholders' Equity	\$415.27B	\$640.48B	+54.2%

Alphabet's balance sheet expansion was driven by several factors:

- New debt and equity financing;
- Unrealized investment gains;
- Continued technical-infrastructure investment;
- The Wiz acquisition;
- Higher TPU inventory;
- Growth in deferred tax liabilities associated with investment gains.

The balance sheet remains strong, but it has become substantially more capital-intensive, more investment-sensitive and more dependent on active financing decisions.

7.3 Cash and Marketable Securities

Alphabet reported **\$242.5 billion** of cash, cash equivalents and short-term marketable securities at June 30, 2026.

Liquidity Composition

Liquidity Component	June 30, 2026
Cash	\$12.82B
Money Market Funds	\$14.52B
Time Deposits	\$3.79B
Government Bonds Held as Cash Equivalents	\$24.77B
Other Cash Equivalents	\$0.01B
Total Cash and Cash Equivalents	\$55.91B
Marketable Equity Securities	\$87.06B
Government Bonds	\$51.82B
Corporate Debt Securities	\$26.16B
Mortgage- and Asset-Backed Securities	\$21.52B
Total Marketable Securities	\$186.56B

Liquidity Component	June 30, 2026
Total Reported Liquidity	\$242.47B

Approximately **\$87.1 billion**, or 35.9% of Alphabet's reported liquidity, consisted of marketable equity securities.

This distinction is important.

Cash, money-market instruments and high-quality debt securities are generally more appropriate for funding predictable operating and capital requirements. Marketable equity securities may be liquid, but their value can fluctuate materially with equity-market and company-specific valuations.

Alphabet also held **\$131.5 billion of non-marketable securities**, primarily equity investments. These assets may have substantial strategic or valuation value but should not be treated as equivalent to cash because they may be concentrated, difficult to sell and dependent on private-market valuation events.

Headline Net Cash

Alphabet's current and long-term debt consisted of:

- Approximately **\$2.0 billion** of the current portion of long-term debt;
- Approximately **\$98.2 billion** of long-term debt.

This produces an analytical headline net-cash position of:

\$242.5B reported liquidity – \$100.2B total debt = approximately \$142.3B net cash

Alphabet therefore remains in a substantial headline net-cash position.

Conservative Net Liquidity

A more conservative calculation excludes the \$87.1 billion of marketable equity securities:

\$155.4B cash and debt securities – \$100.2B total debt = approximately \$55.2B conservative net liquidity

This measure still indicates a strong financial position, but it produces a significantly different conclusion from the headline \$142.3 billion net-cash figure.

The two definitions should be interpreted as follows:

Measure	Approximate Amount	Interpretation
Headline Net Cash	\$142.3B	Includes marketable equity securities
Conservative Net Liquidity	\$55.2B	Excludes marketable equity securities
Non-Marketable Securities	\$131.5B	Separate strategic and valuation asset

Alphabet remains highly liquid, but the quality and volatility of the assets included in "net cash" matter increasingly as financing requirements expand.

Short-Term Liquidity Ratios

Liquidity Metric	Q2 2026
Current Ratio	2.72x
Cash and Marketable Securities / Current Liabilities	1.92x
Gross Debt / Reported Liquidity	0.41x

These ratios do not indicate immediate liquidity stress.

Alphabet also stated that existing liquidity, cash flow from operations and financing activities should be sufficient to meet its operating, investing and financing commitments for the foreseeable future.

7.4 Debt and Debt Maturity Profile

Alphabet's long-term debt carrying value increased from **\$46.5 billion to \$98.2 billion** during the first half of 2026.

The company issued:

- \$20.0 billion of U.S. dollar-denominated fixed-rate senior notes;
- \$31.8 billion equivalent of Sterling-, Swiss franc-, euro-, Canadian dollar- and Japanese yen-denominated fixed-rate notes.

2026 Debt Issuance

Currency	Principal Issued	Weighted-Average Coupon	Weighted-Average Maturity
U.S. Dollar	\$20.0B	4.80%	15 years
Sterling	£5.5B	5.31%	31 years
Swiss Franc	CHF3.1B	1.06%	10 years
Euro	€9.0B	3.90%	13 years
Canadian Dollar	C\$8.5B	4.35%	15 years
Japanese Yen	¥576.5B	2.65%	8 years

The new debt was issued primarily at fixed rates and with long maturities. The 2026 securities mature between 2029 and 2126, depending on the issuance.

This financing structure has several benefits:

- It reduces near-term refinancing risk;
- It locks in financing costs over long periods;
- It better matches long-lived infrastructure assets;
- It diversifies funding across currencies and investor markets;

- It prevents the full cost of the AI build-out from being concentrated in short-term funding.

The multi-currency issuance may also help match portions of Alphabet's global asset and cash-flow base. This is an analytical inference rather than an explicitly stated company objective.

Existing Debt Profile

Debt Vintage	Carrying or Face Amount	Maturity Range
2016 U.S. Dollar Notes	\$2.0B	2026
2020 U.S. Dollar Notes	\$9.0B	2027–2060
2025 U.S. Dollar Notes	\$22.5B	2028–2075
2025 Euro Notes	\$15.1B	2028–2064
2026 U.S. Dollar Notes	\$20.0B	2029–2066
2026 Sterling Notes	\$7.3B equivalent	2029–2126
2026 Swiss Franc Notes	\$3.8B equivalent	2029–2051
2026 Euro Notes	\$10.2B equivalent	2030–2063
2026 Canadian Dollar Notes	\$6.0B equivalent	2031–2056
2026 Japanese Yen Notes	\$3.6B equivalent	2029–2066
Other Long-Term Debt	\$1.7B	Various

Only approximately \$2.0 billion was classified as the current portion of long-term debt. The majority of Alphabet's debt therefore matures after 2028.

Alphabet's financing risk is not primarily near-term refinancing.

The larger risks are:

- A structurally higher interest burden;
- The possibility that AI investments earn less than the cost of capital;
- Greater exposure to foreign-currency debt;
- Lower future financial flexibility if commitments continue increasing;
- The possibility that debt must remain high while free cash flow is weak.

7.5 Interest Income and Interest Expense

Alphabet's interest profile changed materially as debt increased.

Interest Metric	Q2 2025	Q2 2026	H1 2025	H1 2026
Interest Income	\$1.05B	\$1.43B	\$2.05B	\$2.81B
Reported Interest Expense	\$0.26B	\$1.28B	\$0.30B	\$1.81B
Net Interest Income	\$0.79B	\$0.15B	\$1.76B	\$1.00B

Alphabet remained in a modest net-interest-income position during Q2, but the advantage narrowed substantially because interest expense increased nearly fivefold year over year.

Capitalized Interest

Reported interest expense is net of interest that Alphabet capitalizes into infrastructure under construction.

Alphabet capitalized:

- \$363 million of interest in Q2 2026;
- \$628 million during H1 2026.

Therefore, Q2 total interest incurred before capitalization was approximately:

\$1.28B reported interest expense + \$0.36B capitalized interest = approximately \$1.64B

Capitalized interest does not disappear economically.

It is added to the cost of the related infrastructure asset and will generally affect future earnings through depreciation after the asset is placed into service.

The implication is:

Current reported interest expense understates the full financing cost associated with Alphabet's infrastructure construction program.

Interest Coverage

Alphabet's interest coverage remained extremely strong:

- Q2 operating income / reported interest expense: approximately **31.9x**;
- H1 operating income / reported interest expense: approximately **44.4x**.

The company can currently service its debt comfortably.

The relevant concern is therefore not debt-service capacity. It is whether the incremental operating return on AI infrastructure exceeds the combined cost of:

- Interest expense;
- Capitalized interest;
- Preferred dividends;
- Common-stock dilution;
- Future replacement CapEx.

7.6 Common Equity Financing

Alphabet raised approximately **\$30.5 billion** through common-stock issuance during Q2 2026.

The financing consisted of:

- An underwritten public offering generating approximately \$20.5 billion of net proceeds;
- A private placement to an affiliate of Berkshire Hathaway generating approximately \$10.0 billion of net proceeds.

Alphabet stated that the proceeds would be used for general corporate purposes, including capital expenditures to scale AI infrastructure and global compute.

Advantages of Common Equity Financing

- No mandatory interest payment;
- No contractual maturity;
- Greater balance-sheet flexibility;
- Supports long-duration and uncertain-return investments;
- Reduces reliance on debt markets;
- Preserves cash for acquisitions and commitments.

Disadvantages

- Immediate shareholder dilution;
- Lower future EPS per share than would otherwise occur;
- Potential pressure on the stock price;
- Existing shareholders bear part of the infrastructure investment risk;
- Equity is generally a more expensive source of capital than senior debt.

The decision to issue common equity while holding significant cash suggests that management views the AI investment cycle as unusually large, long-duration and strategically important.

It also indicates that management is unwilling to fund the entire program by drawing down cash and investment balances.

7.7 Mandatory Convertible Preferred Stock

Alphabet issued approximately **\$19.0 billion** of 6.25% mandatory convertible preferred stock in June 2026.

The securities:

- Carry a 6.25% cumulative annual dividend;
- Are scheduled to convert into Class A or Class C common shares around May 15, 2029;
- Cannot generally be redeemed by Alphabet before mandatory conversion;
- May pay dividends in cash, common shares or a combination;
- Are subject to capped-call transactions intended to reduce potential dilution.

Preferred Dividend Cost

The approximate annual dividend obligation is:

\$19.0B liquidation preference × 6.25% = approximately \$1.19B per year

This is a meaningful recurring claim on earnings available to common shareholders.

Only \$86 million of preferred dividends affected Q2 because the securities were issued late in the quarter. Future quarters will reflect a substantially larger full-period burden.

Economic Characteristics

Mandatory convertible preferred stock combines elements of debt and equity.

Debt-Like Feature	Equity-Like Feature
Fixed cumulative dividend	Mandatory common-stock conversion
Priority over common dividends	No conventional principal repayment at maturity
Recurring financing cost	Future dilution
Liquidation preference	Share-price-linked conversion rate

Alphabet paid approximately \$1.0 billion for capped-call transactions designed to reduce potential common-share dilution at conversion.

The preferred issuance allows Alphabet to:

- Delay part of the common-stock dilution;
- Obtain long-duration financing;
- Avoid increasing conventional debt by the full amount;
- Preserve current cash;
- Spread the cost of the investment cycle across future periods.

However, it also creates both a fixed dividend burden and eventual dilution risk.

7.8 Commercial Paper, Credit Facilities and the ATM Program

Alphabet retains substantial additional short-term and contingent funding capacity.

Commercial Paper

Alphabet has a commercial-paper program of up to **\$25.0 billion**.

No commercial paper was outstanding at June 30, 2026.

The absence of commercial-paper borrowing is an important positive signal:

Alphabet is not currently relying on short-term unsecured funding to finance long-duration infrastructure.

This reduces refinancing and liquidity mismatch risk.

Credit Facilities

Alphabet had:

- \$11.7 billion of total credit facilities;
- \$1.3 billion outstanding;
- Facilities expiring through April 2030;
- Interest rates of SOFR plus 1.5% to 2.25% on outstanding amounts.

These facilities provide additional liquidity but are more expensive than much of Alphabet's fixed-rate debt.

At-the-Market Equity Program

Alphabet established an at-the-market program allowing the issuance of up to **\$40.0 billion** of Class A and Class C stock.

No shares had been sold through the program at June 30, 2026, leaving the entire amount available. Alphabet stated that proceeds would primarily be used to meet tax obligations associated with employee equity grants.

The ATM program creates a flexible source of equity funding but also creates potential dilution and stock-price overhang.

7.9 Data Center Leases

Alphabet's recorded lease liabilities do not capture the full scale of future data-center lease commitments.

Existing Recorded Lease Obligations

Lease Metric	June 30, 2026
Operating Lease Liabilities	\$18.04B
Finance Lease Liabilities	\$2.59B
Total Undiscounted Operating Lease Payments	\$21.33B
Total Undiscounted Finance Lease Payments	\$2.92B

In addition, Alphabet had entered into data-center leases that had not yet commenced, with future lease payments of **\$85.2 billion**. These leases are scheduled to commence between 2026 and 2031 and have non-cancelable terms ranging from one to 26 years.

Alphabet also entered into a separate short-term lease commitment of approximately **\$5.8 billion**, scheduled to commence in Q3 2026.

These future leases are not yet fully represented as liabilities on the balance sheet because the leases have not commenced.

The economic implication is:

Alphabet's future fixed infrastructure burden is materially larger than the currently recorded debt and lease-liability balances suggest.

7.10 Data Center and Energy Backstops

Alphabet has entered into arrangements designed to support third-party data-center construction and energy infrastructure.

As of June 30, 2026:

- Financial guarantees had a maximum potential payment exposure of **\$7.6 billion**;
- Credit-derivative backstops had a maximum potential payment exposure of **\$43.8 billion**;
- Alphabet had agreed to provide an estimated **\$24.1 billion** of additional future backstops, subject to final terms.

These exposures are contingent rather than conventional funded debt.

If counterparties default, Alphabet may have the right or obligation to:

- Assume the underlying data-center lease;
- Use the facility internally;
- Sublease the facility;

- Make a contractual termination payment;
- Receive certain equipment or infrastructure assets.

Strategic Benefit

The arrangements help third-party developers finance and procure:

- Data-center facilities;
- Long-lead-time equipment;
- Energy infrastructure;
- Power-generation capacity.

This allows Alphabet to secure capacity more quickly without funding every project directly from the beginning.

Financial Risk

The arrangements can still create:

- Future cash payments;
- Additional lease obligations;
- Excess-capacity exposure;
- Counterparty credit risk;
- Fixed-cost commitments;
- Lower flexibility if AI demand slows.

These exposures should not be classified as current funded debt, but they should be included in a broader assessment of Alphabet's infrastructure-adjusted financial obligations.

7.11 Purchase and Energy Commitments

Alphabet reported **\$811.0 billion** of material purchase commitments and other contractual obligations at June 30, 2026, including **\$200.7 billion** classified as short-term.

These obligations primarily relate to:

- Technical-infrastructure supply agreements;
- Inventory and component purchases;
- Open purchase orders;
- Energy contracts;
- Content-licensing commitments.

Alphabet separately disclosed **\$707.0 billion** of expected fixed or guaranteed commitments under long-term supply, energy and content agreements.

The majority relates to long-term supply agreements expected to be fulfilled through 2030. Energy-service agreements may extend through 2054 and generally contain minimum-purchase, take-or-pay or termination provisions.

The \$811 billion and \$707 billion amounts should not be added together because the categories overlap.

Similarly, data-center leases, guarantees, credit derivatives and future backstops may overlap with other contractual-exposure categories.

The correct interpretation is not that Alphabet owes the combined total immediately.

The correct interpretation is:

Alphabet has committed to a much larger long-term infrastructure and supply ecosystem than is visible through current CapEx and funded debt alone.

Take-or-Pay Risk

Take-or-pay agreements require Alphabet to pay for minimum quantities of energy or capacity even if actual usage is lower.

This creates a risk that:

- AI demand is lower than expected;
- Infrastructure utilization is below plan;
- Alphabet pays for unused energy or capacity;
- Contracted costs remain fixed while Cloud pricing declines.

The scale of these commitments increases the importance of backlog conversion and infrastructure utilization.

7.12 Wiz Acquisition and Acquisition Funding

Alphabet completed the Wiz acquisition in March 2026 for approximately **\$29.5 billion**, excluding post-combination compensation arrangements.

Wiz is included within Google Cloud and is intended to strengthen Alphabet's multicloud and AI-driven security capabilities.

Total H1 cash payments for acquisitions and intangible assets were approximately **\$33.7 billion**, indicating that Wiz represented the majority of acquisition-related cash deployment.

The acquisition contributed to:

- Google Cloud goodwill increasing to \$31.5 billion;
- Total Alphabet goodwill increasing to \$57.8 billion;
- Intangible assets increasing to \$9.1 billion;
- Higher future amortization expenses;
- Additional Cloud integration and operating costs.

Capital Allocation Implication

The Wiz acquisition occurred during the same period in which Alphabet was:

- Doubling CapEx;
- Building TPU inventory;
- Increasing strategic investments;
- Maintaining common dividends;
- Assuming significant infrastructure commitments.

Even a company with \$242.5 billion of reported liquidity cannot evaluate a \$29.5 billion acquisition in isolation.

Management must preserve cash for the combined funding needs of:

- AI CapEx;
- Acquisitions;
- Working capital;
- Contractual obligations;
- Legal liabilities;
- Dividends;
- Debt maturities;
- Strategic investments.

External financing allows Alphabet to complete major acquisitions without reducing liquidity by the full purchase price.

7.13 Strategic Investments

Alphabet used approximately **\$22.1 billion** to purchase non-marketable securities during H1 2026 and received approximately \$1.7 billion from maturities and sales.

Net purchases of non-marketable securities therefore used approximately **\$20.4 billion** of cash.

Alphabet also had approximately **\$20.0 billion** of future capital-funding commitments to a private company, contingent on operational and financial milestones through 2030.

These investments may provide:

- Strategic partnerships;
- Access to technology;
- Financial returns;
- Ecosystem advantages;
- Potential future liquidity events.

However, they also compete with AI infrastructure for capital and are generally less liquid than government or corporate debt securities.

7.14 Dividends and Share Repurchases

Alphabet paid approximately **\$5.23 billion** of common dividends during H1 2026.

The quarterly common dividend increased by 5%, from \$0.21 to **\$0.22 per share**.

Based on the H1 common-dividend run rate and the preferred stock's annual dividend, Alphabet's approximate annual dividend burden would be:

Approximately \$10.46B common dividends + \$1.19B preferred dividends = approximately \$11.65B annually

This is an analytical run-rate estimate and assumes no material change in share count or dividend policy.

Share Repurchases

Alphabet conducted **no share repurchases** during Q2 or H1 2026.

This compares with approximately **\$28.3 billion** of repurchases during H1 2025. Alphabet still had **\$69.5 billion** remaining under its existing repurchase authorization.

The absence of repurchases is highly relevant.

It indicates that Alphabet's capital-allocation priorities have shifted from:

Returning excess cash through buybacks

toward:

Funding infrastructure, acquisitions and strategic investments

Alphabet is issuing common and preferred equity at the same time that it has suspended buybacks.

From an economic perspective, the company is no longer using repurchases to offset:

- Stock-based compensation;
- Employee stock issuance;

- New common-share financing;
- Future preferred-stock conversion.

This creates greater net dilution risk for existing common shareholders.

7.15 H1 2026 Capital Allocation Bridge

The following table connects Alphabet's operating cash flow and external financing with its principal uses of capital.

Major Sources of Capital

Source	H1 2026
Operating Cash Flow	\$84.86B
Common-Stock Issuance	\$30.50B
Mandatory Convertible Preferred Stock	\$19.06B
Debt Issuance	\$56.23B
Sale of Interests in Consolidated Entities	\$3.76B
Total Major Sources	\$194.41B

Major Uses of Capital

Use	H1 2026
Capital Expenditures	\$80.60B
Acquisitions and Intangible Assets	\$33.70B
Net Purchases of Marketable Securities	\$9.78B
Net Purchases of Non-Marketable Securities	\$20.38B
Stock-Based Award-Related Payments	\$12.06B
Common Dividend Payments	\$5.23B
Debt Repayments	\$5.25B
Other Investing and Financing Uses	\$2.05B
Total Major Uses	\$169.05B

Cash Reconciliation

Item	H1 2026
Total Major Sources	\$194.41B
Less: Total Major Uses	(\$169.05B)
Increase in Cash Before FX	\$25.36B
Foreign-Exchange Impact	(\$0.15B)
Reported Increase in Cash	\$25.20B

Source and use figures are derived from Alphabet's consolidated statement of cash flows.

The capital flow can be summarized as:

Operating cash flow + debt financing + common equity + preferred equity

flowed primarily into:

CapEx + acquisitions + strategic investments + employee-equity obligations + dividends + debt repayment

7.16 Why External Financing Was Necessary

Operating cash flow alone did not cover the full investment program.

Operating Cash Flow Versus CapEx and Acquisitions

\$84.86B operating cash flow – \$80.60B CapEx – \$33.70B acquisitions = negative \$29.44B

After also accounting for dividends and net purchases of marketable and non-marketable securities, the funding gap would have been approximately **\$64.8 billion**, before other financing and operating uses.

Therefore:

Alphabet could have funded CapEx from operating cash flow, but it could not simultaneously fund CapEx, Wiz, strategic investments, dividends and other obligations without either reducing cash balances or raising external capital.

Management chose to raise external capital and increase cash rather than draw down the balance sheet.

7.17 Why Finance While Holding \$242.5 Billion of Liquidity?

1. The Liquidity Is Not Entirely Cash-Like

Approximately \$87.1 billion of reported liquidity consisted of marketable equity securities, while an additional \$131.5 billion was held in non-marketable securities.

Selling these investments could expose Alphabet to:

- Valuation volatility;
- Concentration risk;
- Tax consequences;
- Strategic constraints;
- Unfavorable market timing.

External financing reduces the risk that Alphabet must sell investments to fund infrastructure.

2. CapEx Is Only One Capital Requirement

Alphabet must also fund:

- Wiz and other acquisitions;
- TPU inventory;
- Strategic private investments;
- Energy and supply commitments;
- Data-center leases;
- Legal settlements;
- Taxes;
- Dividends;
- Employee stock-tax obligations.

The relevant comparison is therefore not simply cash versus CapEx.

It is:

Liquid financial resources versus total future capital and contractual requirements

3. Infrastructure Has Long Economic Lives

Data centers, power infrastructure and networks can remain productive for many years.

Long-term fixed-rate debt can be an appropriate funding instrument because it aligns financing duration with asset duration.

4. Management Is Preserving Strategic Optionality

Maintaining liquidity allows Alphabet to:

- Respond to competitive AI investment;
- Secure scarce components and energy;
- Complete acquisitions;
- Support counterparties;
- Manage legal and regulatory liabilities;
- Continue dividends;
- Withstand a market or economic downturn.

5. AI Returns Remain Uncertain

Equity and preferred-equity financing transfer part of the investment risk to capital providers.

This reduces the risk that Alphabet becomes excessively leveraged before the full profitability of the AI infrastructure cycle is demonstrated.

6. The Capital Build-Out Is Front-Loaded

Cash is spent before:

- Data centers become operational;
- TPU systems are delivered;
- Customer contracts convert into revenue;
- Backlog generates cash;

- Depreciation and operational returns fully mature.

External financing bridges this timing gap.

7.18 Capital Structure Trade-Off

Alphabet's current capital structure can be understood through three sources of financing.

Financing Source	Advantage	Cost or Risk
Operating Cash Flow	No external financing cost	Currently almost fully consumed by CapEx
Debt	Lower cost and long duration	Interest expense and fixed obligations
Common Equity	Maximum flexibility	Immediate dilution
Mandatory Convertible Preferred	Delayed dilution and hybrid flexibility	6.25% dividend and eventual conversion
Leases and Third-Party Capacity	Faster capacity delivery	Higher operating cost and fixed commitments
Guarantees and Backstops	Enables partner-built infrastructure	Contingent liabilities and counterparty risk

Management is not selecting one funding source.

It is using a portfolio of financing sources to balance:

- Cost;
- Duration;
- Flexibility;
- Dilution;
- Liquidity;
- Risk.

7.19 Balance Sheet Risks

1. Investment-Value Volatility

A substantial portion of financial assets consists of marketable and non-marketable equity securities.

A downward revaluation could reduce:

- Reported net income;
- Stockholders' equity;
- Headline net cash;

- Financial flexibility.

2. Rising Fixed Costs

Debt, preferred dividends, leases, energy agreements and take-or-pay commitments increase the amount of cash that must be paid regardless of near-term revenue performance.

3. Future Dilution

Dilution may arise from:

- Common-stock issuance;
- Stock-based compensation;
- Mandatory preferred conversion;
- Preferred dividends paid in common shares;
- Potential ATM issuance.

4. Contractual Capacity Risk

Alphabet may secure more data-center, component or energy capacity than it ultimately requires.

5. Funding-Cost Risk

Refinancing future debt may become more expensive if:

- Interest rates remain high;
- Credit spreads widen;
- AI returns disappoint;
- Free cash flow remains weak.

6. Capitalized Interest and Deferred Cost Recognition

Some current financing costs are capitalized and will affect future depreciation rather than current interest expense.

7. Acquisition Risk

Wiz and other acquisitions require revenue growth, customer retention and synergies to justify the cash paid and goodwill recognized.

7.20 Balance Sheet and Financing Scorecard

Dimension	Rating	Rationale
Headline Liquidity	Very Strong	\$242.5B of cash and marketable securities
Conservative Net Liquidity	Strong	Approximately \$55.2B after excluding marketable equity and subtracting debt
Debt-Service Capacity	Very Strong	Interest coverage remains above 30x
Debt Maturity Profile	Strong	Most debt is long-dated and fixed-rate

Dimension	Rating	Rationale
Short-Term Funding Risk	Low	No commercial paper outstanding
Financial Flexibility	Strong	Credit facilities and \$40B ATM remain available
Free-Cash-Flow Funding Capacity	Weakening	CapEx absorbs nearly all operating cash flow
Equity Dilution	Increasing	Common and preferred stock were issued
Preferred Dividend Burden	Moderate	Approximately \$1.19B annual obligation
Infrastructure Commitments	Very High	Large leases, purchase commitments and backstops
Acquisition Deployment	High	\$33.7B of H1 acquisition cash spending
Shareholder Capital Return	Weaker	Buybacks suspended despite remaining authorization
Solvency Risk	Low	Large liquidity and strong operating income
Capital-Allocation Risk	Increasing	Multiple large uses of capital must generate adequate returns

7.21 Overall Balance Sheet and Financing Conclusion

Alphabet remains financially strong, but it should no longer be analyzed as a simple net-cash company with minimal financing considerations.

The company has:

- \$242.5 billion of reported liquidity;
- Approximately \$142.3 billion of headline net cash;
- Approximately \$55.2 billion of conservative net liquidity excluding marketable equity securities;
- \$98.2 billion of long-term debt;
- \$19.0 billion of mandatory convertible preferred stock;
- Significant future lease, supply, energy and infrastructure obligations;
- Increasing common-share dilution.

The financing activity is not evidence that Alphabet has “run out of money.”

It is evidence that:

The AI infrastructure program, acquisitions, investments and contractual commitments have become too large to fund entirely through current free cash flow without materially reducing liquidity.

Alphabet is attempting to preserve a strong balance sheet while accelerating investment.

The current capital-allocation structure is:

Operating cash flow + debt + common equity + preferred equity

funding:

AI CapEx + acquisitions + strategic investments + dividends + working capital + contractual obligations

The strategy is financially manageable as long as:

1. Google Cloud backlog converts into revenue;
2. Search and Cloud operating profit continue growing;
3. Infrastructure utilization remains high;
4. CapEx growth eventually moderates;
5. Free cash flow recovers;
6. AI investment returns exceed the blended cost of capital.

The principal risk is not immediate insolvency.

The principal risk is that Alphabet commits to a high fixed-cost, high-capital infrastructure base before the long-term economic returns of AI have been fully demonstrated.

The correct institutional interpretation is:

Alphabet retains exceptional financial capacity, but it is transitioning from a passive net-cash model to an actively financed, capital-intensive AI infrastructure model.

This makes balance-sheet quality dependent not only on the amount of cash held, but also on:

- The composition of liquidity;
- The cost and maturity of financing;
- Future dilution;
- Contractual capacity commitments;
- Capital-allocation discipline;
- The eventual cash return from AI infrastructure.

8. Guidance and Earnings Call

8.1 Executive Assessment

Alphabet's Q2 2026 earnings call contained a more constructive assessment of long-term AI demand, but a more cautious near-term outlook for margins and free cash flow.

Management's message can be summarized as follows:

AI demand and long-term return expectations became stronger

while

capacity constraints, capital requirements and near-term cash-flow pressure also became more severe

Management raised full-year 2026 capital-expenditure guidance to **\$195–\$205 billion**, reiterated that 2027 CapEx will increase significantly, and stated that free cash flow will remain under pressure. Alphabet also plans to expand its use of higher-cost third-party capacity in Q3, which will create modest near-term pressure on Google Cloud operating margin.

At the same time, management became more optimistic about the size of the AI opportunity. Sundar Pichai stated that Alphabet had become more bullish on AI return opportunities over the preceding year, while the CFO said that capacity demand continued to exceed the substantial infrastructure additions made over the previous three years.

The central contradiction in the outlook is therefore:

Alphabet is seeing enough demand to justify accelerating investment, but the company has not provided a clear timeline for when this investment will produce improving consolidated free cash flow.

8.2 Explicit Guidance Scorecard

Alphabet does not provide conventional quarterly revenue, EPS or operating-margin ranges. Instead, management provides directional guidance on currency, business trends, capital expenditures and costs.

Guidance Item	Management Guidance	Analytical Interpretation
Q3 Foreign Exchange	Slight consolidated revenue headwind	Reversal from a roughly one-point Q2 tailwind
Search Revenue	Q3 begins lapping stronger Q3 2025 growth	Reported Search growth may decelerate despite healthy underlying demand
Google Cloud Revenue	Significant demand expected to drive strong growth	Growth remains high, but no numerical range was provided
TPU System Revenue	Relatively small contribution in 2026, ramping toward year-end	Most contracted revenue remains deferred
TPU Revenue Timing	Vast majority expected in 2027	2027 Cloud growth will include a larger hardware component
Third-Party Capacity	Usage will expand in Q3	Supports revenue but pressures Cloud margin
FY2026 CapEx	\$195–\$205B	Midpoint increased by \$15B from prior guidance
FY2027 CapEx	Expected to increase significantly	No dollar amount or percentage provided
Depreciation	Expected to continue increasing	Current CapEx will create delayed P&L pressure
Data-Center Operating Costs	Expected to rise, including energy	Additional cost-of-revenue and operating-margin pressure
Hiring	Continued investment in AI and Cloud talent	R&D and compensation growth likely remains elevated
Marketing	Continued support for Gemini and AI products	Sales and marketing expense remains an investment area

Guidance Item	Management Guidance	Analytical Interpretation
Free Cash Flow	Expected to remain under pressure	No recovery date or minimum level provided

Management's prepared remarks provided clear directional guidance but no explicit Q3 revenue, EPS, consolidated operating-margin, Cloud-margin or free-cash-flow targets.

8.3 Foreign-Exchange Guidance

Alphabet expects a slight foreign-exchange headwind to consolidated revenue in Q3, compared with an approximately 100 bps tailwind in Q2. Management expects the effect to be seen primarily in Search and YouTube advertising.

The quarterly change in currency contribution is therefore slightly greater than 100 bps:

Q2: approximately +100 bps

to

Q3: slight negative contribution

This means that reported Q3 growth may decelerate even if constant-currency business momentum remains stable.

For modeling purposes, analysts should distinguish:

- Reported revenue growth;
- Constant-currency growth;
- Organic operating momentum.

A lower reported growth rate caused partly by foreign exchange would be less concerning than a similar decline caused by weaker queries, advertising demand or Cloud consumption.

8.4 Search Revenue Guidance

Management did not indicate that Search demand was weakening. Instead, it warned that Q3 will begin lapping the acceleration in Search performance that started in Q3 2025.

This creates two separate effects:

1. A more difficult year-over-year comparison;
2. A less favorable foreign-exchange environment.

The implied Search outlook is therefore:

Healthy underlying operating performance, but a likely moderation in reported year-over-year growth.

This is not equivalent to a demand warning.

Management continued to describe AI as additive to Search usage. In Q2, AI Mode exceeded one billion monthly active users, AI-powered features continued to generate incremental queries, and Alphabet reported encouraging monetization on searches containing AI Overviews. The company also said that the cost of AI Mode responses had declined to its lowest level since launch.

The main Search debate for Q3 is therefore not whether AI is currently reducing total usage. The key questions are:

- How much Search growth slows because of the comparison base;
- Whether commercial-query growth remains strong;
- Whether AI query monetization continues improving;
- Whether lower inference costs offset greater AI usage;
- Whether Search margin remains resilient as AI adoption expands.

Search Implied Guidance

Search Factor	Q2 Position	Q3 Implication
Query Growth	Strong and AI-supported	Expected to remain positive
Commercial Queries	Expanding	Continued monetization opportunity
AI Advertising	Encouraging but early	Further testing and deployment
Comparison Base	Relatively favorable	Becomes more difficult
Foreign Exchange	Approximately 1-point benefit	Slight headwind
Reported Growth Direction	17% in Q2	Likely to moderate
Fundamental Demand Signal	Strong	No explicit deterioration indicated

The correct institutional interpretation is:

Management is preparing investors for slower reported Search growth without signaling a fundamental weakening in Search demand.

8.5 YouTube Revenue Guidance

Alphabet did not provide specific YouTube revenue guidance.

Q2 YouTube advertising benefited from:

- Direct Response demand;
- Brand advertising;
- Connected-TV growth;
- Significant World Cup-related advertising and engagement.

Management noted that foreign exchange will become a slight headwind to YouTube advertising in Q3.

YouTube's Q3 comparison should also be evaluated carefully because some Q2 strength was associated with the FIFA World Cup. The event attracted more than 1.7 billion unique YouTube viewers, including more than 550 million television viewers, and supported World Cup-related advertising formats.

The event contribution does not mean that underlying YouTube growth was temporary. Management continued to identify structural growth opportunities in:

- Connected television;
- Direct Response;
- Demand Gen;
- Shorts;
- Shopping;
- Creator partnerships;
- YouTube Music and Premium.

However, the absence of numerical guidance makes it difficult to separate structural momentum from the Q2 event benefit.

8.6 Google Cloud Revenue Outlook

Management stated that significant demand is expected to continue driving strong Google Cloud growth. Q2 Cloud revenue increased 82%, while backlog rose by more than \$50 billion sequentially to approximately \$514 billion. Slightly more than half of the backlog is expected to be recognized over the following 24 months.

The Cloud demand outlook is supported by:

- Rapid customer acquisition;
- Greater usage among existing customers;
- Enterprise AI adoption;
- Paid token growth;
- AI infrastructure demand;
- Data analytics and security;
- TPU system contracts.

However, management did not provide a numerical Q3 Cloud growth range.

The implicit guidance is:

Google Cloud should remain Alphabet's fastest-growing major business, but the 82% Q2 growth rate should not automatically be treated as the new permanent run rate.

Several factors could affect the reported growth path:

- Scale-related normalization;
- TPU system delivery timing;
- Backlog conversion;
- Available internal and third-party capacity;
- Wiz's revenue contribution;
- Customer usage beyond contractual commitments;
- Hardware versus recurring-service revenue mix.

Cloud Revenue Signal

Indicator	Q1 2026	Q2 2026	Interpretation
Cloud Revenue Growth	63%	82%	Material acceleration
Cloud Revenue	\$20.0B	\$24.8B	Strong sequential growth
Cloud Backlog	\$462B	\$514B	Continued contracted-demand growth
Backlog Expected Within 24 Months	Just over 50%	Just over 50%	Significant forward visibility

Indicator	Q1 2026	Q2 2026	Interpretation
Capacity Position	Constrained	Still constrained	Demand continues to exceed supply
Management Outlook	Strong demand	Strong growth expected	Constructive

Q1 and Q2 management commentary indicates that Cloud demand strengthened rather than weakened during the quarter.

8.7 Third-Party Capacity and Cloud Margin

The most important new near-term Cloud guidance was Alphabet's decision to expand the use of third-party capacity in Q3.

Management described the capacity as a bridge while Alphabet constructs more internally owned infrastructure. The additional capacity allows the company to support customers and capture revenue that might otherwise be delayed or lost. However, management explicitly stated that its higher cost will pressure Cloud operating margins.

The economic sequence is:

Insufficient internal capacity

leads to

Use of higher-cost third-party capacity

which allows

Immediate customer service and revenue capture

but causes

Near-term Cloud margin pressure

Management argued that this can still be economically attractive when several months of elevated costs secure a multiyear customer relationship with positive lifetime returns.

Margin Interpretation

Google Cloud's Q2 operating margin reached 35.6%, but this level should not be treated as a guaranteed Q3 run rate.

Q3 margin faces at least two identified headwinds:

1. Third-party capacity costs;
2. Continued Wiz integration costs.

Alphabet had already stated in Q1 that Wiz would create a approximately 100-300 bps headwind to Cloud operating margin for the remainder of 2026. In Q2, management added third-party capacity as another explicit margin pressure.

The company did not quantify:

- The amount of third-party capacity;
- The cost premium relative to internal infrastructure;
- The exact Cloud-margin impact;
- How long the bridge period will last;

- Whether all large contracts will have positive lifetime margins;
- The portion of Cloud revenue dependent on external capacity.

The appropriate conclusion is:

Cloud operating income can continue increasing even if the operating-margin percentage declines from Q2's exceptionally strong level.

8.8 TPU Revenue Recognition

Alphabet began recognizing revenue from TPU systems delivered to customer data centers in Q2.

Management expects:

- A relatively small portion of existing TPU agreement revenue to be recognized during 2026;
- Revenue to ramp as Alphabet exits 2026;
- The vast majority of contracted TPU system revenue to be recognized in 2027.

The Q2 call therefore moved the TPU sales story from planning into initial execution.

TPU Revenue Recognition Timeline

Period	Expected TPU System Revenue Contribution
Q1 2026	No material recognized revenue; deliveries expected later
Q2 2026	Initial revenue recognition began
H2 2026	Continued ramp, but relatively small total contribution
Exit Rate of 2026	Expected to increase
2027	Vast majority of existing agreement revenue expected

TPU system sales are included in Cloud backlog when agreements are signed. Alphabet then builds inventory ahead of delivery, which consumes operating cash. Revenue is generally recognized when systems are delivered and control transfers to the customer.

This creates a lag between:

- Contract signing;
- Inventory production;
- Cash usage;
- Delivery;
- Revenue recognition;
- Customer cash collection.

The key modeling risk is that TPU revenue may be uneven because recognition depends on the timing of large system deliveries.

Alphabet also declined to disclose TPU product margins. Management characterized the business as an expansion of Cloud's total addressable market rather than providing a specific profitability profile.

Consequently, higher TPU revenue in 2027 may increase Cloud revenue without necessarily maintaining Q2's 35.6% blended segment margin.

8.9 Capital Expenditure Guidance

Alphabet raised full-year 2026 CapEx guidance from **\$180–\$190 billion** to **\$195–\$205 billion**.

CapEx Guidance	Low End	Midpoint	High End
Previous Guidance	\$180B	\$185B	\$190B
Updated Guidance	\$195B	\$200B	\$205B
Increase	+\$15B	+\$15B	+\$15B
Percentage Increase at Midpoint	8.3%	Approximately 8.1%	7.9%

Management attributed the increase primarily to faster delivery of capacity required to meet growing demand.

Alphabet spent approximately **\$80.6 billion** during the first half of 2026. The updated guidance therefore implies second-half CapEx of:

FY2026 Scenario	Full-Year CapEx	Implied H2 CapEx	Average Quarterly H2 CapEx
Low	\$195.0B	\$114.4B	\$57.2B
Midpoint	\$200.0B	\$119.4B	\$59.7B
High	\$205.0B	\$124.4B	\$62.2B

The implied midpoint quarterly CapEx of approximately \$59.7 billion is materially above Q2 CapEx of \$44.9 billion.

This means that Q2 was not necessarily the peak of Alphabet's 2026 capital-spending cycle.

8.10 2027 CapEx Guidance

Management reiterated that 2027 CapEx will increase significantly from 2026 but did not provide a numerical range.

This is directionally consistent with Q1, when Alphabet first stated that 2027 CapEx would increase significantly. However, the Q2 statement carries greater weight because it followed:

- An additional \$15 billion increase in 2026 guidance;
- Continued supply constraints;
- Accelerating Cloud revenue;
- Larger backlog;
- Greater TPU demand;
- Training of larger frontier models;
- Explicit use of third-party bridge capacity.

The implied message is:

Even a \$195–\$205 billion 2026 investment program is not expected to eliminate Alphabet's capacity shortage before 2027.

Management said that it takes a multiyear view of capacity requirements and will continue investing as long as it sees attractive returns. It also said demand continued to exceed capacity despite significant additions during the preceding three years.

What Management Did Not Disclose

Alphabet did not provide:

- A 2027 CapEx range;
- A percentage growth assumption;
- The server versus data-center mix;
- The amount required for Gemini 4 training;
- Cloud-specific versus internal-product CapEx;
- Maintenance versus growth CapEx;
- A target year for CapEx-to-revenue normalization.

This remains one of the largest forecast uncertainties in Alphabet's investment case.

8.11 Cost and Depreciation Guidance

Management expects technical-infrastructure investment to continue increasing:

- Depreciation;
- Data-center operating expenses;
- Energy costs;
- AI and Cloud compensation;
- Marketing spending for AI products.

These expenses will not all appear at the same time.

Expense Timing

Investment or Cost	Cash-Flow Timing	Income-Statement Timing
Server Purchase	Immediate CapEx	Depreciated after entering service
Data-Center Construction	Immediate and staged CapEx	Depreciation begins after completion
Energy Usage	As capacity operates	Current cost of revenue or operating expense
Third-Party Capacity	Contract-dependent	More immediate margin impact
AI Hiring	Immediate payroll expense	Current R&D or operating expense
Marketing	Immediate cash and P&L expense	Current sales and marketing expense
Capitalized Interest	Cash interest incurred	Recognized later through asset depreciation

The depreciation warning is particularly important because Alphabet had a large amount of infrastructure that had not yet been placed into service at the end of Q2.

Therefore:

Future depreciation may continue rising even if the growth rate of new CapEx eventually begins to moderate.

Management did not provide a numerical depreciation forecast, making it necessary for analysts to estimate depreciation using:

- Property and equipment entering service;
 - Historical useful lives;
 - Server deployment timing;
 - Data-center completion schedules;
 - Capitalized interest;
 - Asset retirements.
-

8.12 Free Cash Flow Guidance

Alphabet explicitly stated that free cash flow will remain under pressure because of technical-infrastructure investment.

This statement is more cautious than management's Q1 commentary.

In Q1:

- Operating cash flow was \$45.8 billion;
- CapEx was \$35.7 billion;
- Free cash flow was positive \$10.1 billion.

In Q2:

- Operating cash flow was \$39.1 billion;
- CapEx was \$44.9 billion;
- Free cash flow was negative \$5.9 billion.

Management did not provide:

- Full-year free-cash-flow guidance;
- A quarterly FCF floor;
- A timeline for FCF recovery;
- A target FCF margin;
- A target operating-cash-flow-to-CapEx ratio;
- A date for the resumption of share repurchases.

The implied free-cash-flow outlook is therefore:

Near-term FCF remains structurally pressured, and a clear recovery is unlikely until operating cash flow grows faster than CapEx.

Because implied H2 CapEx is substantially above the Q2 level, free cash flow could remain volatile or negative unless Cloud, Search and working-capital cash generation accelerate meaningfully.

8.13 When Will AI Improve Free Cash Flow?

Management did not provide a direct answer.

It described AI investments as enabling attractive returns and emphasized that the company is investing only while it sees attractive long-term opportunities. However, it did not define:

- The expected payback period;
- The target AI ROIC;
- The point at which CapEx growth will fall below revenue growth;
- The year in which FCF margin should recover;
- The amount of revenue required per unit of installed capacity.

Sundar stated that management had become more bullish on the AI opportunity over the preceding year and described the consumer and enterprise opportunities as being in their early stages.

This is a positive demand signal, but it is not a quantified cash-return forecast.

The likely return sequence is:

Capacity investment

followed by

Backlog conversion and higher AI usage

followed by

Revenue and operating-income growth

followed eventually by

CapEx normalization and free-cash-flow recovery

Alphabet has demonstrated the middle stages of this sequence in Google Cloud, but not yet the final stage at the consolidated level.

8.14 AI ROIC Commentary

Management's ROIC language became more confident in Q2.

In Q1, Alphabet pointed to:

- Search growth;
- Cloud growth;
- Cloud backlog;
- The scale benefits of its TPU strategy;

as evidence that AI investment was generating returns.

In Q2, Sundar stated that Alphabet had become more bullish on the AI return opportunity over the preceding year. Management also argued that short-term use of expensive third-party capacity can be rational when it secures attractive multiyear customer economics.

However, the company still did not provide:

- AI revenue;
- AI operating income;
- AI invested capital;
- AI after-tax return;
- Customer-level lifetime return;
- Search AI contribution margin;
- Fully loaded Cloud AI margin.

The call therefore provided stronger qualitative confidence without a corresponding increase in quantitative ROIC disclosure.

The correct interpretation is:

Management's conviction in AI returns increased, but investors still cannot independently calculate those returns from disclosed financial data.

8.15 Q1-to-Q2 Earnings-Call Tone Comparison

Topic	Q1 2026 Management Message	Q2 2026 Management Message	Change in Tone
Cloud Demand	Demand described as unprecedented; Cloud accelerated 63%; backlog reached \$462B	Cloud growth accelerated to 82%; backlog reached \$514B; demand remained broad and strong	More bullish and supported by stronger results
Capacity Constraints	Compute was acutely constrained; Cloud revenue could have been higher with more capacity	Demand still exceeded capacity despite major additions; third-party capacity will be expanded	Constraint became more operationally significant
Search AI Monetization	AI Overviews and AI Mode increased queries, including commercial queries; monetization opportunity remained early	AI Mode exceeded 1B users; cost per response fell; commercial monetization remained encouraging; new formats showed early traction	More concrete evidence of monetization and efficiency
CapEx	2026 guidance raised to \$180–\$190B; 2027 expected to increase significantly	2026 guidance raised again to \$195–\$205B; 2027 significant increase reiterated	Materially more aggressive investment stance
Cloud Margin	Cloud margin reached 32.9%; Wiz expected to create a low-single-digit-point headwind	Cloud margin reached 35.6%; third-party capacity and Wiz expected to pressure near-term margin	Current performance stronger, forward margin message more cautious
TPU Revenue	Initial revenue expected later in 2026; vast majority in 2027	Initial revenue recognized; ramp expected through 2026; vast majority still expected in 2027	Execution progressed, but timing remained back-end loaded
Depreciation and Costs	Higher depreciation, energy, AI hiring and marketing expected	Same cost pressures reiterated after another CapEx increase	Greater confidence that cost pressure will persist
Free Cash Flow	Q1 FCF remained positive at \$10.1B; no	Q2 FCF turned negative; management	Clearly more cautious

Topic	Q1 2026 Management Message	Q2 2026 Management Message	Change in Tone
	explicit recovery warning	explicitly said FCF will remain under pressure	
AI ROIC	Search and Cloud growth presented as proof points; investment described as responsible	Management said it had become more bullish and described extraordinary return opportunities	More optimistic qualitatively
Capital Structure	CapEx primarily discussed in relation to demand and backlog	CFO explicitly described funding hierarchy: operating cash flow, then debt, then equity	Greater focus on financing the investment cycle

The Q1 statements are supported by Alphabet's Q1 call, which described unprecedented AI compute demand, a \$462 billion Cloud backlog, a 32.9% Cloud margin and 2026 CapEx guidance of \$180–\$190 billion.

The Q2 call added stronger Cloud results but also more explicit warnings regarding third-party capacity, margin pressure, FCF pressure and a further increase in CapEx.

8.16 Management Tone by Topic

Cloud Demand: More Confident

Management's Cloud message became more confident because the qualitative demand commentary was supported by:

- 82% revenue growth;
- More than \$50 billion of sequential backlog growth;
- Strong paid token usage;
- Greater customer acquisition;
- Customers exceeding contractual commitments;
- Continued supply constraints.

The demand signal appears genuine rather than promotional.

The main analytical concern is no longer whether Cloud demand exists. It is whether Alphabet can deploy sufficient capacity at attractive economics.

Capacity Constraints: More Urgent

Q1 management described compute constraints as acute and said Cloud revenue would have been higher with more capacity. Q2 management took an additional operational step by announcing greater reliance on third-party capacity.

This represents an escalation:

Q1: Capacity constraint identified

Q2: Higher-cost external capacity required to bridge the constraint

The constraint is positive for demand visibility but negative for near-term margin and capital intensity.

Search AI Monetization: Stronger Evidence

Q1 Search commentary focused primarily on:

- Increased usage;
- Commercial-query growth;
- Better query understanding;
- Potential expansion of advertising coverage.

Q2 commentary added:

- More than one billion AI Mode users;
- Lower cost per AI Mode response;
- Encouraging AI Overview monetization;
- More specific commercial queries;
- Active deployment of new advertising formats.

The Search AI monetization thesis therefore strengthened between Q1 and Q2.

However, Alphabet still does not disclose revenue or contribution profit per AI query.

CapEx: Significantly More Aggressive

Alphabet raised CapEx guidance in consecutive quarters.

Guidance Date	FY2026 CapEx Range	Midpoint
Initial 2026 Guidance	\$175–\$185B	\$180B
Q1 Updated Guidance	\$180–\$190B	\$185B
Q2 Updated Guidance	\$195–\$205B	\$200B

From the initial range to Q2, the midpoint increased by approximately **\$20 billion**, or roughly **11.1%**.

The Q2 increase was not described as an isolated accounting adjustment. Management attributed it primarily to accelerated capacity delivery in response to growing demand.

Free Cash Flow: More Cautious

Management's tone became explicitly more cautious after Q2 FCF turned negative.

The company did not attempt to characterize negative FCF as temporary within a specified period. Instead, it stated that free cash flow will remain under pressure.

This wording suggests that investors should not expect an immediate return to Alphabet's previous level of FCF generation.

AI ROIC: More Bullish but Still Unquantified

Management argued that:

- Search growth;
- Cloud growth;
- Backlog;
- Customer demand;
- Full-stack efficiencies;

provide evidence of attractive AI returns.

However, management did not provide a numerical threshold for continuing or stopping investment.

The practical capital-allocation rule appears to be:

Continue increasing investment while observable customer and internal demand supports an attractive long-term return.

This is directionally reasonable but difficult for external investors to independently verify.

8.17 Guidance Credibility Assessment

Areas With Strong Supporting Evidence

Cloud Demand

Management's optimism is supported by reported revenue growth, backlog, operating income and capacity constraints.

Search Resilience

Search revenue remained strong while AI features increased usage, reducing the immediate credibility of the thesis that generative AI is rapidly destroying Google Search economics.

Near-Term Margin Pressure

The warning is credible because Alphabet is explicitly adding higher-cost third-party capacity and absorbing Wiz integration costs.

Free-Cash-Flow Pressure

The warning is supported by negative Q2 FCF and the implied acceleration in second-half CapEx.

Areas With Insufficient Disclosure

2027 Capital Expenditure

"Significant increase" is too broad to model precisely.

AI ROIC

Management provides proof points but no consolidated return calculation.

Third-Party Capacity Economics

Alphabet asserts attractive lifetime economics without disclosing contract margins, duration or upfront losses.

TPU Profitability

Revenue timing is disclosed, but product margin is not.

FCF Recovery

No timeline or target level has been provided.

AI Search Unit Economics

Adoption, monetization and efficiency are discussed without revenue or cost per query.

8.18 Earnings-Call Risk Map

Management Statement	Positive Interpretation	Risk Interpretation
Demand continues to exceed capacity	Strong product demand and revenue visibility	Alphabet may be chronically underbuilt or unable to monetize demand efficiently
Third-party capacity is a bridge	Temporary solution that secures major customers	Higher costs may persist longer than expected
2027 CapEx will increase significantly	Management sees a large long-term opportunity	Investment may continue rising before returns are proven
TPU revenue will ramp in 2027	Future contracted revenue provides visibility	Revenue is delayed and may carry lower margins
Search AI monetization is encouraging	AI is expanding the core franchise	Product-level profitability remains undisclosed
Management is more bullish on AI ROIC	Stronger evidence from operating results	Qualitative confidence may exceed cash-flow evidence
FCF will remain under pressure	Capital is being deployed into growth	Shareholder cash returns may remain weak
Full-stack approach creates efficiency	Custom chips and integrated products improve economics	Internal complexity and capital requirements remain substantial

8.19 Key Forecast Implications

Based on management's explicit and implicit guidance, a reasonable institutional forecast should incorporate the following assumptions.

Revenue

- Search reported growth should moderate because of tougher comparisons and less favorable foreign exchange;
- Underlying Search demand remains healthy;
- YouTube should continue growing, but Q2 included a World Cup benefit;
- Cloud should remain the fastest-growing segment;
- TPU sales remain relatively small in 2026 and become more important in 2027.

Margins

- Q2 Cloud margin should not be treated as the near-term baseline;
- Third-party capacity and Wiz create Q3 and H2 Cloud-margin headwinds;
- Depreciation and energy costs continue increasing;
- AI and Cloud hiring keeps R&D growth elevated;
- Search and Cloud operating leverage may offset part, but not all, of the cost pressure.

Cash Flow

- H2 CapEx should be materially higher than H1 CapEx;
- Free cash flow remains pressured;
- TPU inventory can continue absorbing operating cash before delivery;
- Share repurchases are unlikely to become the primary capital-allocation priority while investment remains elevated.

Capital Structure

Management described operating cash flow as its first funding source, followed by debt and then equity. The CFO stated that Alphabet wanted to maintain a resilient and healthy balance sheet while funding multiyear investment needs.

This suggests that external financing is part of a deliberate multiyear capital plan rather than a one-quarter liquidity response.

8.20 Key Indicators for the Next Earnings Call

Indicator	Positive Signal	Warning Signal
Search Revenue Growth	Moderates only because of FX and comparison base	Underlying query or pricing weakness
AI Search Monetization	More commercial queries and improving ad yield	Higher usage without revenue conversion
AI Search Cost	Continues declining	Cost reductions stall
Cloud Revenue Growth	Remains strong after Q2 acceleration	Sharp normalization not explained by comparisons
Cloud Backlog	Continues growing and converting	Backlog rises while revenue conversion slows
Cloud Margin	Remains resilient despite external capacity	Material compression without operating-income growth

Indicator	Positive Signal	Warning Signal
Third-Party Capacity	Temporary and customer-specific	Becomes a permanent broad-based cost
TPU Revenue	Ramps according to schedule	Delivery delays or lower-than-expected revenue
TPU Margin	Economically attractive	Hardware mix dilutes Cloud economics
2027 CapEx	Supported by specific contracted demand	Increases without greater revenue visibility
Depreciation	Grows slower than revenue	Grows faster than consolidated revenue
Operating Cash Flow	Accelerates	Weakens despite strong reported earnings
Free Cash Flow	Stabilizes or improves	Remains deeply negative
AI ROIC Disclosure	More quantitative evidence	Continued reliance on qualitative claims

8.21 Guidance and Earnings-Call Scorecard

Dimension	Rating	Rationale
Search Demand Outlook	Positive	AI engagement and advertising monetization remain strong
Search Reported Growth Outlook	Moderating	Tougher comparison and FX headwind
Cloud Demand Outlook	Very Strong	Growth, backlog and capacity constraints support visibility
Cloud Revenue Visibility	Strong	More than half of backlog expected within 24 months
Cloud Margin Outlook	Cautious	Third-party capacity and Wiz create pressure
TPU Revenue Outlook	Positive but Back-End Loaded	Majority of current contracts expected in 2027

Dimension	Rating	Rationale
2026 CapEx Outlook	Materially Higher	Guidance raised to \$195–\$205B
2027 CapEx Visibility	Low	Significant increase expected, but unquantified
Cost Outlook	Negative	Depreciation, energy, hiring and marketing continue rising
Free-Cash-Flow Outlook	Weak	Management expects continued pressure
AI ROIC Confidence	Higher	Management became more bullish
AI ROIC Disclosure	Weak	No quantitative return metric
Overall Management Tone	Long-Term Bullish, Near-Term Financially Cautious	Strong demand but higher investment and cost pressure

8.22 Overall Guidance and Earnings-Call Conclusion

Alphabet's Q2 earnings call strengthened the long-term AI demand thesis but weakened the near-term free-cash-flow outlook.

Management's most important positive messages were:

- Search AI adoption is increasing queries;
- AI Search monetization remains encouraging;
- Google Cloud demand remains exceptionally strong;
- Backlog continues to grow;
- Enterprise AI usage is expanding;
- Management has become more bullish about long-term AI returns.

The most important cautionary messages were:

- Search faces a more difficult comparison;
- Q3 foreign exchange becomes less favorable;
- Cloud remains supply-constrained;
- Higher-cost third-party capacity will pressure Cloud margin;
- TPU revenue is weighted toward 2027;
- CapEx was raised again;
- 2027 CapEx will increase significantly;
- Depreciation, energy and AI talent costs will rise;
- Free cash flow will remain under pressure.

The quarter's guidance can be summarized as:

Stronger AI demand

plus

Greater confidence in long-term monetization

minus

Higher CapEx, margin pressure and delayed cash returns

Management is effectively asking investors to accept weaker near-term free cash flow in exchange for a larger future Search, Cloud and AI earnings opportunity.

The core investment question after the call is:

Is Alphabet's worsening near-term cash-flow profile simply the temporary funding cost of a highly attractive, capacity-constrained growth opportunity, or is AI permanently increasing the capital intensity required to protect and grow the business?

Current operating evidence, especially within Google Cloud, supports management's demand thesis.

However, management has not yet provided enough quantitative disclosure to confirm:

- The long-term AI ROIC;
- The normalized Cloud margin after third-party capacity and TPU sales;
- The peak year of the investment cycle;
- The timing of free-cash-flow recovery.

The most accurate institutional conclusion is:

Management became more bullish about AI demand and long-term returns, but also more explicit that shareholders should expect higher capital spending, near-term Cloud-margin pressure and sustained free-cash-flow weakness before those returns are fully realized.

9. Forecast Revision

9.1 Revenue Forecast Model

Executive Assessment

This revenue forecast uses a traditional segment-level growth-extrapolation methodology.

Each revenue category is forecast separately using the corresponding quarter from the prior year as the base:

Forecast Revenue = Prior-Year Same-Quarter Revenue × (1 + Assumed Year-over-Year Growth Rate)

The model preserves Alphabet's historical quarterly seasonality while gradually reducing unusually high growth rates toward more sustainable long-term levels.

The forecast is based on:

- Eight quarters of reported segment revenue from Q3 2024 through Q2 2026;
- The most recent four-quarter year-over-year growth trend;
- Management's Q2 commentary regarding foreign exchange, Search comparisons, Cloud demand and TPU revenue timing;
- A controlled growth-decay assumption rather than a mechanical continuation of the latest quarter;
- Separate assumptions for Search, YouTube Ads, Network, Subscriptions, Google Cloud and Other Bets.

The model should be interpreted as a base-case analytical forecast rather than company guidance or formal Wall Street consensus.

9.1.1 Historical Revenue Dataset

Quarterly Revenue by Business

USD billions

Revenue Category	Q3 2024A	Q4 2024A	Q1 2025A	Q2 2025A	Q3 2025A	Q4 2025A	Q1 2026A	Q2 2026A
Google Search & Other	49.39	54.03	50.70	54.19	56.57	63.07	60.40	63.27
YouTube Ads	8.92	10.47	8.93	9.80	10.26	11.38	9.88	11.06
Google Network	7.55	7.95	7.26	7.35	7.35	7.83	6.97	7.30
Subscriptions, Platforms and Devices	10.66	11.63	10.38	11.20	12.87	13.58	12.38	12.91
Google Cloud	11.35	11.96	12.26	13.62	15.16	17.66	20.03	24.77
Other Bets	0.39	0.40	0.45	0.37	0.34	0.37	0.41	0.38
Revenue Hedging	0.02	0.02	0.26	-0.11	-0.21	-0.07	-0.18	0.11
Total Revenue	88.27	96.47	90.23	96.43	102.35	113.83	109.90	119.80

Historical revenue values are taken from Alphabet's official quarterly earnings releases and Form 10-Q disclosures.

9.1.2 Recent Historical Growth Profile

The following table shows the most recent four reported year-over-year growth rates.

Revenue Category	Q3 2025A	Q4 2025A	Q1 2026A	Q2 2026A	Four-Quarter Average
Google Search & Other	14.5%	16.7%	19.1%	16.8%	16.8%
YouTube Ads	15.0%	8.7%	10.7%	12.9%	11.8%
Google Network	-2.6%	-1.6%	-3.9%	-0.7%	-2.2%
Subscriptions, Platforms and Devices	20.8%	16.7%	19.3%	15.2%	18.0%
Google Cloud	33.5%	47.8%	63.4%	81.8%	56.6%
Other Bets	-11.3%	-7.5%	-8.7%	2.4%	-6.3%

The historical profile shows three distinct trends:

1. Search growth has remained consistently in the mid-to-high teens.
2. YouTube and consumer subscriptions have maintained double-digit growth.
3. Google Cloud has experienced an exceptional acceleration that should not be extrapolated indefinitely without a growth-decay assumption.

9.1.3 Forecast Methodology

Same-Quarter Growth Extrapolation

For each revenue category:

Q3 2026 forecast = Q3 2025 actual × (1 + Q3 2026 assumed growth)

Q4 2026 forecast = Q4 2025 actual × (1 + Q4 2026 assumed growth)

The same approach is applied recursively for subsequent periods.

This method has three advantages:

- It preserves Alphabet's normal quarterly seasonality;
- It avoids using sequential growth during highly seasonal periods;
- It allows different businesses to follow different growth paths.

Controlled Growth Decay

The latest reported growth rate is not applied unchanged to all future quarters.

Instead:

- Mature businesses gradually revert toward lower long-term growth;
- Structurally declining businesses remain negative;
- Cloud growth remains elevated but decelerates as its revenue base expands;
- TPU revenue expected in 2027 prevents Cloud growth from normalizing too quickly;
- Hedging gains and losses are not forecast.

Management expects a slight Q3 foreign-exchange headwind affecting Search and YouTube, tougher Search comparisons beginning in Q3, continued strong Cloud demand, and the majority of existing TPU system revenue to be recognized in 2027. These comments support a lower Search growth rate and a still-elevated Cloud growth path.

9.1.4 Base-Case Growth Assumptions

Revenue Category	Q3 2026E	Q4 2026E	Q1 2027E	Q2 2027E	Q3 2027E	Q4 2027E	Q1 2028E	Q2 2028E
Google Search & Other	15.0%	15.5%	14.5%	14.0%	13.0%	12.5%	12.0%	11.5%
YouTube Ads	11.0%	10.5%	10.5%	9.5%	10.0%	10.0%	9.5%	9.5%
Google Network	-2.5%	-2.5%	-3.0%	-3.0%	-3.0%	-3.0%	-3.0%	-3.0%

Revenue Category	Q3 2026E	Q4 2026E	Q1 2027E	Q2 2027E	Q3 2027E	Q4 2027E	Q1 2028E	Q2 2028E
Subscriptions, Platforms and Devices	16.0%	15.5%	15.0%	14.5%	14.0%	13.5%	13.0%	12.5%
Google Cloud	75.0%	70.0%	60.0%	55.0%	45.0%	40.0%	35.0%	32.0%
Other Bets	0.0%	0.0%	5.0%	5.0%	10.0%	10.0%	10.0%	10.0%
Revenue Hedging	0	0	0	0	0	0	0	0

The assumptions are deliberately more conservative than simply carrying forward the Q2 2026 growth rates.

9.1.5 Google Search & Other Forecast

Forecast Rationale

Search revenue grew between 14.5% and 19.1% during the most recent four quarters.

The model assumes:

- 15.0% growth in Q3 2026;
- Modest stabilization in Q4;
- Gradual deceleration toward approximately 11% by 2028.

The forecast reflects:

- Continued query expansion from AI Overviews and AI Mode;
- Healthy advertiser spending;
- Strong commercial-query monetization;
- Improving AI ad formats;
- Tougher year-over-year comparisons;
- A slight Q3 foreign-exchange headwind;
- Increasing revenue scale.

Alphabet reported that Q2 Search growth was supported by higher queries, advertiser spending and improvements to ad formats and delivery. Paid clicks increased 13% and cost per click increased 3%.

Search Revenue Forecast

Period	Search Revenue	YoY Growth
Q3 2026E	\$65.05B	15.0%
Q4 2026E	\$72.85B	15.5%
Q1 2027E	\$69.16B	14.5%
Q2 2027E	\$72.13B	14.0%
Q3 2027E	\$73.51B	13.0%

Period	Search Revenue	YoY Growth
Q4 2027E	\$81.95B	12.5%
Q1 2028E	\$77.46B	12.0%
Q2 2028E	\$80.42B	11.5%

The model does not assume that generative AI causes a near-term structural collapse in Search revenue. Instead, it assumes that AI supports usage while growth gradually slows because of scale and comparison effects.

9.1.6 YouTube Advertising Forecast

Forecast Rationale

YouTube advertising grew 12.9% in Q2 2026, supported by Direct Response, Brand advertising, connected television and World Cup-related demand.

The model assumes:

- Normalization from the Q2 World Cup benefit;
- Continued connected-TV growth;
- Improving Shorts monetization;
- Stable Direct Response demand;
- Gradual long-term growth in the high-single-digit to low-double-digit range.

YouTube Advertising Forecast

Period	YouTube Ads Revenue	YoY Growth
Q3 2026E	\$11.39B	11.0%
Q4 2026E	\$12.58B	10.5%
Q1 2027E	\$10.92B	10.5%
Q2 2027E	\$12.11B	9.5%
Q3 2027E	\$12.53B	10.0%
Q4 2027E	\$13.84B	10.0%
Q1 2028E	\$11.96B	9.5%
Q2 2028E	\$13.26B	9.5%

The Q2 2027 forecast uses a lower growth rate because it compares against the World Cup-supported Q2 2026 period.

9.1.7 Google Network Forecast

Forecast Rationale

Network revenue has declined in each of the most recent four year-over-year periods.

Alphabet attributed the decline primarily to weaker AdSense revenue, partly offset by growth in AdMob. Q2 Network impressions declined 12%, while cost per impression increased 13%.

The model assumes that:

- Impression volume continues declining;
- Pricing partly offsets the decline;
- Network remains a mature and structurally challenged business;
- Annual revenue contracts approximately 2.5% to 3.0%.

Google Network Forecast

Period	Network Revenue	YoY Growth
Q3 2026E	\$7.17B	-2.5%
Q4 2026E	\$7.63B	-2.5%
Q1 2027E	\$6.76B	-3.0%
Q2 2027E	\$7.08B	-3.0%
Q3 2027E	\$6.95B	-3.0%
Q4 2027E	\$7.40B	-3.0%
Q1 2028E	\$6.56B	-3.0%
Q2 2028E	\$6.87B	-3.0%

Network is not expected to become a meaningful source of consolidated revenue growth.

9.1.8 Subscriptions, Platforms and Devices Forecast

Forecast Rationale

Subscriptions, Platforms and Devices has grown between approximately 15% and 21% during the most recent four quarters.

Alphabet stated that growth was primarily driven by paid subscriptions across YouTube services and Google One, including demand for AI subscription plans.

The model assumes:

- Continued growth in YouTube Music and Premium;
- Continued Google One and consumer AI-plan adoption;
- Stable platform revenue;
- Some product-cycle volatility from Pixel and devices;
- Gradual growth normalization as the subscriber base expands.

Subscriptions Forecast

Period	Revenue	YoY Growth
Q3 2026E	\$14.93B	16.0%
Q4 2026E	\$15.68B	15.5%
Q1 2027E	\$14.24B	15.0%
Q2 2027E	\$14.78B	14.5%
Q3 2027E	\$17.02B	14.0%
Q4 2027E	\$17.80B	13.5%
Q1 2028E	\$16.09B	13.0%
Q2 2028E	\$16.63B	12.5%

This model treats consumer subscriptions as a durable double-digit growth business, but does not assume permanent growth near 20%.

9.1.9 Google Cloud Forecast

Forecast Rationale

Google Cloud requires the largest adjustment to a simple historical-growth model.

Reported Cloud growth accelerated from:

- 33.5% in Q3 2025;
- 47.8% in Q4 2025;
- 63.4% in Q1 2026;
- 81.8% in Q2 2026.

A purely mechanical average or continuation of the Q2 rate would produce an unrealistic forecast because Cloud's revenue base is expanding rapidly.

The base-case model therefore applies a controlled deceleration:

- 75% growth in Q3 2026;
- 70% in Q4 2026;
- 60% to 55% during the first half of 2027;
- 40% to 45% in the second half of 2027;
- Low-30% growth by 2028.

This remains an aggressive growth path, supported by:

- Strong enterprise AI demand;
- A \$514 billion Cloud backlog;
- More than half of backlog expected to be recognized within 24 months;
- Continued capacity constraints;
- Additional third-party capacity;
- Rapid growth in core GCP;

- Initial TPU system revenue;
- The majority of contracted TPU system revenue expected in 2027.

Google Cloud Forecast

Period	Cloud Revenue	YoY Growth
Q3 2026E	\$26.52B	75.0%
Q4 2026E	\$30.03B	70.0%
Q1 2027E	\$32.04B	60.0%
Q2 2027E	\$38.39B	55.0%
Q3 2027E	\$38.46B	45.0%
Q4 2027E	\$42.04B	40.0%
Q1 2028E	\$43.26B	35.0%
Q2 2028E	\$50.68B	32.0%

The model expects Cloud growth to remain exceptionally high through 2027 but does not assume that the current 82% rate is sustainable.

9.1.10 Other Bets Forecast

Other Bets revenue remains small and volatile.

The model assumes:

- Flat revenue during the second half of 2026;
- 5% growth in the first half of 2027;
- 10% growth thereafter as Waymo expands.

Other Bets Forecast

Period	Other Bets Revenue	YoY Growth
Q3 2026E	\$0.34B	0.0%
Q4 2026E	\$0.37B	0.0%
Q1 2027E	\$0.43B	5.0%
Q2 2027E	\$0.40B	5.0%
Q3 2027E	\$0.38B	10.0%
Q4 2027E	\$0.41B	10.0%

Period	Other Bets Revenue	YoY Growth
Q1 2028E	\$0.48B	10.0%
Q2 2028E	\$0.44B	10.0%

Because Other Bets represents less than 1% of Alphabet revenue, forecast errors in this category have little effect on the consolidated revenue estimate.

9.1.11 Eight-Quarter Revenue Forecast

USD billions

Revenue Category	Q3 2026E	Q4 2026E	Q1 2027E	Q2 2027E	Q3 2027E	Q4 2027E	Q1 2028E	Q2 2028E
Google Search & Other	65.05	72.85	69.16	72.13	73.51	81.95	77.46	80.42
YouTube Ads	11.39	12.58	10.92	12.11	12.53	13.84	11.96	13.26
Google Network	7.17	7.63	6.76	7.08	6.95	7.40	6.56	6.87
Subscriptions, Platforms and Devices	14.93	15.68	14.24	14.78	17.02	17.80	16.09	16.63
Google Cloud	26.52	30.03	32.04	38.39	38.46	42.04	43.26	50.68
Other Bets	0.34	0.37	0.43	0.40	0.38	0.41	0.48	0.44
Revenue Hedging	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Revenue	125.41	139.14	133.56	144.89	148.85	163.44	155.80	168.30
Total Revenue Growth	22.5%	22.2%	21.5%	20.9%	18.7%	17.5%	16.7%	16.2%

The model produces continued total-revenue growth above 20% through the first half of 2027, followed by gradual normalization.

The principal driver of the deceleration is not a collapse in Search or YouTube. It is the gradual reduction in Google Cloud's growth rate as the business becomes substantially larger.

9.1.12 Three-Year Revenue Forecast

USD billions

Revenue Category	2024A	2025A	2026E	2027E	2028E
Google Search & Other	198.08	224.53	261.57	296.75	330.04
YouTube Ads	36.15	40.37	44.91	49.39	53.95
Google Network	30.36	29.79	29.08	28.20	27.36
Subscriptions, Platforms and Devices	40.34	48.03	55.91	63.84	71.63
Google Cloud	43.23	58.71	101.35	150.94	197.75
Other Bets	1.65	1.54	1.51	1.62	1.78
Revenue Hedging	0.21	-0.13	-0.07	0.00	0.00
Total Revenue	350.02	402.84	494.24	590.74	682.50
Total Revenue Growth	13.9%	15.1%	22.7%	19.5%	15.5%

The 2028 annual model includes additional assumptions for Q3 and Q4 2028:

Revenue Category	Q3 2028E Growth	Q4 2028E Growth
Search	11.0%	10.5%
YouTube Ads	9.0%	9.0%
Network	-3.0%	-3.0%
Subscriptions	12.0%	11.5%
Google Cloud	30.0%	28.0%
Other Bets	10.0%	10.0%

9.1.13 Revenue-Mix Evolution

Revenue Category	2025A Mix	2026E Mix	2027E Mix	2028E Mix
Search & Other	55.7%	52.9%	50.2%	48.4%
YouTube Ads	10.0%	9.1%	8.4%	7.9%
Network	7.4%	5.9%	4.8%	4.0%
Subscriptions	11.9%	11.3%	10.8%	10.5%

Revenue Category	2025A Mix	2026E Mix	2027E Mix	2028E Mix
Google Cloud	14.6%	20.5%	25.6%	29.0%
Other Bets	0.4%	0.3%	0.3%	0.3%

The most important forecast result is the continued shift toward Google Cloud.

Google Cloud's share of Alphabet revenue is projected to increase from approximately 14.6% in 2025 to approximately 29% by 2028.

This would materially change Alphabet's financial profile:

- Revenue growth would become less dependent on advertising;
- Revenue visibility would improve through contracted Cloud demand;
- Capital intensity would remain elevated;
- Hardware and infrastructure revenue could represent a larger portion of sales;
- Alphabet's consolidated margin would become increasingly dependent on Cloud economics.

9.1.14 External Consensus Cross-Check

A pre-Q2 2026 market-consensus dataset showed projected Alphabet revenue of approximately:

Fiscal Year	External Consensus	This Model	Difference
2026	\$489.25B	\$494.24B	+1.0%
2027	\$586.40B	\$590.74B	+0.7%
2028	\$690.76B	\$682.50B	-1.2%

The external estimates were published immediately before the Q2 results and may not yet reflect subsequent analyst revisions. They are used only as a reasonableness check and are not inputs into the model.

The close relationship indicates that the traditional extrapolation model produces consolidated forecasts broadly consistent with professional market expectations, despite using independently constructed segment assumptions.

9.1.15 Key Forecast Risks

Upside Risks

- Search AI generates more incremental commercial queries than expected;
- Search cost-per-click remains stronger for longer;
- YouTube connected-TV monetization accelerates;
- Paid Gemini and Google One penetration exceeds assumptions;
- Cloud backlog converts faster;
- Third-party capacity enables more Cloud revenue;
- TPU system deliveries exceed expectations;
- Enterprise AI spending remains exceptionally strong.

Downside Risks

- Search growth slows more sharply under tougher comparisons;
 - AI Search changes advertising click behavior;
 - YouTube loses the benefit of event-driven advertising;
 - Consumer subscription growth normalizes faster;
 - Cloud growth is inflated by lower-margin hardware revenue;
 - TPU deliveries are delayed;
 - Cloud supply constraints prevent revenue recognition;
 - Backlog conversion is slower than expected;
 - Wiz or other acquired revenue creates misleading organic comparisons;
 - Foreign exchange becomes more unfavorable.
-

9.1.16 Limitations of Traditional Growth Extrapolation

The traditional growth-extrapolation method is transparent and easy to update, but it has several limitations.

It Does Not Explicitly Model Volume and Price

The model does not separately forecast:

- Search paid clicks and CPC;
- YouTube impressions and CPM;
- Paid subscribers and ARPU;
- Cloud customers and consumption;
- TPU units and average selling prices.

It May Miss Structural Inflection Points

The model may understate or overstate revenue when:

- A new AI product becomes material;
- TPU system revenue begins in large increments;
- A major acquisition changes the consolidation scope;
- Advertising pricing changes sharply;
- Capacity constraints are resolved;
- A business experiences structural disruption.

It Is Most Sensitive to Cloud Assumptions

Because Cloud is projected to become nearly 30% of total revenue, small changes in Cloud growth create large changes in the consolidated forecast.

For example, a 1,000 bps change in annual Cloud growth could alter Alphabet's annual revenue estimate by several billion dollars.

Therefore, the growth-extrapolation model should eventually be supplemented with:

- Backlog-conversion analysis;
 - TPU revenue scenarios;
 - Organic versus acquired Cloud growth;
 - Search paid-click and CPC assumptions;
 - Consumer subscription assumptions.
-

9.1.17 Overall Revenue Forecast Conclusion

The base-case model forecasts Alphabet revenue increasing from **\$402.8 billion in 2025** to:

- **\$494.2 billion in 2026;**
- **\$590.7 billion in 2027;**
- **\$682.5 billion in 2028.**

The corresponding growth rates are:

- **22.7% in 2026;**
- **19.5% in 2027;**
- **15.5% in 2028.**

The revenue outlook can be summarized as:

Stable mid-teens Search growth

plus

High-single-digit to low-double-digit YouTube growth

plus

Sustained subscription growth

plus

Exceptional but gradually decelerating Google Cloud growth

minus

Structural weakness in Google Network

The central forecast conclusion is:

Alphabet's consolidated revenue growth is expected to remain elevated because Google Cloud is becoming a much larger component of the business, even as the growth rates of Search, YouTube and subscriptions gradually normalize.

The primary forecast risk is also Google Cloud.

If Cloud growth and TPU revenue are stronger than assumed, Alphabet could materially exceed this revenue path. If growth normalizes faster or capacity prevents backlog conversion, consolidated revenue could fall below the model despite resilient advertising performance.

9.2 Profit Forecast Model

9.2.1 Executive Assessment

The profit model forecasts:

- Consolidated gross margin;
- Google Services operating margin;
- Google Cloud operating margin;
- Research and development expense;
- Depreciation;
- Operating legal expenses;
- Consolidated operating income and operating margin.

The forecast is linked directly to the segment revenue model established in Section 9.1.

The central adjustment is an **AI infrastructure replacement-cost overlay**. The model assumes that the prices of selected constrained AI infrastructure components—including high-bandwidth memory, advanced accelerators and high-speed optical networking components—are materially higher than the historical costs embedded in Alphabet's existing depreciation base.

However, the model does **not** assume that Alphabet's entire data-center construction cost doubles.

Public disclosures support severe supply tightness and rapid price inflation in memory and AI networking, but they do not establish that every HBM module, GPU, TPU and optical transceiver purchased by Alphabet has uniformly doubled in price. Micron reported sequential DRAM pricing growth in the low-60% range and NAND pricing growth in the mid-80% range in its fiscal Q3 2026, while stating that memory supply conditions could remain tight beyond calendar 2027. Broadcom separately reported accelerating demand for custom AI accelerators and AI networking, but did not disclose equivalent unit-price increases.

Therefore, the model treats a two-times component cost as an explicit analyst assumption rather than a reported industry-wide fact.

The base-case conclusion is:

Alphabet's revenue growth should remain strong, but the combination of higher replacement costs, third-party capacity, TPU hardware mix, depreciation, energy expenses and AI R&D is expected to reduce consolidated operating margin during 2027 before utilization and operating leverage begin to offset the pressure in 2028.

9.2.2 Historical Profitability Baseline

Alphabet reported the following Q2 2026 results:

Metric	Q2 2026 Actual
Revenue	\$119.80B
Gross Profit	\$73.85B
Gross Margin	61.6%
Research and Development	\$18.22B
R&D as % of Revenue	15.2%
Google Services Operating Margin	41.8%
Google Cloud Operating Margin	35.6%
Depreciation	\$7.10B
Operating Income	\$40.77B
Operating Margin	34.0%

Q2 cost of revenue increased 18%, driven by depreciation, traffic acquisition costs, TPU inventory, YouTube content acquisition and technical-infrastructure operating expenses. R&D increased 32%, primarily because of AI employee compensation and depreciation.

Google Cloud operating margin reached 35.6%, but management explicitly warned that third-party capacity would create near-term margin pressure. Alphabet also expects higher depreciation, energy expenses and data-center operating costs as new infrastructure enters service.

Q2's profitability should therefore not be extrapolated without adjustment.

9.2.3 AI Infrastructure Replacement-Cost Overlay

A. Alphabet's Infrastructure Mix

Alphabet stated that approximately:

- 60% of technical-infrastructure investment was allocated to servers;
- 40% was allocated to data centers and networking equipment.

Alphabet also reported that approximately 60% of technical-infrastructure assets already in service consisted of servers and network equipment.

B. Cost-Exposure Assumption

The model divides technical-infrastructure spending into two groups.

Infrastructure Category	Share of Technical CapEx	Share Assumed Exposed to 2x Pricing
Servers and AI compute systems	60%	70%
Data centers and networking	40%	20%
Total exposed share	100%	Approximately 50%

The exposed server cost pool includes, conceptually:

- TPU or GPU accelerators;
- HBM and server DRAM;
- Advanced packaging;
- High-speed network-interface components;
- Optical interconnects;
- Power-delivery components.

The networking exposure includes selected:

- Optical modules;
- Switching equipment;
- Network-interface equipment;
- High-speed interconnect components.

These percentages are analyst assumptions because Alphabet does not disclose a detailed technical-infrastructure bill of materials.

C. Gross Replacement-Cost Calculation

The model applies a two-times price assumption to approximately 50% of technical-infrastructure cost.

Gross replacement-cost factor = 50% × 2.0 + 50% × 1.0

Gross replacement-cost factor = 1.50x

This means that equivalent infrastructure would cost approximately 50% more before accounting for improvements in performance and utilization.

D. Performance and Efficiency Offset

New hardware typically delivers improved:

- Compute performance;
- Memory bandwidth;
- Performance per watt;
- Rack density;
- Networking throughput;
- Model-serving efficiency.

The model assumes that architecture and performance improvements offset approximately 15% of the gross replacement-cost increase.

Effective replacement-cost factor = $1.50 \div 1.15$

Effective replacement-cost factor = approximately 1.30x

Therefore:

A two-times increase in the price of selected components produces an estimated 30% increase in effective infrastructure cost per equivalent unit of compute, rather than a 100% increase in total infrastructure cost.

This is the principal cost assumption used in the profit forecast.

9.2.4 Capital Expenditure Assumptions

Alphabet raised its 2026 CapEx guidance to \$195–\$205 billion and expects CapEx to increase significantly again in 2027. It had already spent \$80.6 billion during the first half of 2026.

The profit model uses the following CapEx path:

Fiscal Year	CapEx Forecast	YoY Growth	Key Assumption
2026E	\$200B	Significant increase	Midpoint of management guidance
2027E	\$285B	+42.5%	Higher replacement costs plus capacity expansion
2028E	\$320B	+12.3%	Prices remain elevated, but capacity growth moderates

The 2027 forecast is derived conceptually as:

$1.30x$ effective replacement-cost factor $\times$ approximately $1.10x$ capacity growth = approximately $1.43x$ CapEx

The resulting 43% increase is broadly consistent with the \$285 billion forecast.

This is deliberately more conservative than assuming that \$200 billion of CapEx simply doubles to \$400 billion.

9.2.5 Depreciation Forecast Methodology

Depreciation is forecast separately from CapEx because the two expenses occur at different times.

The accounting sequence is:

Equipment purchased → Asset under construction → Asset placed in service → Depreciation begins

Alphabet reported:

- \$122.8 billion of assets not yet in service;
- \$321.2 billion of net property and equipment;
- \$13.6 billion of depreciation during H1 2026.

Assets not yet in service may take months or years to become operational. Depreciation begins only when the assets are ready for their intended use.

The current income statement therefore reflects historical equipment costs and only part of the current infrastructure build-out.

Depreciation Assumptions

The model assumes:

- Server and network equipment enters service faster than data-center buildings;
- Approximately 60% of CapEx relates to faster-depreciating server and network assets;
- Higher-cost 2027 equipment begins materially affecting depreciation from the second half of 2027;
- Data-center and power assets enter depreciation more gradually;
- No major extension of server useful lives is assumed.

Depreciation Forecast

USD billions

Period	Depreciation Forecast	Depreciation as % of Revenue
Q3 2026E	\$8.3B	6.6%
Q4 2026E	\$9.4B	6.8%
Q1 2027E	\$10.5B	7.9%
Q2 2027E	\$11.7B	8.1%
Q3 2027E	\$12.7B	8.5%
Q4 2027E	\$13.6B	8.3%
Q1 2028E	\$15.0B	9.6%
Q2 2028E	\$16.5B	9.8%

The annual forecast is:

Fiscal Year	Depreciation	YoY Growth	% of Revenue
2026E	\$31.3B	+48.1%	6.3%
2027E	\$48.5B	+55.0%	8.2%
2028E	\$69.0B	+42.3%	10.1%

Depreciation is a memorandum item in this model. It is already included within:

- Cost of revenue;
- R&D;
- Segment technical-infrastructure allocations.

It must not be subtracted a second time when calculating operating income.

9.2.6 Consolidated Gross Margin Forecast

Gross-Margin Drivers

Positive Factors

- Search revenue remains highly scalable;
- TAC benefits from Search growing faster than Network;
- Subscription revenue continues increasing;
- Cloud utilization improves;
- Internal TPU design may provide performance and cost benefits;
- Revenue grows faster than some fixed infrastructure costs.

Negative Factors

- Higher-cost HBM and AI compute systems;
- Higher-cost optical and networking equipment;
- Third-party Cloud capacity;
- Rising depreciation;
- Energy and data-center operating costs;
- TPU product inventory and cost of goods sold;
- YouTube content-acquisition expenses;
- Potentially lower-margin hardware revenue.

Alphabet's cost of revenue explicitly includes depreciation, energy, equipment, network-capacity costs and inventory costs associated with TPU systems.

Gross-Margin Forecast

Period	Gross Margin
Q2 2026A	61.6%
Q3 2026E	59.6%
Q4 2026E	59.1%

Period	Gross Margin
Q1 2027E	58.6%
Q2 2027E	58.2%
Q3 2027E	58.0%
Q4 2027E	58.1%
Q1 2028E	58.5%
Q2 2028E	59.0%

The forecast assumes that gross margin reaches its trough during the middle of 2027 as:

- More expensive equipment enters service;
- Third-party capacity is used;
- TPU system sales increase;
- Depreciation accelerates;
- Energy and operating costs increase.

Gross margin begins recovering during 2028 as capacity utilization improves and revenue scales across the installed infrastructure base.

9.2.7 Google Services Operating Margin Forecast

Google Services margin reached 41.8% in Q2 2026. The segment benefits from Search and subscription operating leverage, but it must also absorb growing AI Search inference, infrastructure and marketing costs.

Services Margin Drivers

Positive

- Search query and advertising growth;
- Higher CPC and paid clicks;
- Subscription growth;
- Lower consolidated TAC mix;
- Improving AI inference efficiency;
- High fixed-cost operating leverage.

Negative

- AI Overviews and AI Mode inference costs;
- Higher depreciation allocated to Search and YouTube;
- Gemini and Search marketing;
- YouTube content costs;
- Consumer AI product investment;
- Recurring legal and regulatory expense.

Services Operating Margin Forecast

Period	Services Margin
Q2 2026A	41.8%
Q3 2026E	41.3%
Q4 2026E	41.0%
Q1 2027E	40.7%
Q2 2027E	40.4%
Q3 2027E	40.0%
Q4 2027E	39.8%
Q1 2028E	39.8%
Q2 2028E	40.0%

The forecast assumes that Services remains highly profitable but does not maintain its Q2 margin indefinitely.

The margin pressure is moderate rather than severe because Search advertising revenue is expected to continue growing faster than incremental inference and infrastructure costs.

9.2.8 Google Cloud Operating Margin Forecast

Google Cloud margin requires the largest downward revision.

Q2's 35.6% margin reflected:

- Exceptional revenue growth;
- Strong internal capacity utilization;
- Scale leverage;
- Limited initial TPU system revenue.

The forward cost structure will be less favorable because:

- Alphabet is expanding higher-cost third-party capacity;
- Wiz integration expenses continue;
- TPU system sales become a larger percentage of revenue;
- Higher-cost equipment enters service;
- Depreciation increases;
- Energy and network-capacity costs increase.

Management explicitly stated that third-party capacity would pressure Cloud margin in the near term and that most existing TPU system revenue would be recognized in 2027.

Cloud Operating Margin Forecast

Period	Cloud Margin
Q2 2026A	35.6%

Period	Cloud Margin
Q3 2026E	33.0%
Q4 2026E	31.5%
Q1 2027E	30.5%
Q2 2027E	30.0%
Q3 2027E	30.5%
Q4 2027E	31.0%
Q1 2028E	31.8%
Q2 2028E	32.5%

The model does not assume that Cloud profitability collapses.

Instead, it assumes that Cloud operating income continues increasing because revenue grows rapidly, even as the margin percentage temporarily declines.

For example:

Period	Cloud Revenue	Cloud Margin	Cloud Operating Income
Q3 2026E	\$26.52B	33.0%	\$8.75B
Q4 2026E	\$30.03B	31.5%	\$9.46B
Q1 2027E	\$32.04B	30.5%	\$9.77B
Q2 2027E	\$38.39B	30.0%	\$11.52B
Q3 2027E	\$38.46B	30.5%	\$11.73B
Q4 2027E	\$42.04B	31.0%	\$13.03B
Q1 2028E	\$43.26B	31.8%	\$13.76B
Q2 2028E	\$50.68B	32.5%	\$16.47B

This distinction is important:

A declining Cloud margin percentage does not necessarily mean declining Cloud economics if absolute operating income continues growing strongly.

9.2.9 Research and Development Forecast

Alphabet's reported R&D includes:

- AI researchers and engineers;
- Cloud and product engineering;
- Technical-infrastructure depreciation;
- Energy and network-capacity usage;
- Third-party development services.

R&D increased 32% in Q2 2026, driven primarily by \$2.9 billion of higher compensation and \$592 million of higher depreciation.

R&D Forecast Assumptions

The model assumes:

- Continued hiring in AI and Cloud;
- High compensation for specialized AI employees;
- Higher depreciation allocated to R&D;
- Continued frontier-model training;
- Gradual operating leverage after 2027;
- No immediate reduction in shared AI research spending.

R&D Forecast

Period	R&D Expense	R&D as % of Revenue
Q3 2026E	\$19.06B	15.2%
Q4 2026E	\$21.29B	15.3%
Q1 2027E	\$20.70B	15.5%
Q2 2027E	\$22.60B	15.6%
Q3 2027E	\$23.07B	15.5%
Q4 2027E	\$25.01B	15.3%
Q1 2028E	\$23.53B	15.1%
Q2 2028E	\$25.08B	14.9%

R&D intensity peaks during the first half of 2027 because higher depreciation and AI talent expense reach the income statement before full operating leverage is realized.

9.2.10 Legal Expense Forecast

Alphabet recorded \$2.1 billion of PriceRunner-related legal charges in Q2 2026:

- \$1.5 billion was recorded in operating expenses;
- \$581 million was recorded below operating income in other income and expense.

Only the \$1.5 billion operating component affected operating margin.

The model forecasts only operating legal charges.

Legal-Expense Methodology

Legal costs are divided into:

1. Recurring legal, regulatory and compliance spending;
2. Large case-specific judgments and settlements.

It would be unrealistic to forecast zero legal expense for Alphabet because the company faces recurring competition, privacy and regulatory matters.

However, it would also be inappropriate to repeat the PriceRunner charge every quarter.

Operating Legal Expense Forecast

Period	Operating Legal Expense
Q3 2026E	\$0.6B
Q4 2026E	\$0.9B
Q1 2027E	\$0.5B
Q2 2027E	\$0.7B
Q3 2027E	\$0.5B
Q4 2027E	\$0.8B
Q1 2028E	\$0.5B
Q2 2028E	\$0.7B

These are normalized analyst estimates rather than company guidance.

A separate legal-risk scenario should be applied in valuation because individual antitrust judgments could be substantially larger.

9.2.11 Other Operating Expense Assumptions

To complete the operating-margin bridge, the model also assumes:

Expense Category	2026 H2	2027	2028
Sales and Marketing as % of Revenue	Approximately 7.0%	Approximately 6.8%–6.9%	Approximately 6.6%–6.7%
Base G&A Excluding Major Legal Charges	Approximately 4.0%	Approximately 3.9%	Approximately 3.8%

Sales and marketing remains elevated because Alphabet is investing in Gemini, Search and other AI products. Management has also indicated that AI product marketing will remain an investment priority.

9.2.12 Eight-Quarter Profit Forecast

USD billions, except margins

Metric	Q3 2026E	Q4 2026E	Q1 2027E	Q2 2027E	Q3 2027E	Q4 2027E	Q1 2028E	Q2 2028E
Revenue	125.41	139.14	133.56	144.89	148.85	163.44	155.80	168.30
Gross Margin	59.6%	59.1%	58.6%	58.2%	58.0%	58.1%	58.5%	59.0%
Services Operating Margin	41.3%	41.0%	40.7%	40.4%	40.0%	39.8%	39.8%	40.0%
Cloud Operating Margin	33.0%	31.5%	30.5%	30.0%	30.5%	31.0%	31.8%	32.5%
R&D Expense	19.06	21.29	20.70	22.60	23.07	25.01	23.53	25.08
R&D as % of Revenue	15.2%	15.3%	15.5%	15.6%	15.5%	15.3%	15.1%	14.9%
Depreciation	8.30	9.40	10.50	11.70	12.70	13.60	15.00	16.50
Operating Legal Expense	0.60	0.90	0.50	0.70	0.50	0.80	0.50	0.70
Operating Income	41.29	44.74	42.64	45.38	46.83	51.66	50.76	55.85
Operating Margin	32.9%	32.2%	31.9%	31.3%	31.5%	31.6%	32.6%	33.2%

The forecast produces an operating-margin trough in Q2 2027.

This reflects the expected overlap of:

- Higher component costs;
- Third-party Cloud capacity;
- TPU hardware growth;
- Higher depreciation;
- AI hiring;
- Energy and network costs.

Operating margin begins improving during 2028 as revenue and capacity utilization grow across the enlarged infrastructure base.

9.2.13 Three-Year Profit Forecast

USD billions, except margins

Metric	2026E	2027E	2028E
Revenue	494.24	590.74	682.50
Gross Profit	299.45	343.83	404.07
Gross Margin	60.6%	58.2%	59.2%
Services Operating Margin	42.3%	40.2%	40.1%
Cloud Operating Margin	33.2%	30.5%	32.7%
R&D Expense	75.60	91.38	101.46
R&D as % of Revenue	15.3%	15.5%	14.9%
Depreciation	31.3	48.5	69.0
Depreciation as % of Revenue	6.3%	8.2%	10.1%
Operating Legal Expense	3.0	2.5	2.4
Operating Income	166.49	186.51	228.90
Operating Margin	33.7%	31.6%	33.5%

The key pattern is:

- Strong revenue growth;
- Substantial operating-income growth;
- Temporary margin compression;
- Rapid depreciation growth;
- Gradual margin recovery after 2027.

9.2.14 Cost-Shock Sensitivity

The following table compares a conventional model with the replacement-cost-adjusted model.

Metric	2027 Conventional Case	2027 Replacement-Cost Case	Difference
Gross Margin	60.0%	58.2%	-180 bps
Cloud Operating Margin	33.0%	30.5%	-250 bps
Depreciation	\$43.0B	\$48.5B	+\$5.5B
Operating Margin	33.3%	31.6%	-170 bps

Metric	2027 Conventional Case	2027 Replacement-Cost Case	Difference
Operating Income	\$196.7B	\$186.5B	-\$10.2B

Metric	2028 Conventional Case	2028 Replacement-Cost Case	Difference
Gross Margin	61.0%	59.2%	-180 bps
Cloud Operating Margin	35.0%	32.7%	-230 bps
Depreciation	\$60.0B	\$69.0B	+\$9.0B
Operating Margin	35.0%	33.5%	-150 bps
Operating Income	\$238.9B	\$228.9B	-\$10.0B

The replacement-cost adjustment reduces projected operating income by approximately \$10 billion annually in both 2027 and 2028.

9.2.15 Severe Cost Scenario

A more severe scenario would assume that the entire server and networking system—not just selected components—costs twice as much, with limited performance offset.

Under that scenario:

- 2027 CapEx could approach \$350–\$400 billion;
- Cloud operating margin could fall into the mid-20% range;
- Consolidated gross margin could fall toward 55%–56%;
- Consolidated operating margin could fall below 30%;
- Depreciation could exceed \$75 billion by 2028;
- Free-cash-flow recovery would be materially delayed.

This scenario is not used as the base case because there is not enough public evidence to conclude that Alphabet's entire infrastructure system cost has doubled.

It should instead be treated as a Bear Case sensitivity.

9.2.16 Key Metrics to Monitor

Indicator	Positive Signal	Negative Signal
HBM and DRAM pricing	Price increases moderate	Prices remain structurally elevated
Accelerator procurement	Better performance per dollar	Unit costs rise faster than performance
Optical networking cost	Higher throughput offsets prices	Cost per delivered bandwidth increases
CapEx guidance	2027 increase below model	CapEx rises toward severe case
Assets not yet in service	Convert rapidly into revenue capacity	Continue accumulating
Depreciation growth	Remains below revenue growth	Exceeds revenue growth
Cloud margin	Remains above 30%	Falls into mid-20% range
Third-party capacity	Temporary bridge	Becomes permanent
TPU revenue mix	Attractive absolute profit	Hardware dilutes margin materially
R&D intensity	Peaks and declines	Remains above 16%
Energy cost per compute unit	Declines	Continues increasing
Operating margin	Troughs during 2027	Continues falling into 2028

9.2.17 Overall Profit Forecast Conclusion

Alphabet's current profit base is supported by infrastructure that was purchased and deployed before the full effect of the latest AI component-price escalation reached the income statement.

The future model must therefore distinguish:

Historical-cost depreciation

from

Current replacement cost

The model applies a two-times cost assumption to the approximately 50% of technical infrastructure judged to be most exposed to constrained AI components. After allowing for performance and efficiency improvements, this produces an approximately 30% increase in effective infrastructure cost per unit of equivalent capacity.

The resulting forecast shows:

- Gross margin declining from 61.6% in Q2 2026 to approximately 58% during 2027;
- Services margin declining moderately but remaining near 40%;
- Cloud margin falling from 35.6% to approximately 30% before recovering;
- R&D remaining near 15% of revenue;
- Depreciation increasing from approximately \$31 billion in 2026 to \$69 billion in 2028;
- Consolidated operating margin falling to approximately 31% during 2027 before recovering toward 34%.

The central conclusion is:

Alphabet can continue growing operating income despite substantially higher AI infrastructure costs, but the market should not assume that Q2 2026's gross margin or Cloud margin is sustainable while higher-cost equipment, third-party capacity and depreciation enter the financial statements.

The most important forecast variable is not simply the amount Alphabet spends.

It is:

Whether the revenue and operating profit generated per dollar of higher-cost AI infrastructure improve quickly enough to offset the rise in depreciation, energy, networking, HBM, accelerator and optical-system costs.

10. Valuation

10.1 Executive Assessment

Alphabet's reported Q2 2026 GAAP EPS should not be used directly for valuation because the quarter included approximately **\$99.0 billion of equity-security gains**, which increased net income by \$77.1 billion and diluted EPS by \$6.26 per share. Almost all of those gains were unrealized and unrelated to recurring operating performance.

The valuation therefore uses four approaches:

1. Normalized P/E;
2. EV/EBITDA;
3. Free Cash Flow Yield;
4. Sum-of-the-Parts.

As of the July 24, 2026 close:

Share Class	Closing Price
GOOG	\$319.09
GOOGL	\$319.74
Blended Reference Price	Approximately \$319.40
Approximate Market Capitalization	\$3.90–\$3.91T

Alphabet's diluted weighted-average common share count was approximately **12.31 billion** in Q2 2026. The valuation model uses **12.55 billion NTM diluted shares** to reflect the full-period effect of the June common-equity issuance, ongoing stock-based compensation and partial preferred-stock dilution. Alphabet issued common shares and mandatory convertible preferred securities during June 2026, so the Q2 weighted-average share count does not yet reflect their full effect.

The four valuation methods produce different conclusions:

Valuation Method	Base-Case Value per Share
Normalized P/E	\$274
EV/EBITDA	\$289
CapEx-Normalized FCF	Approximately \$255–\$295
SOTP	\$317
Composite Base Value	Approximately \$290

The divergence reflects Alphabet's unusual financial structure:

- Recurring operating earnings remain strong;
- Current free cash flow is heavily depressed by AI investment;
- Google Cloud may deserve a higher multiple than the advertising business;
- Alphabet holds a very large equity-investment portfolio;
- Shared AI R&D and infrastructure commitments create significant costs that are not fully captured in Cloud's segment margin.

The overall valuation conclusion is:

Alphabet appears approximately fairly valued under an optimistic SOTP framework, but expensive under normalized earnings and free-cash-flow approaches that incorporate higher AI infrastructure costs.

10.2 Normalized P/E Valuation

10.2.1 Why GAAP EPS Is Not Appropriate

Alphabet reported diluted GAAP EPS of \$9.11 for Q2 2026.

However:

Q2 EPS Measure	Amount
Reported GAAP EPS	\$9.11
Less: Equity-Investment Gain Contribution	(\$6.26)
EPS Excluding Equity Gains	Approximately \$2.85
Illustrative EPS Excluding Equity Gains and PriceRunner Charge	Approximately \$2.99

The appropriate P/E denominator is therefore normalized operating EPS rather than reported GAAP EPS.

Using reported GAAP earnings would artificially reduce Alphabet's apparent P/E because the investment revaluation may not recur and could reverse in a future period. Alphabet explicitly warns that investment-value fluctuations may create significant volatility in other income and expense.

10.2.2 Normalized NTM EPS Bridge

The NTM period covers Q3 2026 through Q2 2027.

The operating-profit forecast from Section 9.2 is:

Period	Forecast Operating Income
Q3 2026E	\$41.29B
Q4 2026E	\$44.74B
Q1 2027E	\$42.64B
Q2 2027E	\$45.38B
NTM Operating Income	\$174.05B

The normalized EPS calculation excludes equity-investment gains and uses the following assumptions:

Normalization Assumption	NTM Estimate
Operating Income	\$174.05B
Normalized Net OI&E	(\$3.00B)
Normalized Pre-Tax Income	\$171.05B
Normalized Tax Rate	19.0%
Normalized Net Income	\$138.55B
Preferred Dividends	(\$1.19B)
Net Income Available to Common	\$137.36B
NTM Diluted Shares	12.55B
Normalized NTM EPS	\$10.95

The normalized OI&E assumption reflects:

- Higher recurring interest expense;
- Interest income on liquidity;
- Foreign-exchange and debt-security movements;
- Exclusion of equity-investment revaluation gains;
- Exclusion of major one-time legal interest expenses.

Alphabet's Q2 interest expense increased to \$1.28 billion, while interest income was \$1.43 billion. Excluding equity gains, Q2 OI&E was approximately negative \$1.05 billion, demonstrating that below-the-line earnings are no longer a major positive contributor.

10.2.3 Forward EPS Forecast

EPS Period	Normalized EPS
NTM: Q3 2026–Q2 2027	\$10.95
FY2027E	\$11.59
FY2028E	\$14.16

The FY2027 and FY2028 estimates incorporate:

- The cost-adjusted operating-margin forecast;
- Rising depreciation;
- Higher recurring interest expense;
- Preferred dividends;
- An increasing diluted share count;
- No recurring contribution from equity-investment gains.

At the current blended reference price of approximately \$319.40, the shares trade at:

Multiple	Current Implied Valuation
Normalized NTM P/E	29.2x
FY2027 P/E	27.6x
FY2028 P/E	22.6x

Market commentary following the results placed Alphabet at approximately 24 times consensus forward earnings, near the upper part of a cited historical range of roughly 12x to 29x. The difference from this model reflects the model's lower cost-adjusted EPS assumptions, particularly higher depreciation and AI infrastructure costs.

10.2.4 Target P/E Framework

Alphabet's justified P/E depends on:

- Search durability;
- Google Cloud growth;
- AI monetization;
- Long-term operating margins;
- Free-cash-flow conversion;
- Capital intensity;
- Dilution;
- Regulatory risk.

Bear Case

Assumptions:

- AI infrastructure costs remain elevated;
- Cloud margin falls below 30%;
- Search growth slows faster;
- CapEx remains structurally high;
- Normalized NTM EPS falls to \$10.20;
- Target P/E contracts to 22x.

Bear value = \$10.20 × 22 = approximately \$224 per share

Base Case

Assumptions:

- Search remains resilient;
- Cloud operating income continues growing;
- Cloud margin moderates toward 30%;
- CapEx remains elevated through 2027;
- Free-cash-flow recovery is gradual;
- Normalized NTM EPS is \$10.95;
- Target P/E is 25x.

Base value = \$10.95 × 25 = approximately \$274 per share

Bull Case

Assumptions:

- Search AI monetization remains highly successful;
- Cloud backlog converts rapidly;
- Cloud margin remains above 32%;
- Infrastructure utilization offsets higher replacement costs;
- Normalized NTM EPS reaches \$11.70;
- Target P/E expands to 28x.

Bull value = \$11.70 × 28 = approximately \$328 per share

Scenario	Normalized NTM EPS	Target P/E	Implied Value
Bear	\$10.20	22x	\$224
Base	\$10.95	25x	\$274
Bull	\$11.70	28x	\$328

Using a probability distribution of 20% Bear, 60% Base and 20% Bull:

Probability-weighted P/E value = approximately \$275 per share

P/E Conclusion

At approximately \$319 per share, Alphabet is already pricing in an outcome close to the Bull Case under this cost-adjusted earnings model.

The current valuation provides limited protection if AI infrastructure costs remain high or if Cloud margin normalizes toward 30%.

10.3 EV/EBITDA Valuation

10.3.1 NTM EBITDA

The NTM EBITDA bridge is:

Metric	NTM Forecast
Operating Income	\$174.05B
Depreciation	\$39.90B
Estimated Amortization	\$1.80B
NTM EBITDA	\$215.75B

EBITDA is useful because it adds back the rapidly increasing depreciation expense associated with the AI infrastructure build-out.

However, EBITDA should not be treated as cash flow because it excludes:

- Capital expenditures;
- Working-capital investment;
- Interest;
- Taxes;
- Preferred dividends;
- TPU inventory spending;
- Infrastructure lease obligations.

10.3.2 Current Enterprise Value

Alphabet reported:

- \$242.5 billion of cash, cash equivalents and marketable securities;
- Approximately \$100.2 billion of current and long-term debt;
- \$19.0 billion of mandatory convertible preferred stock.

Headline Enterprise Value

Using all reported cash and marketable securities:

Headline EV = Market capitalization + Debt + Preferred stock – Cash and marketable securities

Headline EV = \$3.91T + \$0.10T + \$0.02T – \$0.24T

Headline EV = approximately \$3.79T

Headline NTM EV/EBITDA = \$3.79T ÷ \$215.75B = approximately 17.5x

Conservative Operating Enterprise Value

Approximately \$87.1 billion of Alphabet's marketable securities consisted of equity securities rather than cash or fixed-income securities.

Excluding marketable equities from operating liquidity:

Cash and debt securities = \$242.5B – \$87.1B = approximately \$155.4B

Conservative operating EV = approximately \$3.87T

Conservative NTM EV/EBITDA = approximately 17.9x

This is the more appropriate operating multiple before separately valuing Alphabet's equity-investment portfolio.

10.3.3 EV/EBITDA Target Valuation

The target-value bridge separately recognizes the investment portfolio:

Non-Operating Item	Base-Case Treatment
Cash and Debt Securities	\$155.4B
Marketable Equity Securities	$\$87.1\text{B} \times 85\% = \74.0B
Non-Marketable Securities	$\$131.5\text{B} \times 60\% = \78.9B
Debt	(\$100.2B)
Preferred Stock	(\$19.0B)
Accrued Legal and Regulatory Obligations	(\$17.4B)

Alphabet's non-marketable securities had a carrying value of \$131.5 billion, primarily concentrated in an investment in a private company. Because of liquidity, concentration, tax and valuation risk, the base model applies a 40% haircut.

Scenario	NTM EV/EBITDA	Operating EV	Equity Value per Share
Bear	14x	\$3.02T	\$254
Base	16x	\$3.45T	\$289
Bull	18x	\$3.88T	\$323

The base case uses a 16x NTM EBITDA multiple.

This represents a premium to a mature advertising business but a discount to a pure high-growth AI infrastructure or software company.

EV/EBITDA Conclusion

The EV/EBITDA method produces a base value of approximately **\$289 per share**.

At the current price, Alphabet is trading close to the 18x Bull Case result.

10.4 Free Cash Flow Yield

10.4.1 Reported FCF Yield

Alphabet reported trailing-12-month operating cash flow of **\$185.7 billion**, CapEx of **\$132.4 billion**, and free cash flow of **\$53.3 billion**.

Reported TTM FCF yield = $\$53.3\text{B} \div \3.91T

Reported TTM FCF yield = approximately 1.36%

Metric	Reported TTM
Operating Cash Flow	\$185.7B
Capital Expenditures	(\$132.4B)
Free Cash Flow	\$53.3B
Market Capitalization	Approximately \$3.91T
Reported FCF Yield	1.36%

This yield is low relative to Alphabet's historical cash-generation profile.

It reflects the beginning of the current AI investment cycle but does not yet include the entire expected effect of the higher 2026 and 2027 CapEx program.

10.4.2 Normalized Near-Term FCF Yield

The model forecasts NTM operating cash flow of approximately **\$200 billion**.

The estimate reflects:

- Normalized operating earnings;
- Depreciation and amortization;
- Stock-based compensation;
- Working-capital requirements;
- Higher interest and tax payments;
- No cash contribution from unrealized investment gains.

The NTM CapEx forecast is approximately **\$262 billion**, based on:

- Approximately \$119 billion of H2 2026 CapEx;
- Approximately \$143 billion of H1 2027 CapEx.

Normalized NTM FCF = \$200B – \$262B = approximately negative \$62B

Normalized NTM FCF yield = negative \$62B ÷ \$3.91T

Normalized NTM FCF yield = approximately negative 1.6%

Metric	Normalized NTM
Normalized Operating Cash Flow	Approximately \$200B
Forecast Capital Expenditures	Approximately (\$262B)
Normalized Free Cash Flow	Approximately (\$62B)
Normalized FCF Yield	Approximately –1.6%

This estimate is more negative than reported TTM FCF because the model incorporates the expected acceleration in capital expenditures.

Alphabet's Q2 FCF was already negative \$5.9 billion, and management increased 2026 CapEx guidance while indicating that spending will increase significantly again in 2027.

10.4.3 CapEx-Normalized FCF Yield

Current CapEx includes an unusually large capacity-build phase.

To estimate a more mature free-cash-flow level, the model replaces peak investment spending with a long-term AI infrastructure CapEx assumption of **22% of revenue**.

This rate is still substantially higher than Alphabet's historical pre-AI capital intensity and therefore does not assume a return to its former asset-light economics.

NTM revenue is approximately:

$$\text{\$125.4B} + \text{\$139.1B} + \text{\$133.6B} + \text{\$144.9B} = \text{approximately } \text{\$543.0B}$$

Normalized CapEx is:

$$\text{\$543.0B} \times 22\% = \text{approximately } \text{\$119.5B}$$

CapEx-normalized FCF is:

$$\text{\$200.0B normalized operating cash flow} - \text{\$119.5B normalized CapEx}$$

$$\text{CapEx-normalized FCF} = \text{approximately } \text{\$80.5B}$$

$$\text{CapEx-normalized FCF yield} = \text{\$80.5B} \div \text{\$3.91T}$$

$$\text{CapEx-normalized FCF yield} = \text{approximately } 2.1\%$$

FCF Measure	Free Cash Flow	FCF Yield
Reported TTM	\$53.3B	1.36%
Normalized NTM at Planned CapEx	Approximately (\$62B)	Approximately -1.6%
CapEx-Normalized	Approximately \$80.5B	Approximately 2.1%

Maintenance-CapEx Sensitivity

A more generous owner-earnings method could assume that maintenance CapEx is approximately 1.25 times NTM depreciation and amortization:

$$\text{Maintenance CapEx} = \text{approximately } \text{\$52B}$$

This would produce owner earnings of approximately:

$$\text{\$200B} - \text{\$52B} = \text{approximately } \text{\$148B}$$

The implied owner-earnings yield would be approximately **3.8%**.

However, this method is likely too optimistic for a company whose competitive position increasingly depends on continuously upgrading AI servers, memory, networking, optical equipment and power capacity.

FCF Conclusion

The FCF method produces the widest valuation range because the distinction between growth and maintenance CapEx is uncertain.

Using a 2.25%–2.75% required normalized FCF yield produces an approximate equity-value range of:

Target FCF Yield	Implied Value per Share
2.25%	Approximately \$285
2.50%	Approximately \$257
2.75%	Approximately \$233

A reasonable central FCF value is approximately **\$255–\$275 per share**, with a higher value approaching \$295 only if a large part of current CapEx is temporary rather than recurring.

10.5 Sum-of-the-Parts Valuation

10.5.1 Why SOTP Is Appropriate

Alphabet's businesses have increasingly different financial characteristics.

Business	Growth Profile	Margin Profile	Capital Intensity
Google Search and Services	Mid-teens growth	Approximately 40% margin	Moderate but rising
Google Cloud	High growth	Approximately 30% forward margin	Very high
Other Bets and Waymo	Early-stage	Loss-making	High
Equity Investments	Non-operating	Valuation-driven	Financial assets

A single consolidated P/E may undervalue Cloud while simultaneously overvaluing:

- Shared AI research expenses;
- Other Bets losses;
- Investment gains;
- Peak infrastructure spending.

SOTP is therefore useful because it assigns different multiples and risk adjustments to each component.

10.5.2 2027 Segment Forecast Base

Segment	2027E Revenue	2027E Operating Margin	2027E Operating Income
Google Services	\$438.2B	40.2%	\$176.1B
Google Cloud	\$150.9B	30.5%	\$46.0B
Other Bets	\$1.6B	Negative	Loss-making
Consolidated Alphabet	\$590.7B	31.6%	\$186.5B

The difference between Services plus Cloud operating income and consolidated operating income is approximately **\$35.7 billion**.

This represents, conceptually:

- Alphabet-level shared AI R&D;
- Corporate costs;
- Other Bets losses;
- Unallocated activities.

This cost burden must be subtracted in the SOTP model because Cloud's reported segment margin is not fully loaded with all shared AI development costs.

10.5.3 Base SOTP Valuation

Google Services

2027 Services operating income:

\$176.1B

Base multiple:

16x EBIT

Services value:

\$176.1B × 16 = approximately \$2.82T

The multiple reflects:

- Search durability;
- Strong advertising margins;
- Subscription growth;
- AI disruption risk;
- Regulatory risk;
- Increasing Search inference costs.

Google Cloud

2027 Cloud operating income:

\$46.0B

Base multiple:

28x EBIT

Cloud value:

\$46.0B × 28 = approximately \$1.29T

The premium reflects:

- Rapid growth;
- High backlog;
- Enterprise AI adoption;
- Improving scale;
- Strategic infrastructure value.

The multiple is constrained by:

- High capital intensity;
- TPU hardware mix;
- Third-party capacity;
- Replacement-cost inflation;
- Uncertainty regarding fully loaded AI margins.

Other Bets and Waymo

Base estimated strategic value:

\$60B

This value is not derived from current Other Bets revenue or operating profit.

It represents an option value for:

- Waymo;
- Wing;
- Other early-stage businesses.

The valuation remains highly uncertain because Alphabet does not separately disclose Waymo revenue, paid rides, fleet utilization or contribution margins.

Shared AI R&D and Corporate Costs

2027 unallocated operating loss:

Approximately \$35.7B

Base capitalization multiple:

10x

Value adjustment:

Approximately negative \$357B

This adjustment prevents the SOTP from overstating value by applying premium multiples to Services and Cloud while ignoring shared model-development and corporate expenses.

Equity-Investment Portfolio

Investment Asset	Carrying Value	Base Haircut	SOTP Value
Marketable Equity Securities	\$87.1B	15%	\$74.0B
Non-Marketable Securities	\$131.5B	40%	\$78.9B
Total Investment Value	\$218.5B	30.0% weighted average	\$152.9B

The non-marketable portfolio is given a larger haircut because:

- The holdings are less liquid;
- Valuations depend on private transactions;
- The portfolio is concentrated;
- Tax may be payable when investments are realized;
- Large positions may not be saleable at carrying value.

Alphabet reported that the non-marketable portfolio primarily consisted of an investment in a private company and had experienced substantial unrealized valuation gains.

Net Operating Liquidity

Component	Value
Cash and Debt Securities	\$155.4B
Less: Debt	(\$100.2B)
Less: Preferred Stock	(\$19.0B)
Net Operating Liquidity	\$36.2B

The model also subtracts \$17.4 billion of short-term accrued legal and regulatory fines and settlements.

10.5.4 Base SOTP Bridge

Component	Base Value
Google Services	\$2.82T
Google Cloud	\$1.29T
Other Bets and Waymo	\$0.06T
Less: Shared AI R&D and Corporate Burden	(\$0.36T)
Equity-Investment Portfolio	\$0.15T

Component	Base Value
Net Operating Liquidity	\$0.04T
Less: Accrued Legal and Regulatory Obligations	(\$0.02T)
Total Equity Value	Approximately \$3.98T
NTM Diluted Shares	12.55B
SOTP Value per Share	Approximately \$317

The base SOTP value is close to the current share price.

This suggests that the market may already be assigning:

- A premium valuation to Google Cloud;
- Meaningful value to Alphabet's investment portfolio;
- A substantial value to Search durability;
- Only limited discount for current negative FCF.

10.5.5 SOTP Scenario Analysis

Assumption	Bear	Base	Bull
Services EBIT Multiple	14x	16x	18x
Cloud EBIT Multiple	22x	28x	34x
Other Bets Value	\$40B	\$60B	\$120B
Shared Cost Multiple	12x	10x	8x
Marketable Equity Recovery	70%	85%	90%
Non-Marketable Recovery	40%	60%	75%
SOTP Value per Share	\$257	\$317	\$380

The SOTP range is wide because Cloud's appropriate multiple and the realizable value of the investment portfolio are highly uncertain.

SOTP Conclusion

SOTP produces a higher valuation than the P/E and FCF methods because it separately recognizes:

- Cloud's high-growth value;
- Waymo optionality;
- Equity-investment assets.

However, it also depends on optimistic assumptions that those assets can be monetized without:

- Significant taxes;

- Liquidity discounts;
- Lower future valuations;
- Additional capital requirements.

10.6 Composite Valuation

The four valuation methods should not receive equal weight.

Method	Base Value	Weight	Weighted Contribution
Normalized P/E	\$274	30%	\$82
EV/EBITDA	\$289	25%	\$72
FCF Yield	\$275	20%	\$55
SOTP	\$317	25%	\$79
Composite Value	\$289	100%	Approximately \$289

Rounded composite fair value:

Approximately \$290 per share

Relative to the blended reference price of approximately \$319.40:

Base-case downside = approximately 9%

10.7 Valuation Range

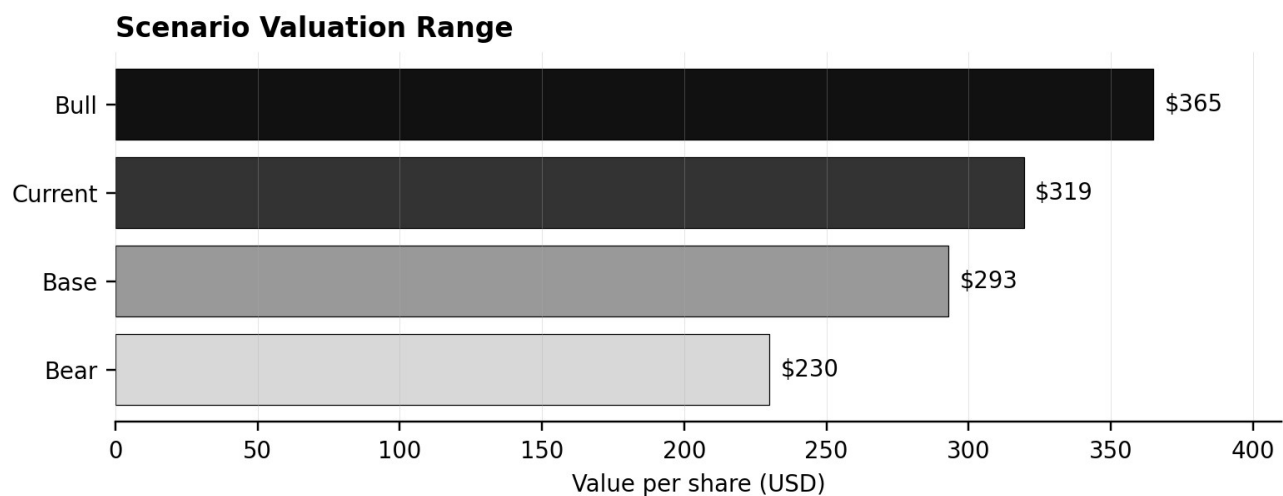


Figure 4. Valuation range across normalized earnings, enterprise-value, free-cash-flow and SOTP methodologies.

Scenario	Fair Value Range	Primary Assumptions
Bear	\$220–\$255	Lower Search growth, Cloud margin below 30%, continued negative FCF
Base	\$275–\$300	Strong growth, but elevated CapEx and margin pressure
Bull	\$325–\$380	Strong backlog conversion, resilient Cloud margin and better AI ROIC

The current price is:

- Above the base P/E valuation;
- Above the base EV/EBITDA valuation;
- Above the base FCF valuation;
- Approximately equal to the base SOTP valuation;
- Below the Bull SOTP outcome.

This means that the stock is currently dependent on successful execution rather than valuation support alone.

10.8 Key Valuation Sensitivities

Normalized EPS Sensitivity

NTM EPS	23x P/E	25x P/E	27x P/E
\$10.00	\$230	\$250	\$270
\$11.00	\$253	\$275	\$297
\$12.00	\$276	\$300	\$324
\$13.00	\$299	\$325	\$351

Cloud Multiple Sensitivity

Every five-turn change in the Cloud EBIT multiple changes the SOTP by approximately:

$$\text{\$46.0B Cloud EBIT} \times 5 = \text{\$230B}$$

Per-share effect:

$$\text{\$230B} \div \text{12.55B shares} = \text{approximately \$18 per share}$$

Investment-Portfolio Sensitivity

Every 10% change in the recoverable value of the \$218.5 billion equity-investment portfolio changes equity value by approximately:

$$\text{\$21.9B, or approximately \$1.74 per share}$$

Operating-Margin Sensitivity

A 100 bps change in 2027 consolidated operating margin changes operating income by approximately:

\$590.7B revenue × 1% = approximately \$5.9B

After tax, the EPS effect is approximately:

\$5.9B × 81% ÷ 12.65B shares = approximately \$0.38 per share

At a 25x P/E, this corresponds to approximately:

\$9.50 per share of valuation impact

10.9 Valuation Risks

Upside Risks

- Search AI increases commercial queries and advertising yield;
- Cloud growth remains above the forecast;
- Cloud margin remains above 32%;
- TPU systems carry attractive margins;
- Third-party capacity is temporary;
- Component prices fall;
- CapEx peaks earlier;
- Waymo disclosure supports a higher standalone value;
- Investment holdings continue appreciating;
- Share repurchases resume.

Downside Risks

- Search growth slows below the low teens;
 - AI queries carry weaker monetization or higher costs;
 - Cloud growth normalizes faster;
 - TPU hardware dilutes margins;
 - Third-party capacity becomes permanent;
 - Higher-cost infrastructure produces more depreciation than expected;
 - CapEx remains above \$300 billion;
 - Negative FCF persists;
 - Additional common equity is issued;
 - Private-equity investments are revalued downward;
 - Regulatory and legal obligations increase.
-

10.10 Overall Valuation Conclusion

Alphabet's Q2 2026 headline GAAP earnings materially overstate recurring profitability because investment gains accounted for most of the reported net income.

The appropriate valuation starting points are:

- Approximately \$10.95 of normalized NTM EPS;
- Approximately \$215.8 billion of NTM EBITDA;
- \$53.3 billion of reported TTM FCF;
- Negative modeled NTM FCF under the current CapEx program;

- A separately valued equity-investment portfolio.

The four valuation methods tell different stories:

Normalized P/E

Alphabet appears expensive because the current share price represents approximately 29x the model's cost-adjusted NTM EPS.

EV/EBITDA

The shares trade close to the upper end of a reasonable operating multiple range.

FCF Yield

Current and forecast FCF yields are weak because the AI infrastructure build-out absorbs most or all operating cash flow.

SOTP

The current valuation is more defensible if investors assign premium value to Google Cloud, Waymo and Alphabet's investment portfolio.

The valuation can be summarized as:

Strong Search franchise

plus

High-value Google Cloud growth

plus

Large financial-investment assets

minus

Peak AI CapEx, negative near-term FCF, rising depreciation and dilution

The base composite value is approximately:

\$290 per share

The current price near \$319 implies that investors are already assuming:

- Continued strong Cloud growth;
- Successful AI Search monetization;
- Eventual CapEx normalization;
- No major deterioration in Cloud margins;
- Meaningful recoverable value from the investment portfolio.

The central investment conclusion is:

Alphabet is not clearly overvalued under a premium SOTP framework, but it is expensive under normalized earnings and cash-flow frameworks that fully recognize the rising cost of AI infrastructure.

A sustainable valuation above the current price requires evidence that:

1. Cloud operating income continues growing despite margin pressure;
2. Search AI monetization remains profitable;
3. CapEx-to-revenue begins declining;
4. Free cash flow recovers;

5. Future equity issuance and dilution remain limited.

11. Bull, Base and Bear Scenario Analysis

11.1 Executive Assessment

Alphabet’s valuation depends increasingly on the interaction between five variables:

1. The durability of Google Search growth;
2. The growth and profitability of Google Cloud;
3. The cost and utilization of AI infrastructure;
4. The timing of free-cash-flow recovery;
5. The valuation multiple investors are willing to apply to a more capital-intensive business.

The three scenarios are designed to capture different outcomes for these variables.

Management’s Q2 outlook supports both the upside and downside cases. Google Cloud demand remains exceptionally strong, Search AI monetization is progressing, and management has become more optimistic about long-term AI returns. At the same time, Alphabet raised 2026 CapEx guidance to \$195–\$205 billion, expects another significant increase in 2027, and warned that third-party capacity will pressure near-term Cloud margins and free cash flow.

The scenario structure is:

- Bull Case:** AI expands Alphabet's addressable market and installed capacity generates attractive returns.
- Base Case:** AI creates meaningful revenue growth, but high capital requirements delay the cash return to shareholders.
- Bear Case:** AI increases the cost of protecting Search and competing in Cloud faster than it creates incremental profit.

11.2 Scenario Summary

Key Variable	Bull Case	Base Case	Bear Case
Search Growth	AI expands queries, commercial intent and monetization	Search remains resilient but gradually normalizes	AI assistants divert traditional queries and reduce monetization
YouTube Growth	CTV, Shorts, commerce and subscriptions accelerate	High-single-digit to low-double-digit growth continues	Advertising demand slows and Shorts monetization disappoints
Consumer Subscriptions	Gemini and YouTube paid conversion accelerates	Recurring subscriptions continue growing steadily	Free AI alternatives limit paid conversion

Key Variable	Bull Case	Base Case	Bear Case
Cloud Growth	Enterprise AI and infrastructure demand remain exceptionally strong	Growth remains high but gradually normalizes	Large contracts slow, utilization disappoints or deliveries are delayed
Cloud Operating Margin	Scale, utilization and custom silicon expand margins	Margin falls toward 30% before partially recovering	Third-party capacity, hardware mix and depreciation materially compress margin
Services Operating Margin	AI monetization exceeds incremental inference cost	Margin remains near 40%	AI query costs rise faster than advertising revenue
AI Infrastructure Costs	Higher costs are offset by performance and utilization	Costs remain elevated but manageable	HBM, accelerators, optics, energy and capacity costs remain structurally high
CapEx	High but increasingly productive	Remains elevated for several years	Continues rising without adequate incremental returns
Free Cash Flow	Recovers strongly after the main build-out	Recovers slowly after prolonged pressure	Remains weak or negative
Financing	Debt and equity issuance stabilize	Financing remains available but dilution persists	Further equity issuance and leverage are required
Valuation Multiple	Expands as AI ROIC becomes visible	Remains broadly stable	Compresses because of lower growth and cash conversion
Equity Outcome	AI platform leader with strong cash recovery	Strong company at a demanding valuation	Capital-intensive growth fails to justify valuation

11.3 Bull Case

11.3.1 Operating Thesis

The Bull Case assumes that Alphabet's AI investments generate both offensive and defensive returns.

AI-powered Search increases:

- Total queries;
- Commercial-query volume;
- Advertising relevance;

- Conversion rates;
- Revenue per user.

Google Cloud remains capacity-constrained because enterprise AI demand continues exceeding available infrastructure. Backlog converts rapidly, internal capacity replaces higher-cost third-party capacity, and TPU economics improve through scale and utilization.

Consumer Gemini, Workspace AI, Vertex AI and security products become material revenue contributors.

The Bull Case does not assume that CapEx becomes low. It assumes that high CapEx produces proportionately higher revenue, operating profit and future cash flow.

11.3.2 Bull-Case Assumptions

Metric	Bull-Case Assumption
2027 Search Growth	15%–17%
2028 Search Growth	13%–15%
2027 YouTube Ads Growth	12%–14%
2027 Subscription Growth	16%–18%
2027 Cloud Growth	55%–65%
2028 Cloud Growth	35%–40%
2027 Services Margin	42%–43%
2027 Cloud Margin	33%–35%
2028 Cloud Margin	35%–37%
2027 Consolidated Operating Margin	34.5%–35.5%
2028 Consolidated Operating Margin	36%–37%
2027 CapEx	\$255B–\$275B
2028 CapEx	\$270B–\$295B
Free Cash Flow	Improves materially after 2027
Target Normalized P/E	28x–30x
Target EV/EBITDA	18x–19x

Search

The Bull Case assumes AI Overviews and AI Mode create incremental queries without reducing traditional advertising economics.

Positive evidence would include:

- Search growth remaining above 15%;
- Continued paid-click growth;
- Stable or rising CPC;
- Lower cost per AI response;
- Wider deployment of advertisements in AI Mode;
- Stable Google Services margins.

Google Cloud

Cloud remains Alphabet's primary upside driver.

The Bull Case assumes:

- Backlog converts faster than expected;
- Third-party capacity allows Alphabet to capture demand immediately;
- Internal capacity comes online on schedule;
- TPU product revenue adds to growth without materially diluting margins;
- Enterprise AI adoption expands beyond pilot programs;
- Cloud operating income continues growing faster than CapEx-related costs.

CapEx and FCF

CapEx remains high but becomes more productive.

The critical relationship becomes:

Incremental AI operating profit growth > Incremental depreciation and infrastructure cost growth

Operating cash flow eventually grows faster than CapEx, allowing:

- FCF recovery;
- Reduced external financing;
- Resumption of share repurchases;
- Lower dilution risk.

11.3.3 Bull-Case Financial Output

Metric	2027E Bull	2028E Bull
Revenue	Approximately \$630B–\$650B	Approximately \$760B–\$790B
Operating Margin	Approximately 35%	Approximately 37%
Operating Income	Approximately \$220B–\$228B	Approximately \$281B–\$292B
Normalized EPS	Approximately \$13.5–\$14.5	Approximately \$18–\$19
Free Cash Flow	Low positive to improving	Approximately \$70B–\$100B

These figures are scenario assumptions rather than company guidance.

11.3.4 Bull-Case Valuation

Method	Bull-Case Value
Normalized P/E	Approximately \$330–\$355
EV/EBITDA	Approximately \$330–\$355
FCF Yield	Approximately \$340–\$380
SOTP	Approximately \$375–\$390
Composite Bull Value	Approximately \$365 per share

The Bull Case requires the market to conclude that Alphabet's current negative or weak free cash flow is temporary rather than structural.

11.4 Base Case

11.4.1 Operating Thesis

The Base Case assumes that Alphabet monetizes AI successfully, but that the financial return develops more slowly than the revenue growth.

Search remains resilient and continues growing in the low-to-mid teens. AI does not destroy the traditional Search franchise, but higher inference, depreciation and infrastructure costs prevent a major expansion in Services margin.

Google Cloud remains the fastest-growing segment, but the growth rate gradually normalizes as the revenue base becomes larger. Cloud operating margin declines from Q2's exceptional 35.6% toward approximately 30% because of:

- Third-party capacity;
- Wiz integration;
- Higher depreciation;
- Energy and networking costs;
- TPU hardware revenue;
- Higher AI talent expense.

Alphabet reported 82% Q2 Cloud growth and a 35.6% segment margin, but management also warned that higher-cost external capacity would pressure margins.

The Base Case assumes that absolute Cloud operating income continues increasing even as the margin percentage temporarily declines.

11.4.2 Base-Case Assumptions

Metric	Base-Case Assumption
2027 Search Growth	12%–14%
2028 Search Growth	10%–12%
2027 YouTube Ads Growth	9%–11%

Metric	Base-Case Assumption
2027 Subscription Growth	13%–15%
2027 Cloud Growth	40%–50%
2028 Cloud Growth	28%–35%
2027 Services Margin	Approximately 40%
2027 Cloud Margin	Approximately 30%–31%
2028 Cloud Margin	Approximately 32%–33%
2027 Consolidated Operating Margin	Approximately 31%–32%
2028 Consolidated Operating Margin	Approximately 33%–34%
2027 CapEx	Approximately \$285B
2028 CapEx	Approximately \$320B
Free Cash Flow	Weak in 2027; gradual recovery in 2028
Target Normalized P/E	24x–26x
Target EV/EBITDA	15x–17x

Search

AI features expand engagement, but reported growth gradually moderates because of:

- Higher comparison bases;
- Increasing scale;
- Foreign-exchange variability;
- Partial substitution of traditional search behavior;
- More cautious advertiser pricing.

Cloud

Cloud growth remains high but becomes increasingly mixed.

Revenue includes:

- Recurring GCP consumption;
- Enterprise AI services;
- Workspace;
- Security and Wiz;
- TPU product sales.

The Base Case assumes that TPU revenue increases total Cloud growth but carries lower margins than recurring platform and software revenue.

CapEx and FCF

CapEx remains elevated through at least 2027 because Alphabet continues building:

- Servers;
- TPUs and GPUs;
- HBM-intensive systems;
- Networking and optical capacity;
- Data centers;
- Power infrastructure.

Free cash flow recovers only gradually because higher operating cash flow is partly offset by:

- Increased CapEx;
- Working-capital requirements;
- TPU inventory;
- Interest expense;
- Preferred dividends.

11.4.3 Base-Case Financial Output

Metric	2027E Base	2028E Base
Revenue	Approximately \$591B	Approximately \$683B
Operating Margin	Approximately 31.6%	Approximately 33.5%
Operating Income	Approximately \$187B	Approximately \$229B
Normalized EPS	Approximately \$11.6	Approximately \$14.2
Free Cash Flow	Near breakeven to modestly negative	Gradually improves

These results are based on the revenue and cost-adjusted profit models developed in Section 9.

11.4.4 Base-Case Valuation

Method	Base-Case Value
Normalized P/E	Approximately \$274
EV/EBITDA	Approximately \$289
FCF Yield	Approximately \$255–\$295
SOTP	Approximately \$317
Composite Base Value	Approximately \$290 per share

The Base Case assumes that Alphabet remains a high-quality compounder, but that its greater capital intensity prevents a return to the premium cash-flow profile investors previously associated with the company.

11.5 Bear Case

11.5.1 Operating Thesis

The Bear Case does not require Alphabet's revenue to collapse.

It assumes that AI changes the economic structure of the business in an unfavorable way.

Search may continue generating substantial revenue, but:

- Users increasingly begin discovery through competing AI assistants;
- AI queries produce fewer traditional paid clicks;
- Commercial queries become more expensive to serve;
- Advertising revenue per query fails to offset inference costs;
- TAC and distribution costs remain elevated.

Cloud demand also weakens from current exceptional levels.

Large customers may:

- Delay deployments;
- Shift workloads across multiple providers;
- Negotiate lower token and compute prices;
- Reduce excess contractual commitments;
- Favor competing models or infrastructure.

At the same time, Alphabet remains committed to large, long-duration investments and contractual capacity obligations.

11.5.2 Bear-Case Assumptions

Metric	Bear-Case Assumption
2027 Search Growth	6%–9%
2028 Search Growth	4%–7%
2027 YouTube Ads Growth	3%–6%
2027 Subscription Growth	7%–10%
2027 Cloud Growth	20%–30%
2028 Cloud Growth	15%–22%
2027 Services Margin	35%–38%
2027 Cloud Margin	23%–27%
2028 Cloud Margin	22%–27%
2027 Consolidated Operating Margin	27%–29%
2028 Consolidated Operating Margin	26%–29%
2027 CapEx	\$325B–\$350B

Metric	Bear-Case Assumption
2028 CapEx	\$340B–\$375B
Free Cash Flow	Negative or structurally weak
Target Normalized P/E	19x–22x
Target EV/EBITDA	12x–14x

Search

The Bear Case assumes AI engagement is real but less economically valuable than traditional Search.

A negative unit-economic relationship emerges:

Incremental AI advertising revenue < Incremental inference, depreciation and model-development costs

Search remains a large business, but operating leverage declines.

Google Cloud

Cloud growth slows before Alphabet's capacity program reaches full utilization.

The principal risks are:

- Slower backlog conversion;
- Excess capacity;
- Permanent dependence on expensive third-party infrastructure;
- Lower compute and token pricing;
- Lower-margin TPU hardware mix;
- Customer concentration;
- Higher energy costs.

CapEx and FCF

CapEx remains high because Alphabet must complete projects already under construction and satisfy supply, energy and lease commitments.

This creates the most dangerous timing mismatch:

Revenue growth slows before capital spending and depreciation decline

Free cash flow remains negative, and Alphabet may require:

- Additional debt;
- Further common-equity issuance;
- Greater use of preferred securities;
- Prolonged suspension of repurchases.

11.5.3 Bear-Case Financial Output

Metric	2027E Bear	2028E Bear
Revenue	Approximately \$520B–\$550B	Approximately \$570B–\$610B

Metric	2027E Bear	2028E Bear
Operating Income	Approximately \$146B–\$154B	Approximately \$154B–\$165B
Normalized EPS	Approximately \$8.5–\$9.5	Approximately \$9–\$10
Free Cash Flow	Negative	Negative or very weak

11.5.4 Bear-Case Valuation

Method	Bear-Case Value
Normalized P/E	Approximately \$190–\$210
EV/EBITDA	Approximately \$210–\$230
FCF Yield	Approximately \$190–\$225
SOTP	Approximately \$250–\$260
Composite Bear Value	Approximately \$230 per share

SOTP remains higher than the other methods because Search, Cloud, Waymo and the investment portfolio retain significant asset value even in a weaker operating environment.

11.6 Detailed Scenario Comparison

Metric	Bull	Base	Bear
2027 Revenue	\$630B–\$650B	Approximately \$591B	\$520B–\$550B
2027 Revenue Growth	Approximately 25%–29%	Approximately 19.5%	Approximately 5%–12%
2027 Search Growth	15%–17%	12%–14%	6%–9%
2027 Cloud Growth	55%–65%	40%–50%	20%–30%
2027 Services Margin	42%–43%	Approximately 40%	35%–38%
2027 Cloud Margin	33%–35%	30%–31%	23%–27%
2027 Operating Margin	Approximately 35%	Approximately 31.6%	Approximately 28%

Metric	Bull	Base	Bear
2027 CapEx	\$255B–\$275B	Approximately \$285B	\$325B–\$350B
2028 FCF	\$70B–\$100B	Gradual recovery	Negative or weak
Normalized P/E	28x–30x	24x–26x	19x–22x
Composite Value	\$365	\$290	\$230

11.7 Probability-Weighted Valuation

The probability-weighted value is calculated as:

Probability-Weighted Value = Bull Probability × Bull Value + Base Probability × Base Value + Bear Probability × Bear Value

The model assigns the following probabilities:

Scenario	Probability	Target Value	Weighted Contribution
Bull	20%	\$365	\$73
Base	60%	\$290	\$174
Bear	20%	\$230	\$46
Probability-Weighted Value	100%	\$293 per share	\$293 per share

Calculation:

$$20\% \times \$365 + 60\% \times \$290 + 20\% \times \$230 = \$293$$

Based on the July 24, 2026 closing prices of approximately \$319 for GOOG and GOOGL, the probability-weighted valuation implies approximately **8% downside**.

This does not mean the stock must decline to \$293.

It means:

At the current price, the market is assigning either a higher probability to the Bull Case or a higher valuation multiple to the Base Case than this model assumes.

11.8 Market-Implied Scenario

The current share price of approximately \$319 is close to:

- The Base SOTP value;
- The upper end of the Base valuation range;
- The lower end of the Bull valuation range.

The price can be reverse-engineered using the scenario values.

Assuming Bear and Base probabilities remain 20% and 60%, a price near \$319 would require a Bull Case value or probability materially above the current assumptions.

Alternatively, if the scenario values remain unchanged, the market-implied probabilities would need to shift approximately toward:

Scenario	Model Probability	Illustrative Market-Implied Probability
Bull	20%	Approximately 35%–40%
Base	60%	Approximately 50%–55%
Bear	20%	Approximately 10%

This is an approximate inference rather than an observable market statistic.

The implication is:

The market is pricing Alphabet as though successful AI monetization and eventual cash-flow recovery are considerably more likely than a prolonged capital-return disappointment.

11.9 Scenario Upgrade and Downgrade Triggers

Indicators Supporting a Shift Toward the Bull Case

Indicator	Bull Trigger
Search Growth	Remains above 15% despite tougher comparisons
AI Search Economics	Cost per response falls and ad monetization expands
Cloud Growth	Remains above 55% for multiple quarters
Cloud Margin	Remains above 33% despite third-party capacity
Backlog Conversion	Revenue growth remains consistent with contracted demand
TPU Revenue	Ramps without meaningful margin dilution
2027 CapEx	Below approximately \$280B
Operating Cash Flow	Grows faster than CapEx
Free Cash Flow	Turns sustainably positive
Capital Returns	Buybacks resume without additional equity issuance

Indicators Supporting a Shift Toward the Bear Case

Indicator	Bear Trigger
Search Growth	Falls below 10%
Paid Clicks and CPC	Both weaken materially
AI Search Cost	Fails to decline
Cloud Growth	Falls below 35% earlier than expected
Cloud Margin	Falls below 28%
Third-Party Capacity	Becomes a permanent broad-based requirement
TPU Sales	Large revenue but low margin or delayed delivery
2027 CapEx	Exceeds approximately \$325B
Depreciation Growth	Consistently exceeds revenue growth
Free Cash Flow	Remains deeply negative through 2028
Financing	Additional major common-equity issuance
Backlog	Growth slows or conversion disappoints

11.10 Scenario Risk Matrix

Variable	Bull Outcome	Base Outcome	Bear Outcome
AI Search	Expands revenue and margins	Preserves Search growth	Cannibalizes traditional economics
Gemini Consumer	Material paid subscription business	Modest recurring revenue	High usage but weak conversion
Enterprise AI	Production adoption expands rapidly	Gradual deployment	Pilots fail to convert
Cloud Capacity	High utilization	Adequate utilization	Excess capacity
TPU Economics	Strong proprietary advantage	Mixed hardware and service economics	Low-margin product exposure
AI Component Prices	Moderate or offset by performance	Remain elevated	Rise faster than efficiency

Variable	Bull Outcome	Base Outcome	Bear Outcome
Depreciation	Absorbed by revenue growth	Temporary margin pressure	Structural margin compression
Contract Commitments	Secure profitable capacity	Manageable fixed obligations	Excess-cost burden
Financing	Stabilizes	Remains manageable	Dilution and leverage increase
FCF	Strong recovery	Slow recovery	Persistently weak

11.11 Overall Scenario Conclusion

Alphabet's investment case has meaningful upside and downside asymmetry because the company is simultaneously:

- Protecting the world's largest digital advertising franchise;
- Building a rapidly expanding enterprise Cloud platform;
- Funding one of the largest infrastructure programs in corporate history;
- Carrying substantial financial investment assets;
- Assuming significant supply, lease, energy and data-center commitments.

The Bull Case requires evidence that:

AI revenue growth and infrastructure utilization exceed the growth in depreciation, energy, model-development and financing costs.

The Base Case assumes that:

AI creates significant value, but cash returns are delayed by a prolonged capital-investment cycle.

The Bear Case assumes that:

AI increases Alphabet's competitive cost structure faster than it increases sustainable per-share earnings and free cash flow.

The probability-weighted fair value is approximately:

\$293 per share

compared with a current market price near \$319.

The most important conclusion is:

Alphabet's current valuation leaves limited room for execution errors. The stock can justify or exceed its current price if Search AI monetization remains strong, Cloud margins remain resilient and free cash flow begins recovering after 2027. If CapEx and depreciation continue increasing without corresponding cash returns, valuation compression becomes the primary downside risk.

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Financial forecasts, scenario values and valuation ranges are analytical estimates based on assumptions stated in the report. Actual results may differ materially because of changes in advertising demand, Cloud consumption, AI infrastructure costs, capital expenditures, regulation, litigation, financing, foreign exchange, competition and other factors.

Reported company data should be read together with Alphabet Inc. filings, earnings releases and investor-relations materials. Consensus figures and market-derived estimates may change after publication. All monetary amounts are in U.S. dollars unless otherwise stated.

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